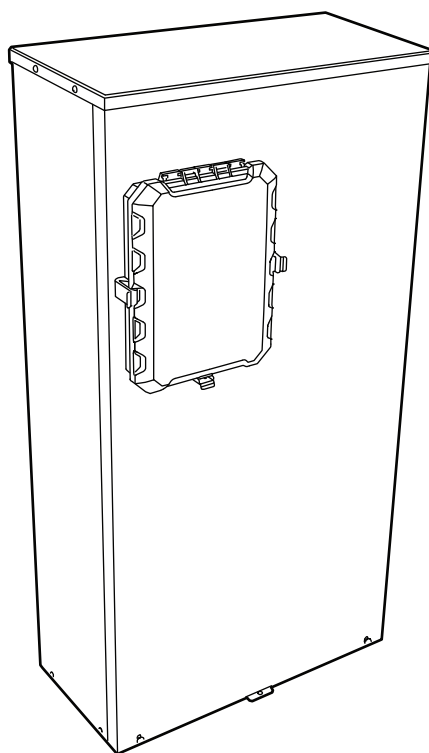


**TX Series Transfer Switches and Accessories
100–1600 Amp**



models:

100–400 Amp
600 Amp
800–1600 Amp

TRANSFER SWITCHES

Safety

Throughout this publication and on tags and decals affixed to the generator, DANGER, WARNING, and CAUTION blocks are used to alert personnel to special instructions about a particular operation that may be hazardous if performed incorrectly or carelessly. Observe them carefully. Their definitions are as follows:

DANGER

Indicates a hazardous situation which, if not avoided, will result in death or serious injury.

(000001)

WARNING

Indicates a hazardous situation which, if not avoided, could result in death or serious injury.

(000002)

CAUTION

Indicates a hazardous situation which, if not avoided, could result in minor or moderate injury.

(000003)

NOTE: Notes provide additional information important to a procedure or component.

These safety alerts cannot eliminate the hazards they indicate. Observing safety precautions and strict compliance with the special instructions while performing the action or service are essential to preventing accidents.

WARNING

CANCER AND REPRODUCTIVE HARM

www.P65Warnings.ca.gov

(000393a)

Read This Manual Thoroughly

This diagnostic manual has been written and published by Generac to aid dealer technicians and company service personnel when servicing the products described herein.

It is assumed that these personnel are familiar with the servicing procedures for these products, or like or similar products manufactured and marketed by Generac, and that they have been trained in the recommended servicing procedures for these products, including the use of common hand tools and any special Generac tools or tools from other suppliers.

Generac could not possibly know of and advise the service trade of all conceivable procedures by which a service might be performed and of the possible hazards and/or results of each method. We have not undertaken any such wide evaluation. Therefore, anyone who uses a procedure or tool not recommended by Generac must first satisfy themselves that neither his nor the products safety will be endangered by the service procedure selected.

All information, illustrations and specifications in this manual are based on the latest product information available at the time of publication.

When working on these products, remember that the electrical system and engine ignition system are capable of violent and damaging short circuits or severe electrical shocks. If you intend to perform work where electrical terminals could be grounded or touched, the battery cables should be disconnected at the battery.

Any time the intake or exhaust openings of the engine are exposed during service, they should be covered to prevent accidental entry of foreign material. Entry of such materials will result in extensive damage when the engine is started.

During any maintenance procedure, replacement fasteners must have the same measurements and strength as the fasteners that were removed. Metric bolts and nuts have numbers that indicate their strength. Customary bolts use radial lines to indicate strength while most customary nuts do not have strength markings. Mismatched or incorrect fasteners can cause damage, malfunction and possible injury.

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Section 1 TX Controller

The TX transfer switch is designed to operate independently with any standard generator utilizing 2-wire start. The transfer switch is programmed locally using the integrated display and touchpad.

The TX transfer switch controller, controls all transfer switch operation, generator function, and set-point/timing considerations.

The controller:

- monitors voltages and frequencies against set-points
- initiates generator function
- controls in-phase or time delay transfers
- manages generator shutdown
- performs exercise operations
- logs functions

An integrated display is used to change set-point values, perform desired operational changes, and monitor status.

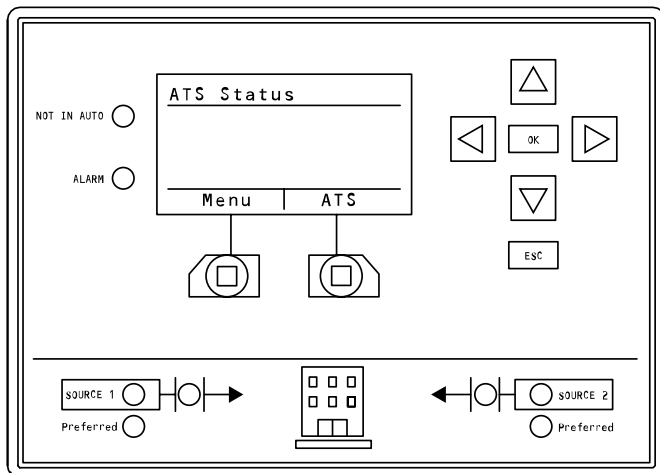


Figure 1-1. TX Switch Controller

A simple dry contact (Form C) is provided as a 2-Wire Start (2WS) for generator control. This allows for control of nearly any generator with a simple 2-wire system. An additional communications link is provided for RS-485 communication, allowing external query of transfer switch status and remote voltage and optional power and current monitoring.

The automatic transfer switch is used for transferring electrical loads from a normal (utility) power source to a standby (emergency) power source. Such a transfer of electrical loads occurs automatically when the normal power source has failed or is substantially reduced and the standby source voltage and frequency have reached a programmable set-point. The transfer switch prevents electrical feedback between two different power sources (such as the normal and standby sources).

Synchronization Limits

Synchronization requirements are:

- Generator frequency within 1Hz
- Voltage within +/- 5% of nominal (based on controller settings).
- Phase angle within +/- 10 degrees

Voltage Limits

The following criteria are used for a local determination of proper Utility.

- Dropout — any phase outside -70 to +130% of nominal (not the average voltage).
- Pickup — all phases > +75% of nominal (programmable).

Phase Rotation

Firmware 1.20 and up provides the option to select phase rotation.

Menu Map

- Maintenance Password: 7123
- Installer Password: 8123
- Dealer Password: 3219 (or 9123 prior to 2022)
- Once the password is entered the screens will be available for 25 minutes.

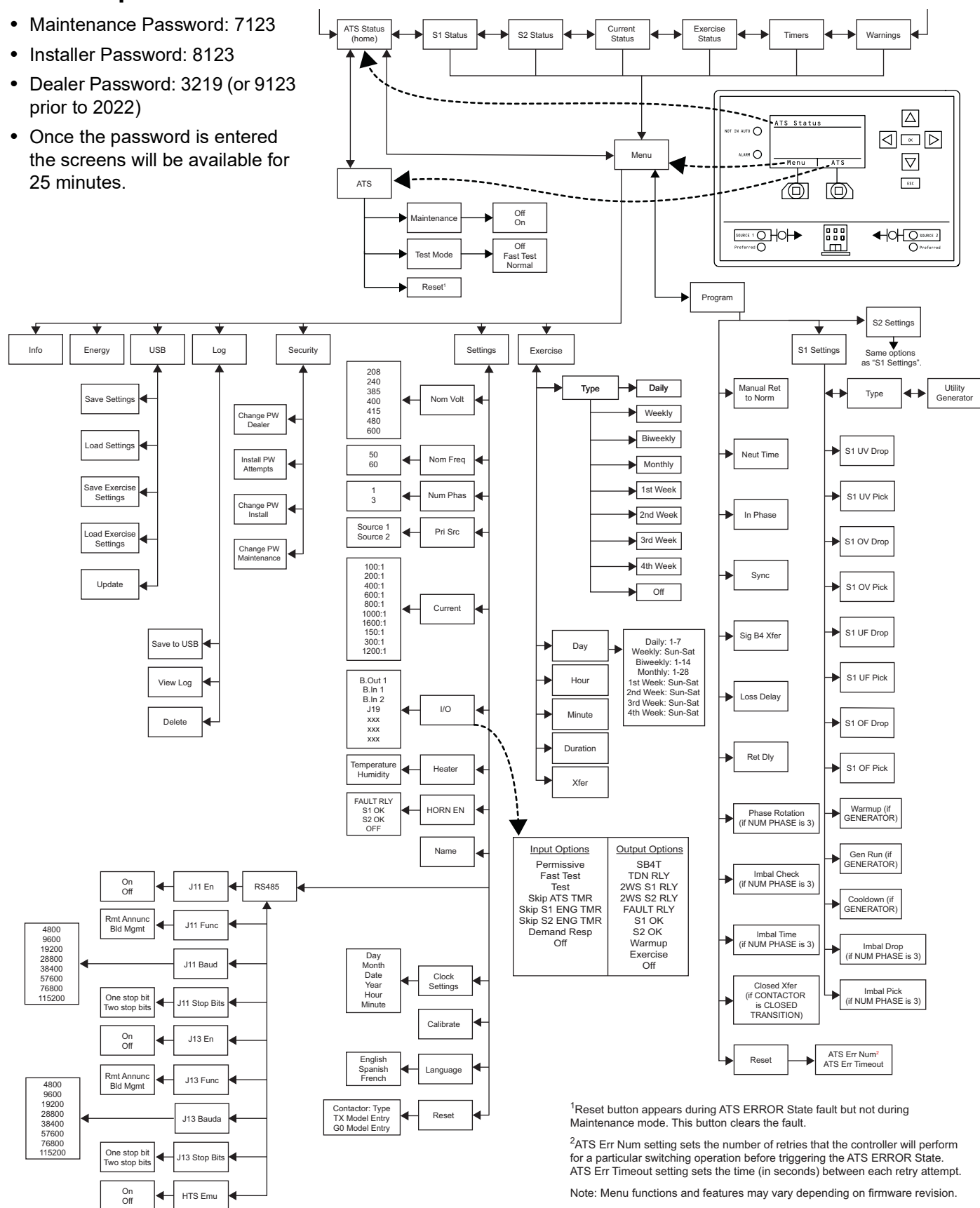


Figure 1-2. TX Menu Map – Overview

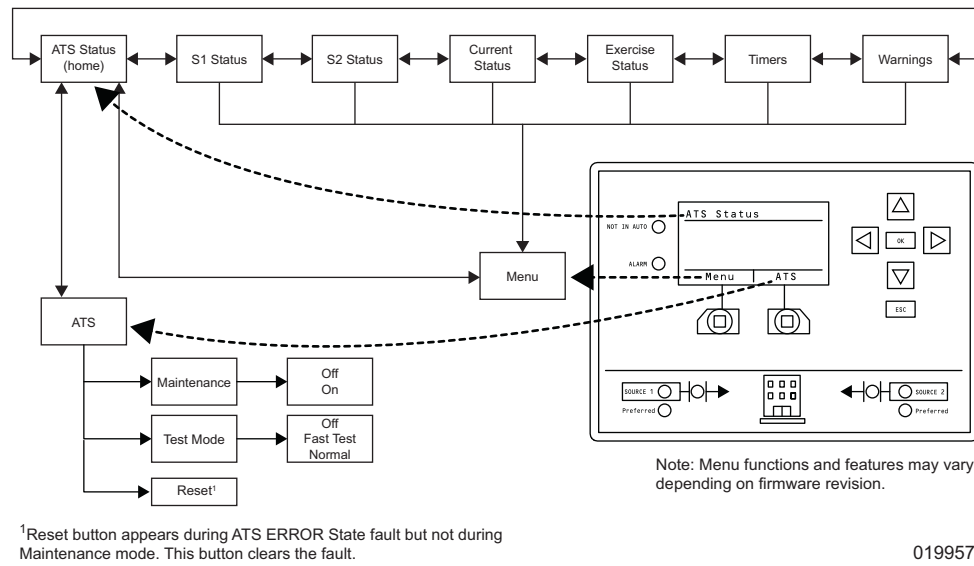


Figure 1-3. TX Menu Map – ATS Status

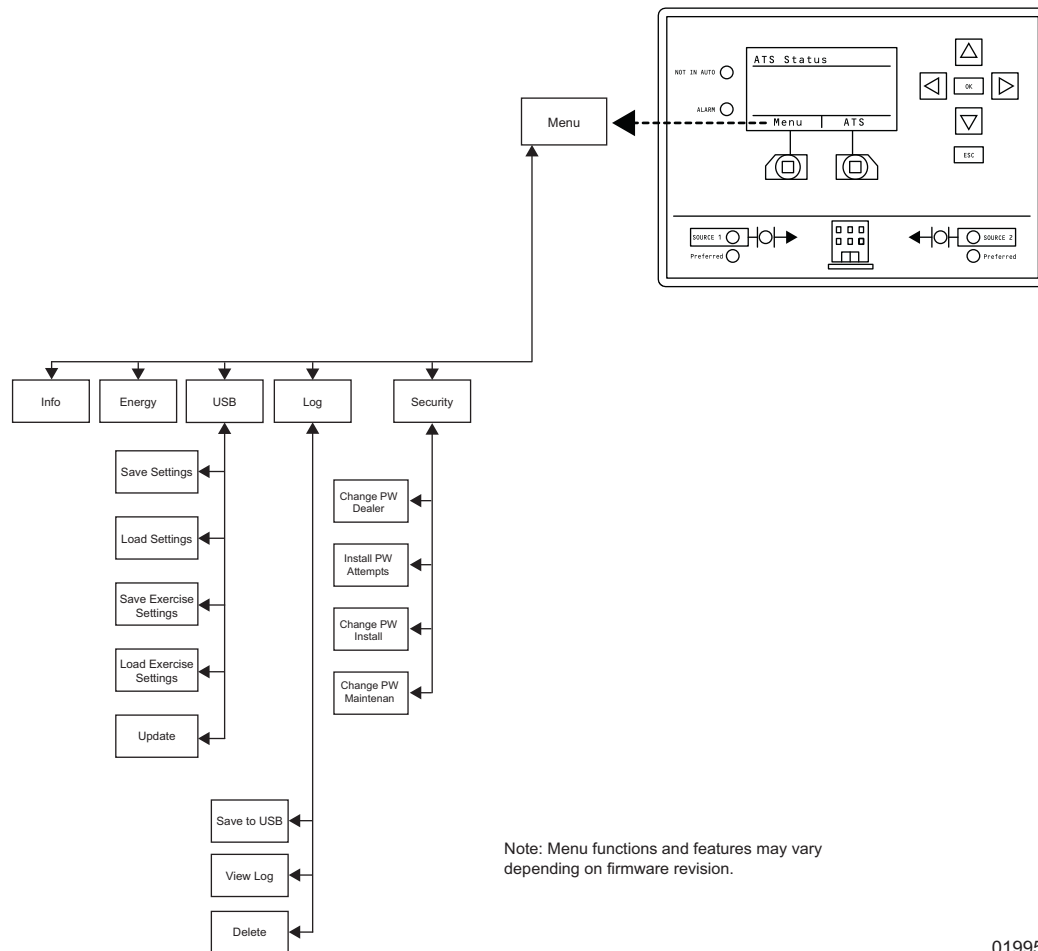
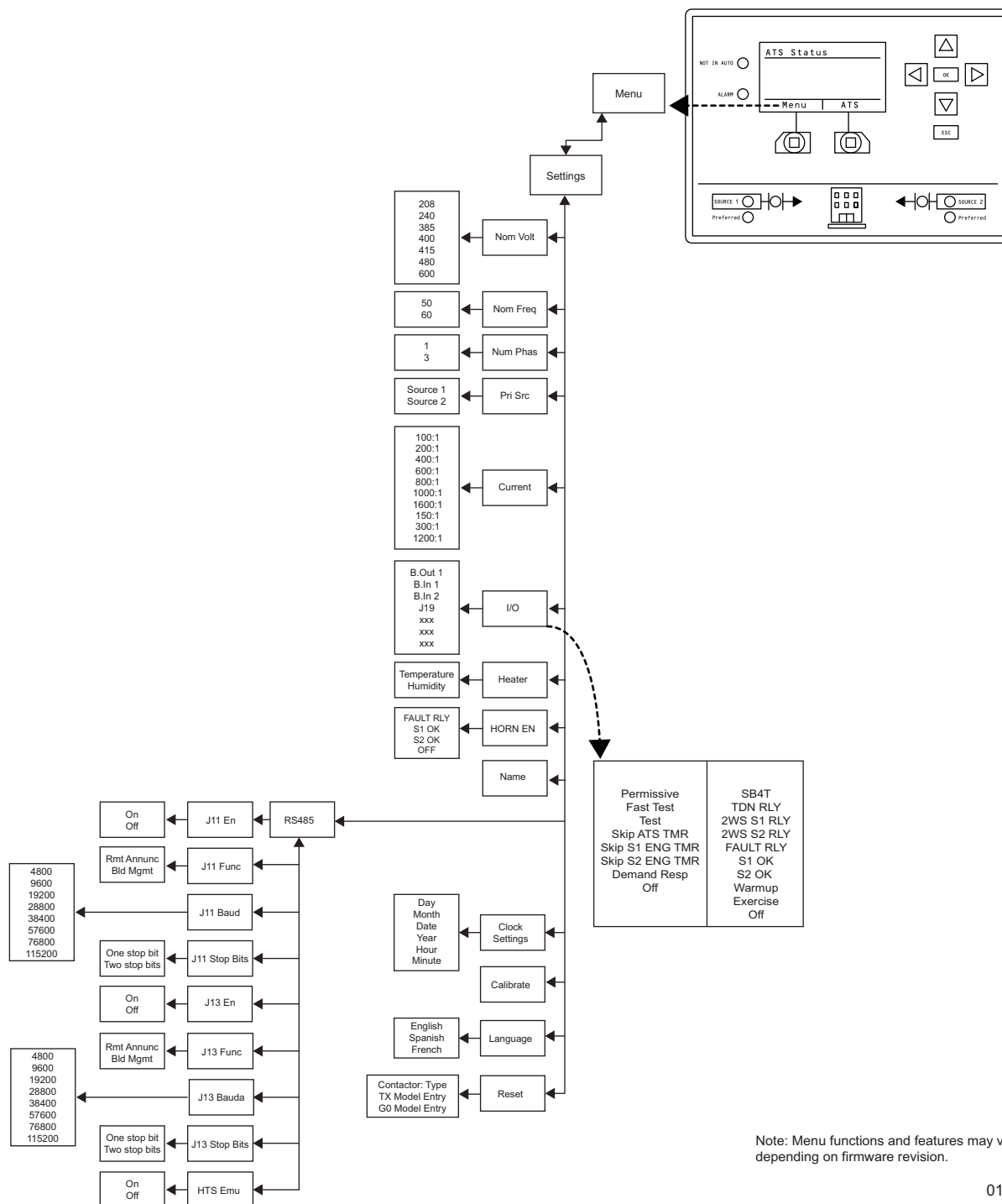
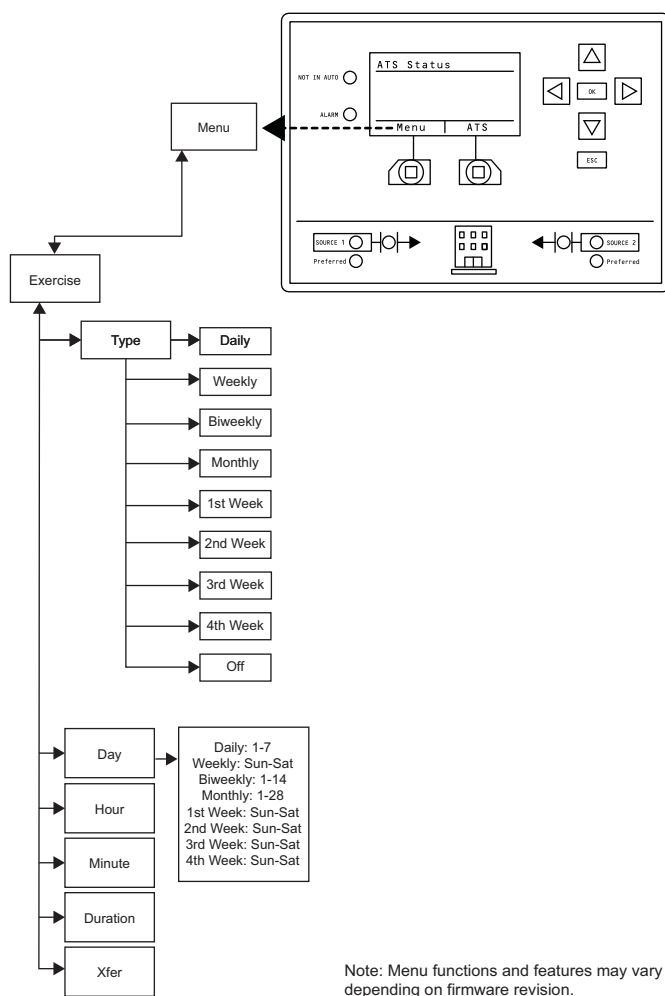


Figure 1-4. TX Menu Map – Info, Energy, USB, Log and Security



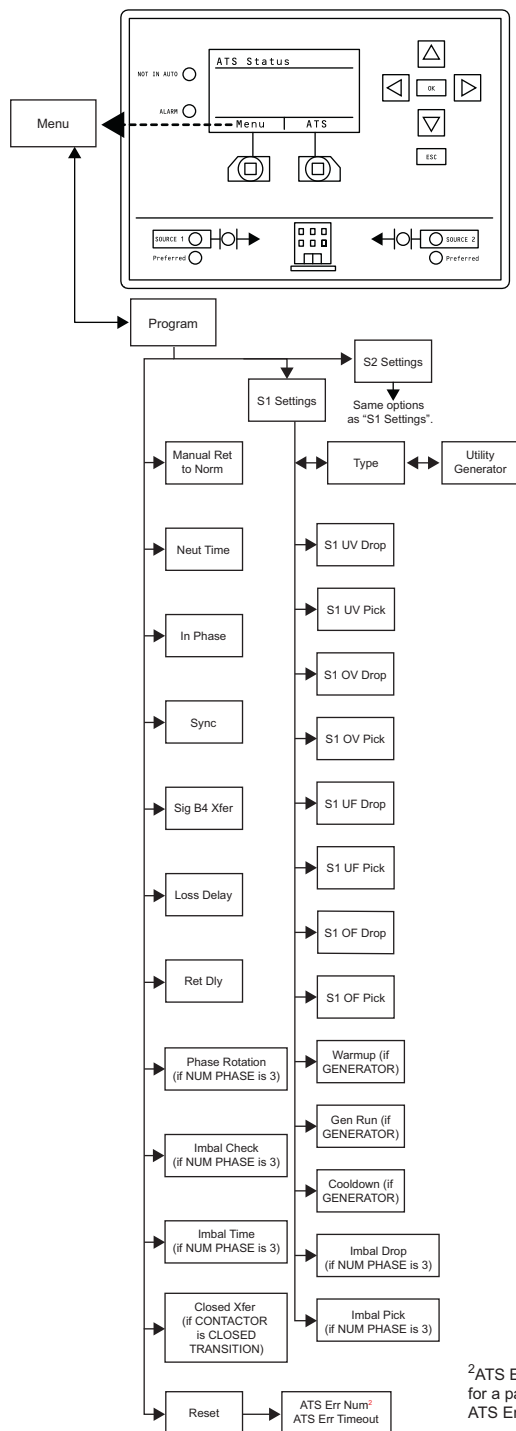
019956

Figure 1-5. TX Menu Map – Settings



019958

Figure 1-6. TX Menu Map – Exercise



019959

Figure 1-7. TX Menu Map – Program

Update Firmware (prior to V 1.19)

Procedure

IMPORTANT NOTE: Firmware update may override current system settings. Record all system settings and parameters prior to performing a firmware update.

1. Navigate to GenService and download the latest firmware (FW_ATS_vX_XX.bin) onto a flash drive.

NOTE: The flash drive must be formatted FAT32.

NOTE: SanDisk flash drives are recommended. Other brands may also work.

2. See [Figure 1-8](#). Insert the flash drive into the USB port beside the battery. An indicator will appear on the display. Use dealer password 3219.

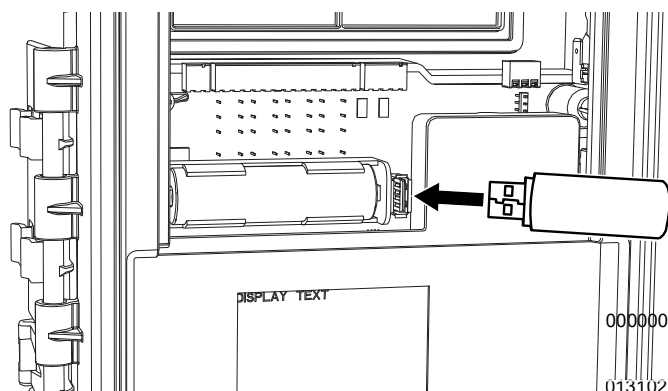


Figure 1-8. Insert Flash Drive

3. Turn generator OFF to disable two-wire start (2WS) capabilities as firmware update process may temporarily command 2WS to activate.
4. Navigate to "Menu" → "Info" → "Update".
5. The controller will search for update files and ask you to select a file for the update.
6. Find the file (FW_ATS_vX_XX.bin) by selecting "NO" to scroll through files and "YES" to accept the correct revision. Six green lights on the bottom of the controller will illuminate while the update is in process.
7. The controller will display the "ATS Status" screen once finished.
8. Remove flash drive from USB port.

NOTE: If the USB drive is not recognized in software revisions 1.18 and prior, the USB interface may lock out additional attempts at loading a new drive. Remove all power sources including battery to restart the control to reset. See Owner's Manual for additional guidance on acceptable USB Owner's drive brands.

Update Firmware (V 1.19 and up)

Procedure

IMPORTANT NOTE: Firmware update may override current system settings. Record all system settings and parameters prior to performing a firmware update.

1. Navigate to GenService and download the latest firmware (FW_ATS_vX_XX.bin) onto a flash drive.

NOTE: The flash drive must be formatted FAT32.

NOTE: SanDisk flash drives are recommended. Other brands may also work.

2. See [Figure 1-8](#). Insert the flash drive into the USB port beside the battery. An indicator will appear on the display. Use dealer password 3219.
3. Turn generator OFF to disable two-wire start (2WS) capabilities as firmware update process may temporarily command 2WS to activate.
4. Navigate to "Menu" → "USB" → "Update".
5. The controller will search for update files and ask you to select a file for the update.
6. Find the file (FW_ATS_vX_XX.bin) by selecting "NO" to scroll through files and "YES" to accept the correct revision. Six green lights on the bottom of the controller will illuminate while the update is in process.
7. The controller will display the "ATS Status" screen once finished.
8. Remove flash drive from USB port.

Change Settings

Procedure

1. Navigate to "Menu" → "Settings".
2. Change all of the settings to the default or the site requirements. The settings prior to firmware update will not remain valid after updating firmware.
3. Navigate to "Menu" → "Program" and repeat step 2.
4. Exit to the "ATS Status" screen.

Calibrate Controller with Calibration Values

NOTE: Upgrades from firmware version 1.04 requires calibration in the field. If calibration values were recorded on the Settings screen, only enter the same values in the following procedure.

Procedure

1. Navigate to "Menu" → "Settings" → "Calibration".

- See **Figure 1-9**. Press the “Z/S” button (A) to highlight the two numbers on the right (B).
- Press the up and down buttons (C) until the first number matches the second number.

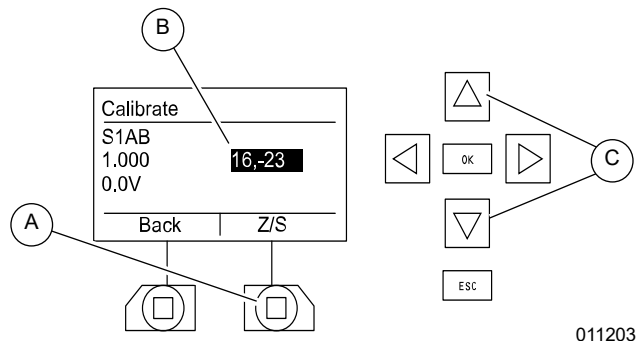


Figure 1-9. Highlight and Change Numbers

Calibrate Controller without Prior Calibration Values

- Use a digital multimeter (DMM) to verify voltage and current values in controller are correct prior to zero voltage and current calibration.
- Remove power from S2 so all input from S2A, S2B, and S2C is zero voltage and zero current.
- Navigate to “Menu” → “Settings” → “Calibration”.
- See **Figure 1-9**. Press the “Z/S” button (A) to highlight the two numbers on the right (B).
- Press the up and down buttons (C) until the first number matches the second number.
- Press the left and right buttons and repeat steps 3 and 4 for the following values: S2A, S2B, S2C.
- Start generator to power S2.
- Remove power from S1 so all input from S1A, S1B, and S1C is zero voltage and zero current.
- Repeat steps 3–5 for S1.
- Press the left and right buttons and repeat steps 3 and 4 for the following values: S1A, S1B, S1C, IA, IB, IC, IN.
- Exit to the “ATS Status” screen.
- Apply normal power to S1.
- Remove S2 power after proper cool down period. The controller is now calibrated.
- Turn OFF Maintenance Modes and Test Modes.

Ready for Use

- Place controller into normal ATS Mode via Menu.
- Verify controller is in normal ATS Mode.
- Perform any additional testing as required.
- Return generator to AUTO to enable 2WS command functionality.

Voltage Imbalance 10-30s max 20s

Configuring Controller for Nominal Voltage

The transfer switch control is compatible with various voltage sources without additional hardware, but must be properly configured to provide nominal 208-277 VAC to the power control circuit and the correct parallel/series load setting to the anti-condensation heater (if equipped). Connectors J5/J6 on the main harness (A000041477) and J23/J24 on the heater harness (A0000405404) are used to make the appropriate connections to allow the transfer switch to be connected to the system voltages as listed in the table below. J5/J6 and J23/J24 are connected by the factory for the voltage selected during product selection. The harness connections must be configured if the product is to be installed at an alternate voltage. To change selections:

- See **Figure 1-10**. Remove the screw and interlock plate (A).

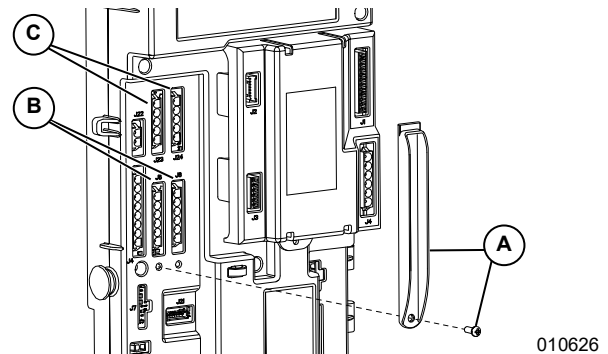


Figure 1-10. Remove/Install Interlock Plate

- Move connectors J5/J6 (B) and J23/J24 (C) to their specified location.
- Install the interlock plate and screw.



DANGER

Electrocution. High voltage is present at transfer switch and terminals. Contact with live terminals will result in death or serious injury.

(000129)

IMPORTANT NOTE: The interlock plate MUST be installed after selection to protect against potentially hazardous voltages and to ensure both heater and control power are correctly configured.

CAUTION

Equipment damage. Exceeding rated voltage and current will damage the auxiliary contacts. Verify that voltage and current are within specification before energizing this equipment.

(000134a)

WARNING

Electric shock. Only a trained and licensed electrician should perform wiring and connections to unit. Failure to follow proper installation requirements could result in death, serious injury, and equipment or property damage.

(000155a)

IMPORTANT NOTE: In applications where the Neutral Block is not connected to a wired Neutral, the control power supply connectors **MUST** be connected to J6/J24 (voltage is 208/240VAC) or a **480 V Delta Kit (P/N G0098680)** **MUST** be utilized. The control power supply source, when configured for 480 VAC GRD-Y (J5/J23), is from Phase A to the Neutral Block. An unconnected Neutral Block and J5 connection will result in hazardous voltages existing on the otherwise unconnected Neutral Block and may pose a shock hazard.

IMPORTANT NOTE: Only the voltages and wiring specified in the table below are acceptable for use with this transfer switch. Voltages such as 120 VAC, 480 V Corner Grounded Delta, 480 V V-Delta, 480 V Ungrounded, 600 V systems, and anything other than listed in the table below are not acceptable without additional hardware. If factory configured for 480 V Delta, additional transformers are included and wired as shown in the schematic diagram. See **480 V Delta Kit (P/N G0098680)**.

Connectors	System Voltages
J5/J23	480 VAC GRD-Y*
J6/J24	208/240 VAC

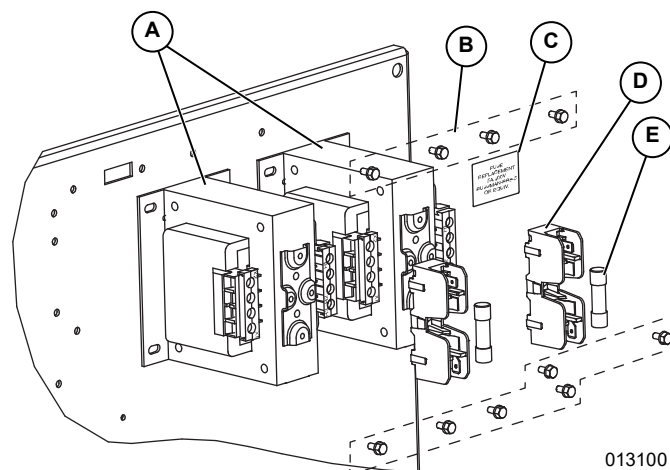
*See Important Notes above for all 3-wire applications

480 V Delta Kit (P/N G0098680)

The 480 V Delta Kit

- See **Figure 1-11**. Install components as shown.
- See **Figure 1-12**. Install wire harness per schematic.
- Connect transformer harness male connector to J6 on controller.

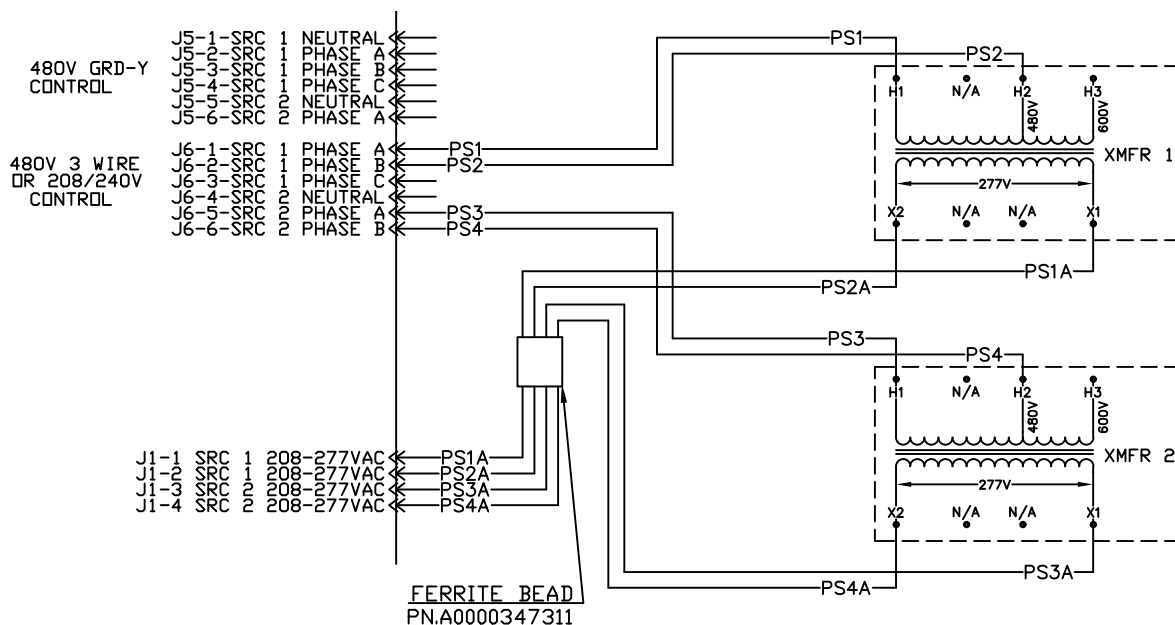
- Connect transformer harness female connector to main wire harness male connector that was removed from J5 or J6.
- Re-connect heater connector (if present) to J23.
- Verify 277 VAC between J1-1 and J1-2 also J1-3 and J1-4.



013100

Figure 1-11. 480 V Delta Kit

ITEM	QTY	DESCRIPTION
A	2	Transformer 600/480:277 200VA TX
B	10	Screw HHTT M4-0.7 X 8 ZP
C	1	Decal, Fuse Rating 5A 600V
D	2	Fuse-block 30A 600V 1POS
E	2	Fuse 5A X Buss
F	1	Harness 600/480 Transformer TX 100-400A (Not Shown)



019145

Figure 1-12. 480 V Neutral Connection Kit With Transformers Schematic

TX Controller Fuses

See [Figure 1-13](#). The control board is equipped with four internal fuses located behind a cover on the back of the controller. These fuses are non-replaceable. If these fuses are open or blown it is recommended to replace the controller.

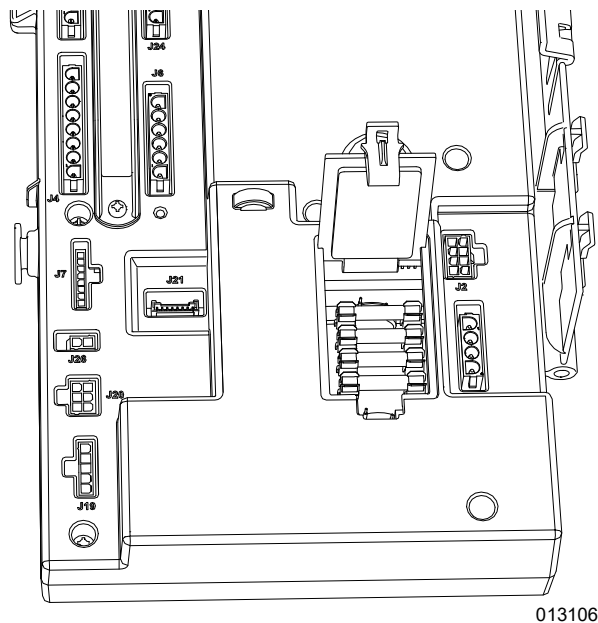


Figure 1-13. TX Controller Internal Fuses

TX Controller Battery

Keep controller powered via external supply, S1, or S2 if possible. Remove old battery and replace with new battery. If controller is reset during battery removal, reset clock time and date after inserting new battery.

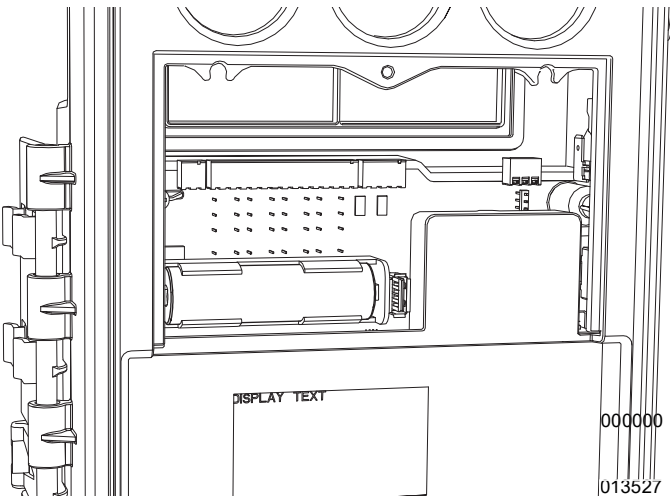


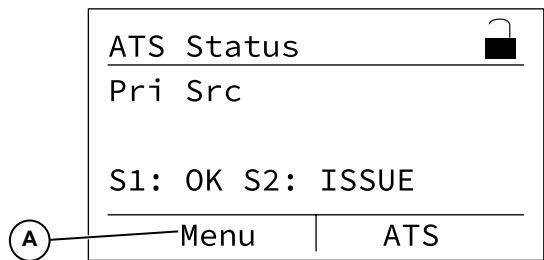
Figure 1-14. Replace Battery

HTS Emulation

HTS Emulation allows a TX transfer switch to be utilized in an application as a replacement for an HTS transfer switch, or in a system where multiple HTS transfer switches exist and an additional ATS is required. This emulation mode allows communication between the TX transfer switch and the H-Panel generator.

Procedure

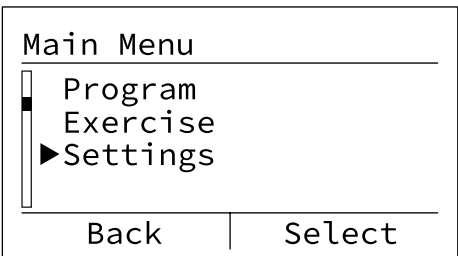
- 1. See [Figure 1-15](#). Select “Menu” (A) from the front screen.



015281

Figure 1-15. Select “Menu”

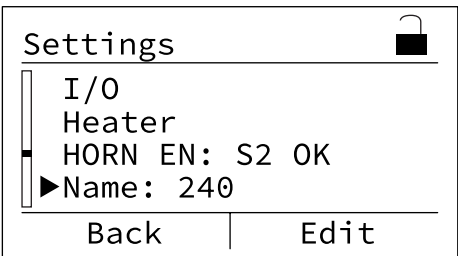
- 2. See [Figure 1-16](#). Navigate to “Settings” and input the Dealer Level Password.



015282

Figure 1-16. Navigate to “Settings”

- 3. See [Figure 1-17](#). Navigate to “Name.”



015283

Figure 1-17. Navigate to “Name”

4. See [Figure 1-18](#). The name is the MODBUS address of the Switch. Change it to 240-243 depending on which switch you want to configure in GenLink.

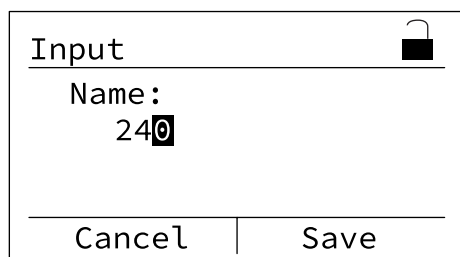


Figure 1-18. Change “Name”

5. See [Figure 1-19](#). Navigate to “RS485 Settings” and press OK.

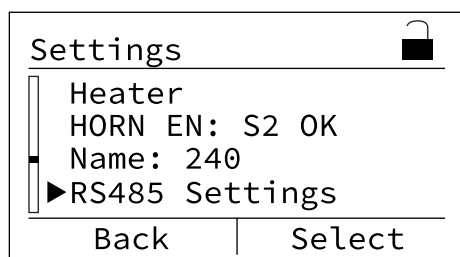


Figure 1-19. Navigate to “RS485 Settings”

6. See [Figure 1-20](#) and [Figure 1-21](#). Configure the RS485 port.
7. Set Enable to “On” (B).
8. Set J13 to “Bld Mgmt” (C).
9. Baud and Stop bits (D) must match the configuration in GenLink.
10. Set HTS Emu to “On” (E). This will cause the TX to respond to H-Panel controllers.

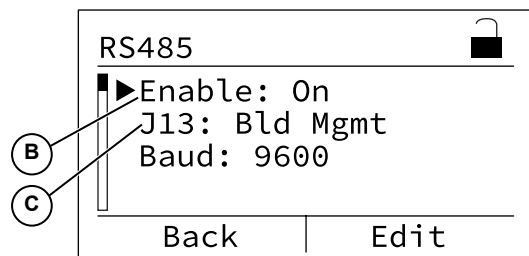
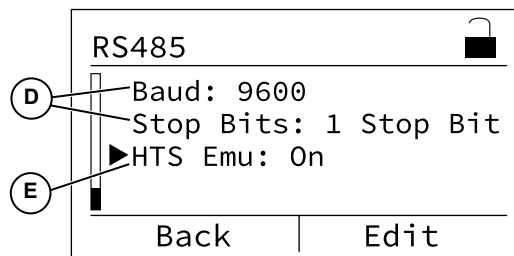


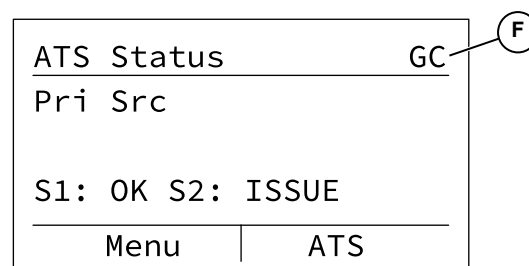
Figure 1-20. Configure RS485 Port



015287

Figure 1-21. Configure RS485 Port

11. Navigate to the ATS Status screen.
12. See [Figure 1-22](#). If the TX is receiving commands from the H-panel, a “GC” icon (F) will appear in the corner for “Generator Connect.”



015288

Figure 1-22. Generator Connect

015286

Summary of Parameters

NOTE: See owner's manual for most current Summary of Parameters.

Summary of Parameters		
Voltage and Frequency Settings	Range	Default Value
Source 1 (Normal) is Genset	Yes or No	No
Source 1 Undervoltage Dropout	50-97%	85%
Source 1 Undervoltage Pickup	52-99%	90%
Source 1 Overvoltage Dropout	105-120%	110%
Source 1 Overvoltage Pickup	103-118%	105%
Source 1 Underfrequency Dropout	90-97%	90%
Source 1 Underfrequency Pickup	91-99%	95%
Source 1 Overfrequency Dropout	103-110%	105%
Source 1 Overfrequency Pickup	101-109%	102%
Source 1 Voltage Imbalance Drop	5-20%	5%
Source 1 Voltage Imbalance Pickup	3-18%	3%
Source 1 Warmup Time	0-1800s	3s
Source 1 Cooldown Time	0-1800s	300s
Source 1 Minimum Run Time	300-1800s	1200s
Source 2 is Generator	Yes or No	Yes
Source 2 Undervoltage Dropout	50-97%	85%
Source 2 Undervoltage Pickup	52-99%	90%
Source 2 Overvoltage Dropout	105-120%	110%
Source 2 Overvoltage Pickup	103-118%	105%
Source 2 Underfrequency Dropout	90-97%	90%
Source 2 Underfrequency Pickup	91-99%	99%
Source 2 Overfrequency Dropout	103-110%	105%
Source 2 Overfrequency Pickup	101-109%	102%
Source 2 Voltage Imbalance Drop	5-20%	5%
Source 2 Voltage Imbalance Pickup	3-18%	3%
Source 2 Minimum Run Time	300-1800s	1200s
Source 2 Warmup Time	0-1800s	3s
Source 2 Cooldown Time	0-1800s	300s
Phase Rotation Check	ABC, CBA, OFF	ABC
Supply Overvoltage	350 VAC	Fixed
Manual Return to Normal	Yes or No	No
Time Delay Settings		
Transfer to Emergency	120s max	3s
Re-transfer to Normal	1,800s max	300s
Time Delay Neutral	120s max	5s
Elevator Signal	120s max	30s
In Phase Transfer	Yes or No	No
In Phase Synchronization Time	60-3600s	300s
Preferred Source	S1, S2	S1
Voltage Imbalance Enable	Yes or No	Yes
Voltage Imbalance Timeout	10-30s max	20s

Exercise Settings		
Type	Daily, Weekly, Monthly, First Day, Second Day, Third Day, Fourth Day, Off	Weekly
Day	Daily: N/A, Weekly: 1-7, Monthly: 1-28, Day: 1-7, Off: N/A	Sun
Hour	0–23	9
Minutes	0–59	0
Duration (minutes)	5–600	30
Transfer on Exercise	Yes or No	No

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Section 2 100–400 Amp Contactor Basics

Basic TX Contactor

The transfer mechanism houses the main current carrying contacts, along with other components required for switch operation.

- Available in 2-pole, 3-pole, or 4-pole configurations.
- The main contacts are of silver alloy or silver alloy composition, to resist welding or sticking.
- The main contacts are electrically operated and mechanically latched in place.

NOTE: If a 4-pole neutral switching is required, the neutral will be on the furthest connection from the A-phase when looking at the switch.

A typical 4-pole transfer mechanism is shown in [Figure 2-1](#).

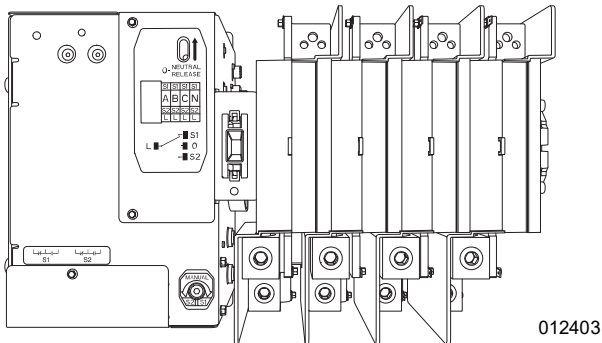


Figure 2-1. Typical 4-pole Transfer Mechanism

Main Contacts

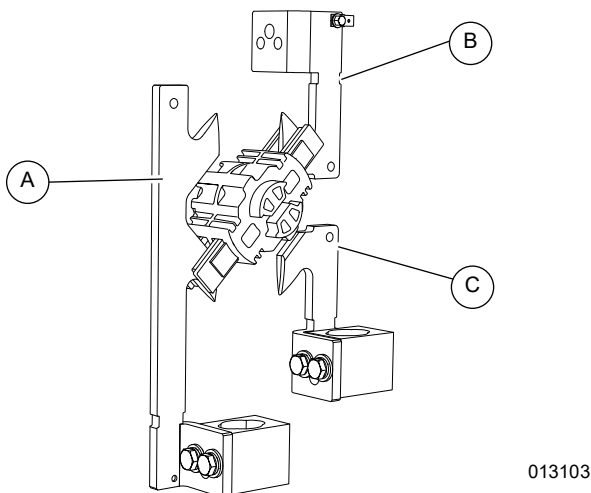


Figure 2-2. Main Contacts

A	Load
B	Source 1 Position
C	Source 2 Position

[Figure 2-3](#) shows the load terminals connected to the Source 1 (utility) terminals. The indicator will display “S1.”

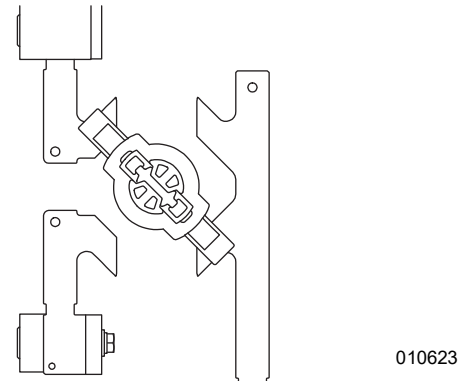


Figure 2-3. Utility (Source 1)

[Figure 2-4](#) shows the contacts disconnected from the load. The indicator will display “0.”

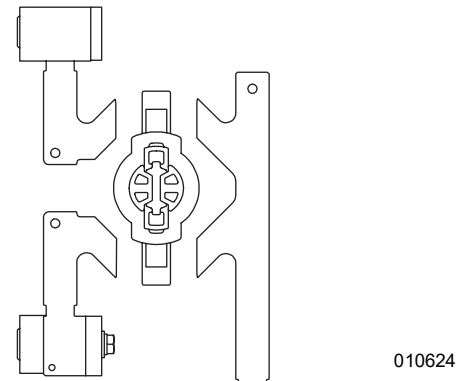


Figure 2-4. Neutral

[Figure 2-5](#) shows the load terminals connected to the Source 2 (emergency) power supply. The indicator will display “S2.”

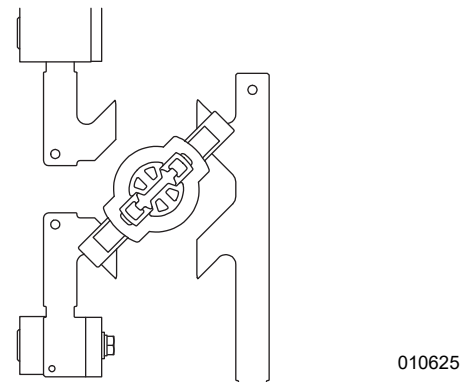


Figure 2-5. Generator (Source 2)

Transfer Mechanism

- The main contacts are actuated by spring force and a high speed drive motor. Larger amperage units will have two motors and smaller units will have one.
- Power to the drive motors comes from the source voltage being transferred to.
- Mechanical latch is used for time delay neutral.

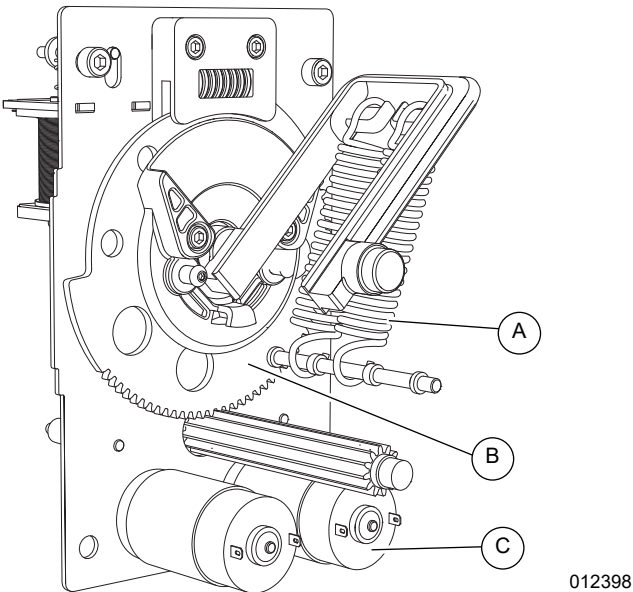


Figure 2-6. Contactor Drive Components

A	Charging Springs
B	Drive Gear
C	Drive Motor

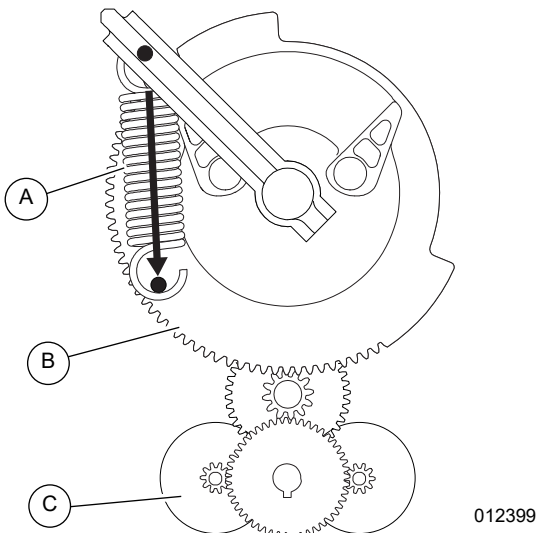


Figure 2-7. Contactor In Source 1

- When the drive motor rotates it will turn the gears to charge the springs.
- Spring tension will draw the mechanism to over-center (*Figure 2-8*).

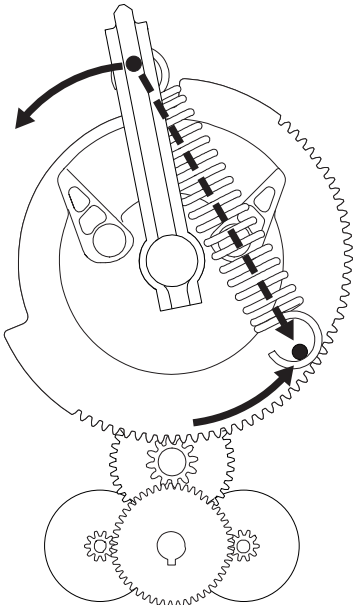


Figure 2-8. Spring Tension Over-Center

- When not equipped with *Time Delay Neutral* (TDN), the mechanism will travel to the opposite source without hesitation (*Figure 2-9*).

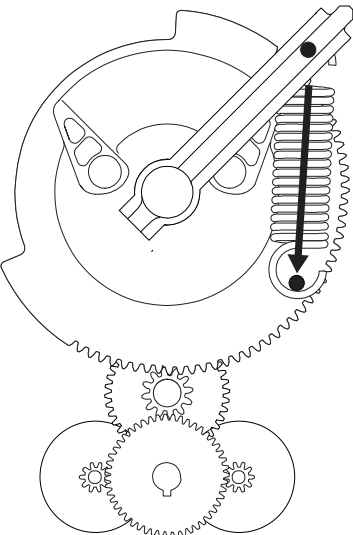


Figure 2-9. Contactor In Source 2

- If equipped with *Time Delay Neutral* the mechanism latches in the neutral position until a timer expires. A solenoid will energize once the TDN timer expires to release the mechanism to the opposite source.

NOTE: The mechanism operates at an extremely high rate of speed. In-phase transfer occurs in approximately 18 ms while the motor reaches speeds as high as 8000 RPM.

Time Delay Neutral

The Time Delay Neutral (TDN) mechanism operates to stop and hold the contactor in the neutral position, preventing it from completing its travel from one source to the other.

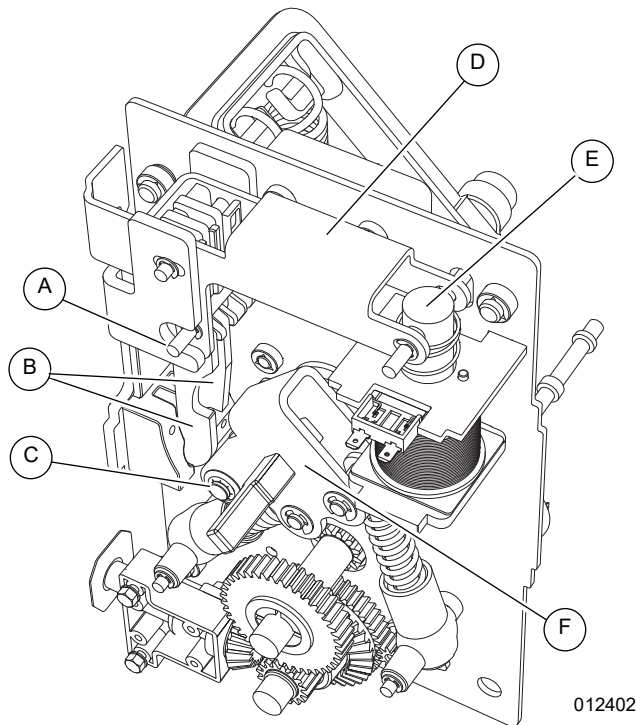


Figure 2-10. Time Delay Neutral

A	Trip Lever Pin
B	Neutral Levers (1 & 2)
C	Neutral Pin
D	Trip Lever
E	Solenoid
F	Toggle Bracket

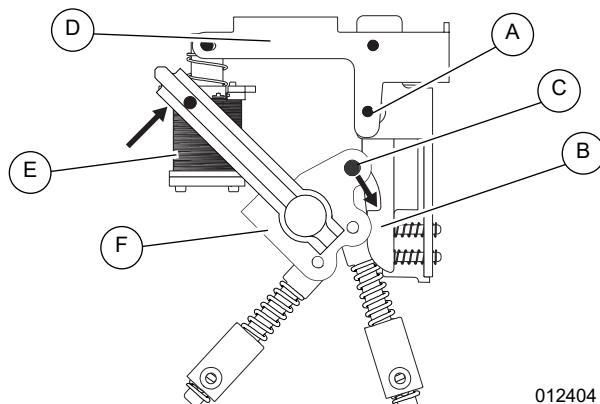
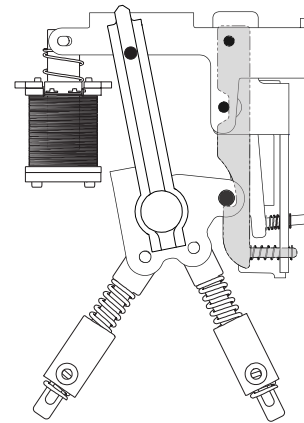


Figure 2-11. TDN Source 1

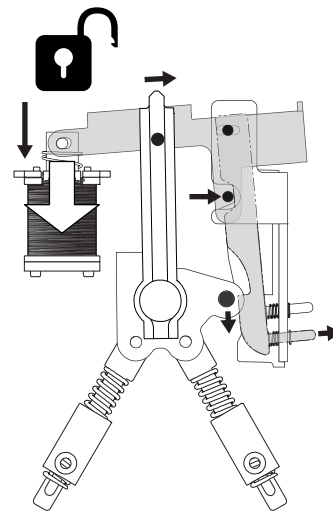
During a transfer, when the toggle bracket reaches the neutral position the toggle pin is latched and held by a neutral lever until a delay timer expires.



012405

Figure 2-12. TDN Neutral – Locked

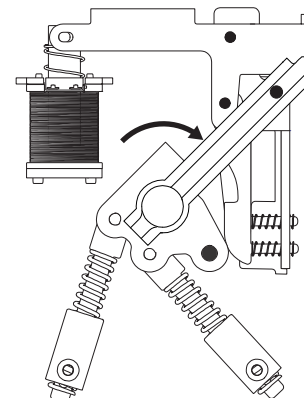
At the end of the delay interval the TDN solenoid is energized. The solenoid activates the TDN trip lever and neutral lever to release the neutral pin.



012406

Figure 2-13. TDN Neutral – Unlocked

The contactor then immediately completes its travel to the destination source.



012407

Figure 2-14. TDN Source 2

Performing Manual Operation



⚠ DANGER

Electrocution. Do not manually transfer under load. Disconnect transfer switch from all power sources prior to manual transfer. Failure to do so will result in death or serious injury, and equipment damage.

(000132)

⚠ WARNING

Sudden start-up. Always set the switch control to MAINTENANCE MODE before working on equipment. Failure to do so could result in death or serious injury.

(000683)



⚠ WARNING

Consult Manual. Read and understand manual completely before using product. Failure to completely understand manual and product could result in death or serious injury.

(000100a)

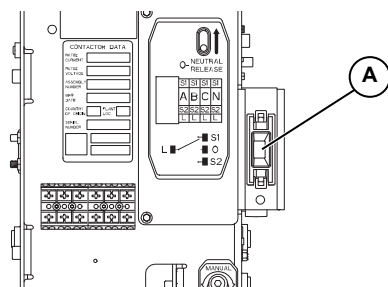
Follow all safety guidelines in Section 1 of the owner's manual before performing manual operation.

1. Remove all power before performing a manual operation.
2. Use the switch controller to turn Maintenance ON before performing manual operation.

IMPORTANT NOTE: Failure to turn the Maintenance function ON before manually operating the contactor will result in the “Cont Pos Error”. This is an active error that constantly checks the position of the contactor. If the contactor is moved out of expected position, this will immediately be displayed. It is not acknowledgeable. Once the contactor position is reported to be in the correct state, the controller will clear the fault. This fault is also reported in maintenance mode.

Source 1 to Source 2 Transition

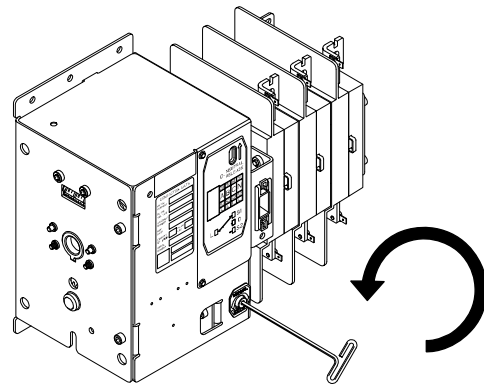
Before proceeding, verify the position of the switch by observing window (A) in [Figure 2-15](#). If “S1” is displayed proceed with Step 1. If “0” is displayed proceed to Step 2. If “S2” is displayed proceed to [Source 2 to Source 1 Transition](#).



010629

Figure 2-15. Switch Position Indicator Window

1. See [Figure 2-16](#). With the manual operator inserted into the lower receptacle, rotate handle counter-clockwise to transfer out of Source 1.
 - Approximately ten rotations will be necessary.
 - Switch will exit from Source 1 and either “0” or “S2” will be displayed in the observing window.
 - If “S2” is displayed, transition is complete and manual handle may be removed.
 - If switch is equipped with time delay option, “0” will be displayed.



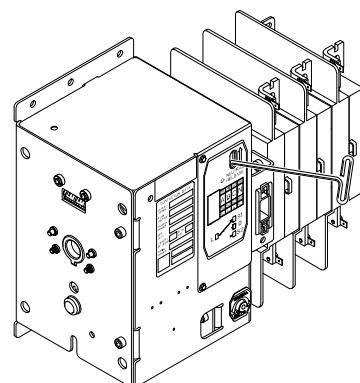
011204

Figure 2-16. Rotate Manual Operator Counter-clockwise

When the observing window displays “0,” the switch is in the Neutral Off position and the neutral catch must be released to complete the desired operation.

2. To release the Neutral Off position, insert the manual operator into the upper receptacle on the mechanism as shown in [Figure 2-17](#).
3. When fully engaged, apply light upward pressure to release the latch.

The contactor will now be in either Source 1 or Source 2 position. Repeat step 1 if necessary.



010634

Figure 2-17. Insert Manual Operator into Upper Receptacle

Source 2 to Source 1 Transition

Before proceeding, verify the position of the switch by observing window (A) in [Figure 2-15](#). If window reads “S2”, proceed with Step 1, and if it reads “0” or “S1,” proceed with Step 2.

1. See [Figure 2-18](#). With the manual operator inserted into the lower receptacle, rotate handle clockwise to transfer out of Source 2.
 - Approximately ten rotations will be necessary.
 - Switch will exit from Source 2 and either “0” or “S1” will be displayed in the observing window.
 - If “S1” is displayed, transition is complete and manual handle may be removed.
 - If switch is equipped with time delay option, “0” will be displayed.

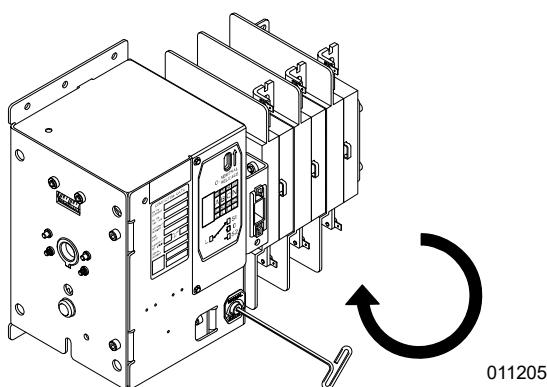


Figure 2-18. Rotate Manual Operator Clockwise

If the observing window displays “0,” the switch is in the Neutral Off position and the neutral catch must be released to complete the desired operation.

2. To release the Neutral Off position, insert the manual operator into the upper receptacle on the mechanism as shown in [Figure 2-17](#).
3. When fully engaged, use light pressure up to release the latch.

The contactor will now be in either Source 1 or Source 2 position. Repeat step 1 if necessary.

Source 1 or Source 2 to Neutral Position

If switch is equipped with a time delay/neutral option, the switch may be mechanically placed into the neutral position. This operation is performed by completing Step 1 in the [Source 1 to Source 2 Transition](#) or [Source 2 to Source 1 Transition](#). This position is exited by completing Step 2 in the applicable section.

NOTE: The transition between sources must be completed once started and it is not recommended to attempt returning to the previous source once in the neutral position.

NOTE: Remove manual operating handle after use and return it to its storage location.

Internal Mechanism

The internal mechanism (P/N A0001344116) is replaceable if damaged or malfunctioning.

TX Controller

Configuring the Controller for Nominal Voltage

- The transfer switch controller can operate with multiple voltages.
- See [Figure 2-19](#). To configure the controller, remove the interlock plate (A) to expose the J24 (B) and J6 (C) connectors.

NOTE: If the voltage applied to the controller is 480 VAC and the harness connector is placed into J5, the controller screen will display “Please move J5 to J6”. This will display whenever the line-to-line voltage exceeds 350 VAC and the harness connector is incorrectly placed into J5.

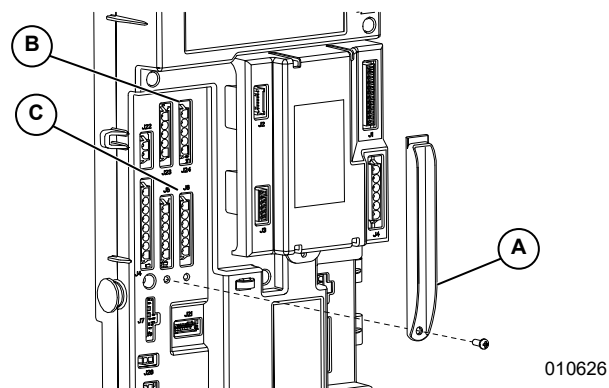


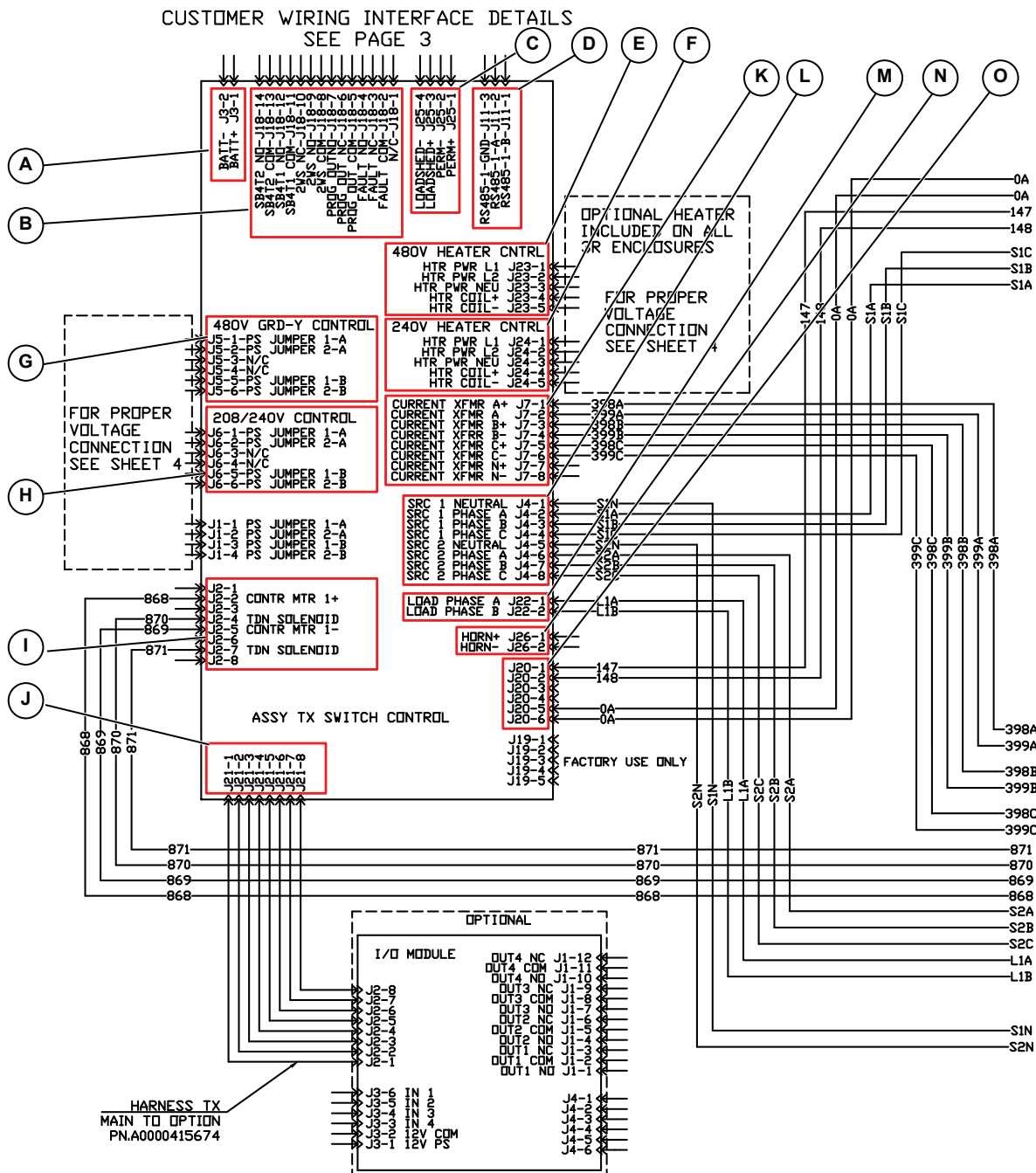
Figure 2-19. TX Control Panel

Wiring Diagram and Schematic Diagram

- | | | |
|--|---------------------------|----------------------|
| A. Battery voltage input | F. 240 V heater control | K. Current I/P |
| B. Signal before transfer and 2WS | G. 480 V control jumper | L. Volt I/P |
| C. Permissive and load shed (see Note) | H. 208/240 control jumper | M. Load I/P |
| D. RS-485 modbus | I. Motor/TDN control | N. Horn |
| E. 480 V heater control | J. Optional | O. Line/Gen position |

NOTE: Permissive and load shed are Programmable Inputs 1 and 2.

NOTE: Diagram in **Figure 1-20** is Version 1. See **Figure 1-22** for Version 2



011378

Figure 1-20. Control Panel Connections (Version 1)

Wiring Diagram and Schematic Diagram (continued)

- P. TDN coil
- Q. Motor for transfer
- R. Aux switch contacts
- S. Limit switch holder

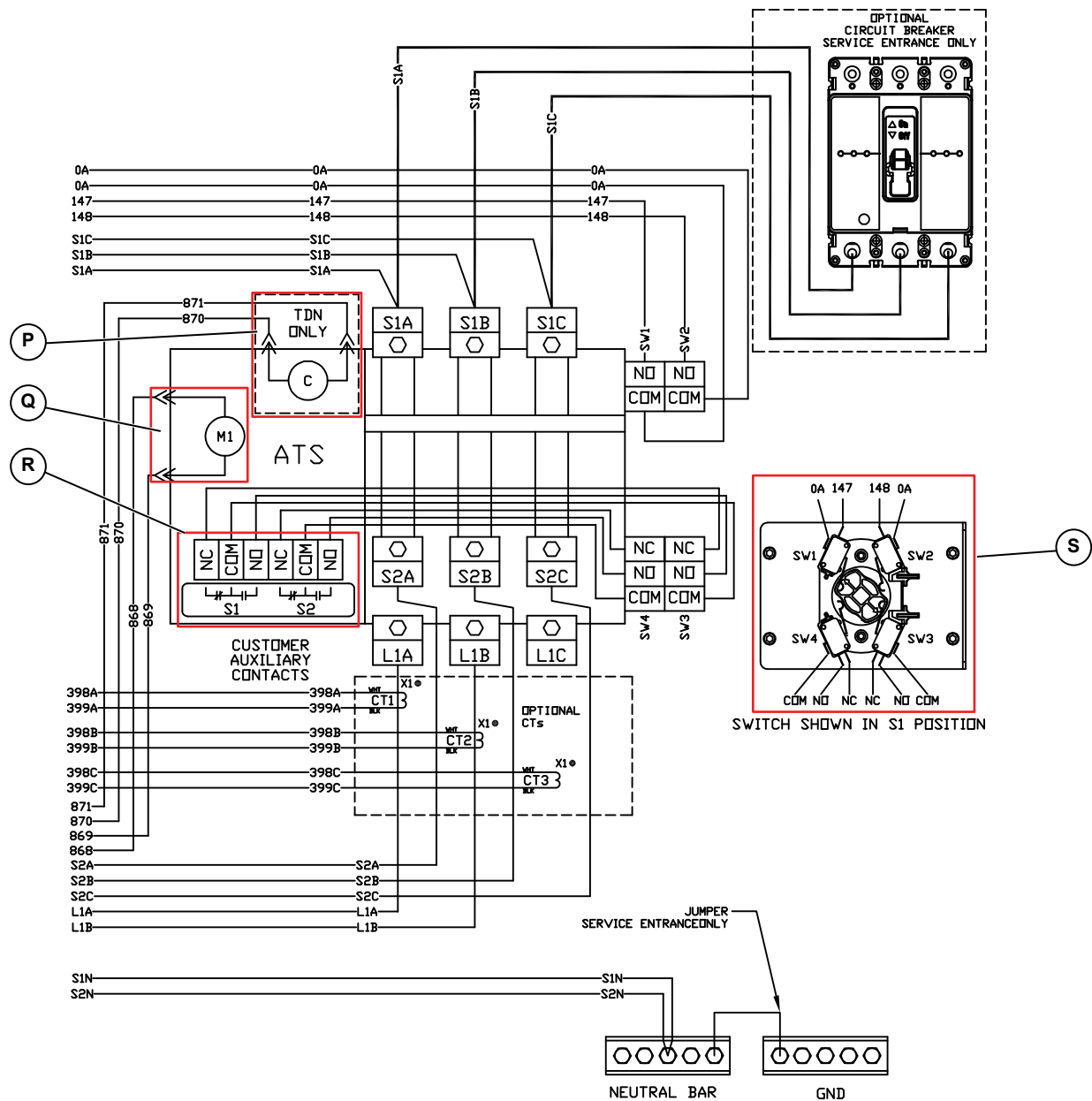
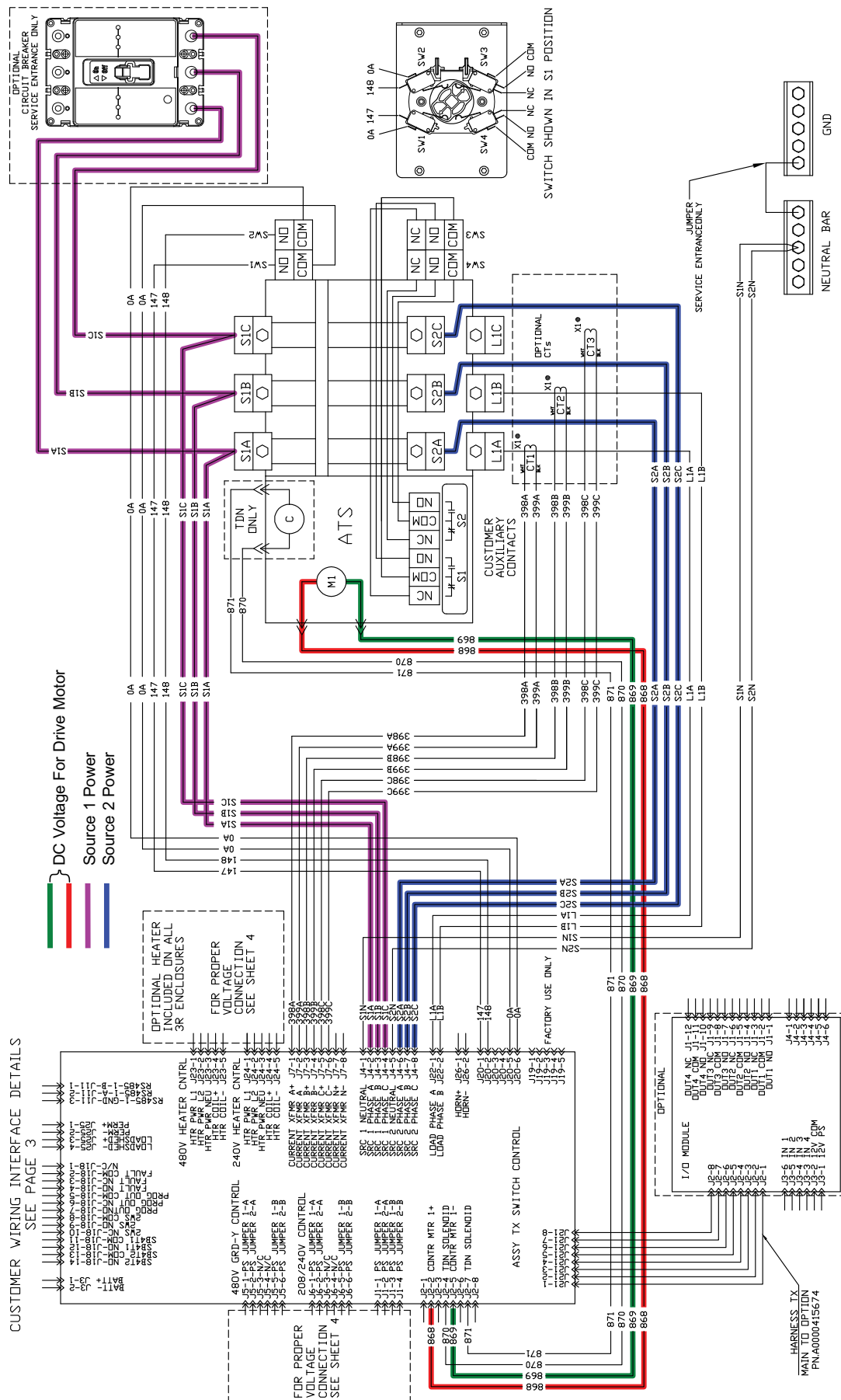


Figure 1-21. Control Panel Connections (continued)

011378

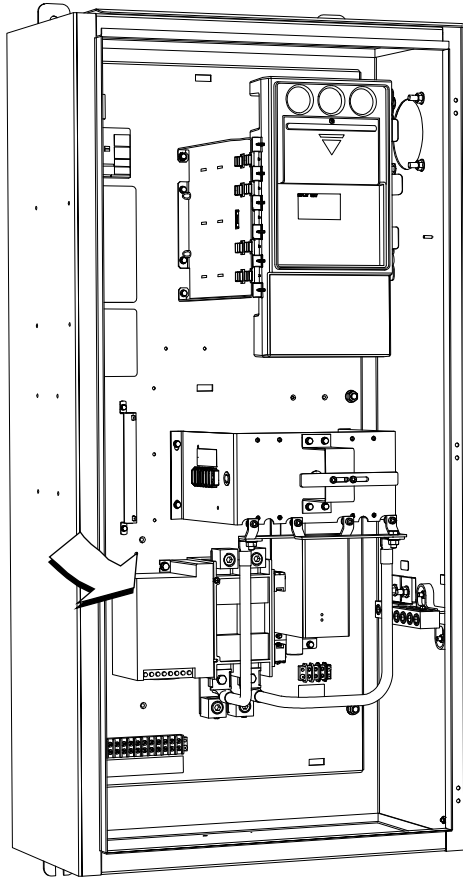


Diagnostic Repair Manual



TX TTS

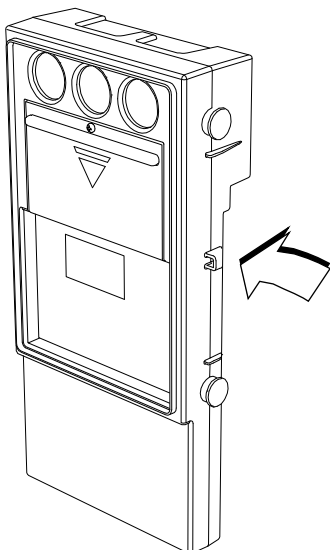
- See [Figure 2-24](#). The TTS Switches currently utilize the W-Style contactor.



013104

Figure 2-24. W-Style Contactor

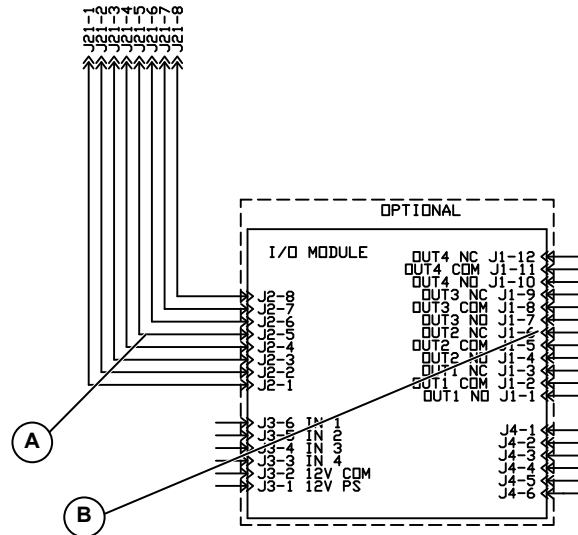
- See [Figure 2-25](#). When using the W-Style contactor the I/O Module must be used to control the contactor.



013107

Figure 2-25. I/O Module

- See [Figure 2-26](#). The I/O Module communicates with the TX controller and is pre-programmed for operation (A).
- Voltage control for W-Style Contactor 208/240 VAC (B).



011382

Figure 2-26. Wiring for TTS Switch

- The TX Controller will control the transfer power by closing the contacts of the relay in the I/O module.
- When 208/240 VAC is applied across the A1/A2 or B1/B2 and the limit switches are in the closed position the transfer will occur.

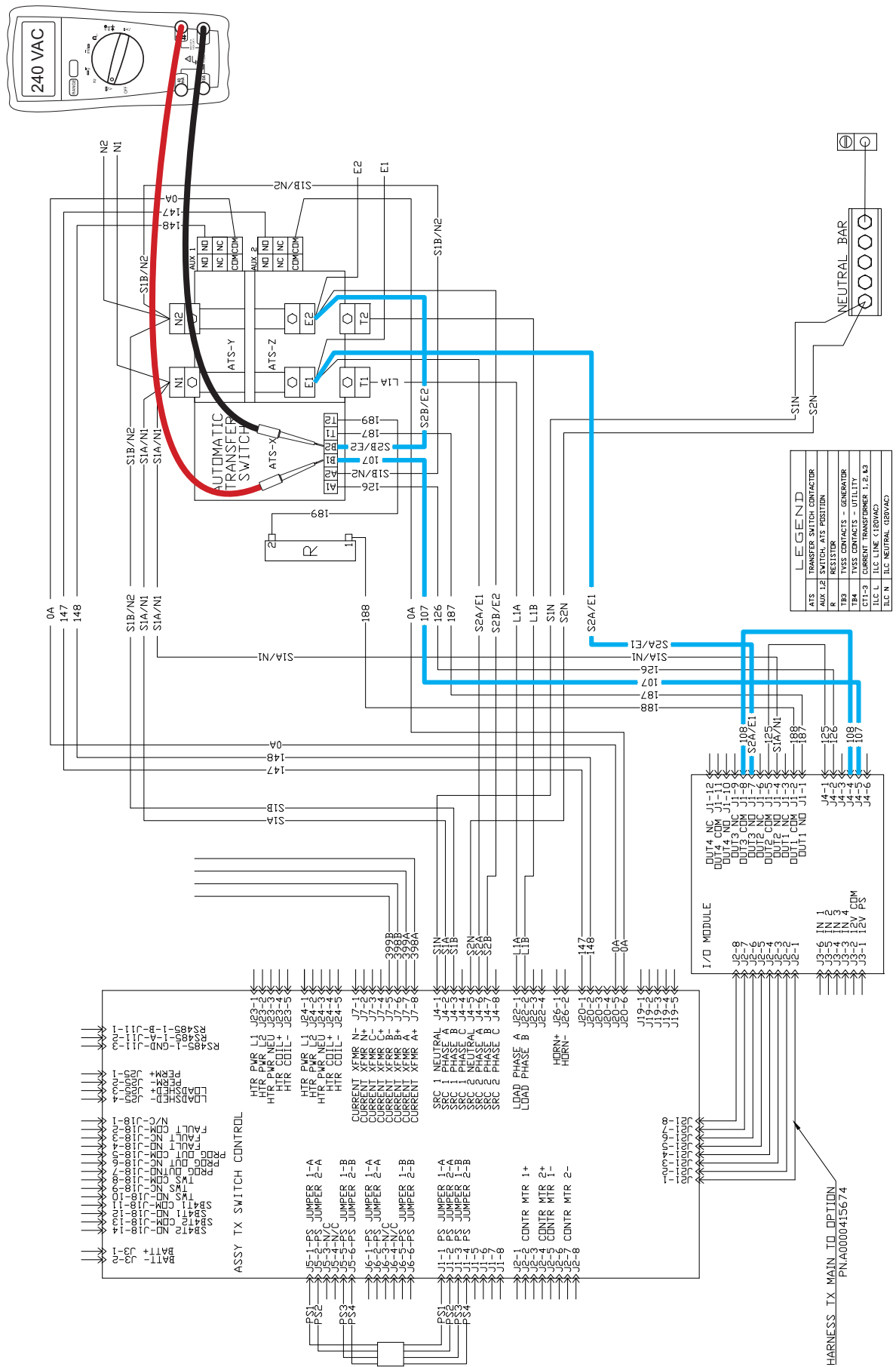


Figure 1-27. S1 and S2 Source Sensing

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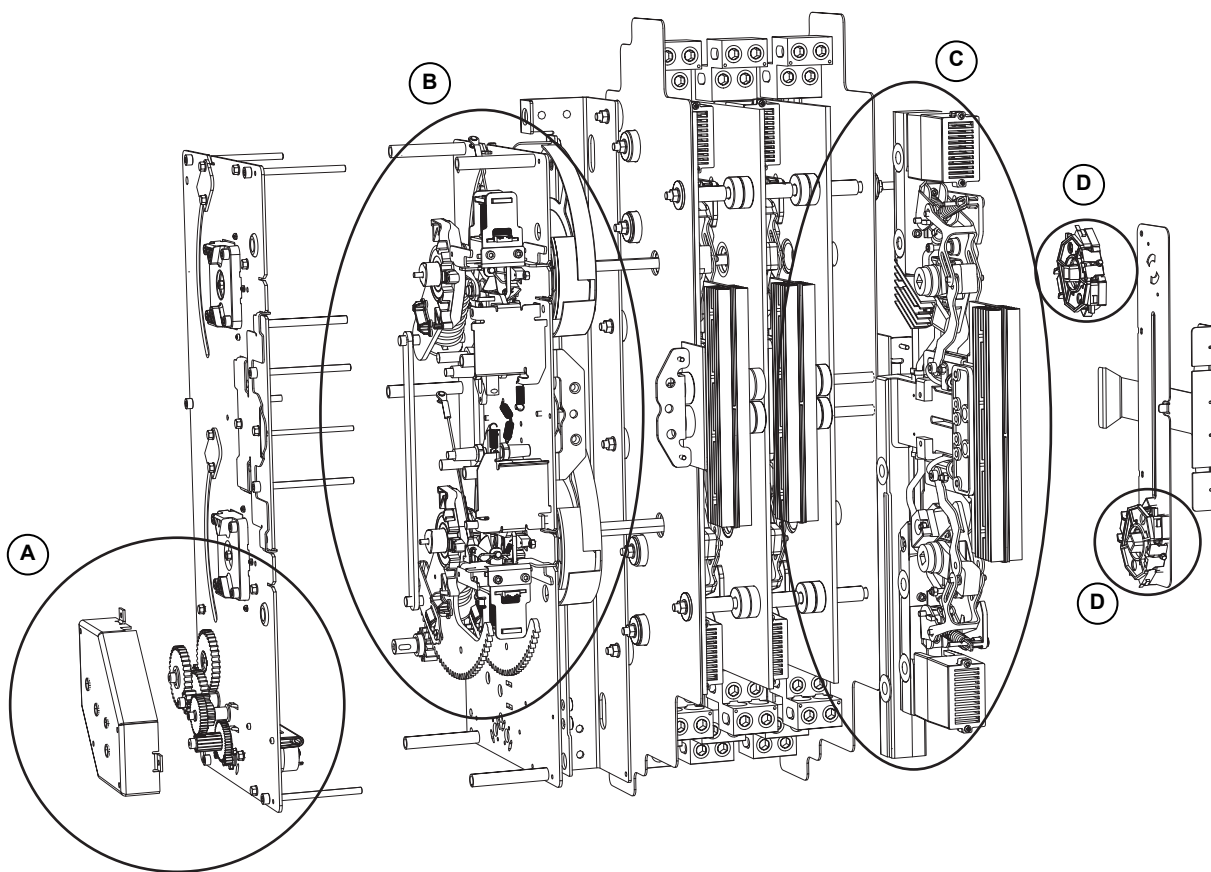
Section 3 800–1600 Amp TX Contactor Basics

800–1600 Amp TX Contactor

The transfer mechanism houses the main current carrying contacts, along with other components required for switch operation.

- Available in 2-pole, 3-pole, or 4-pole configurations.
- The main contacts are of silver alloy or silver alloy composition, to resist welding or sticking.
- The main contacts are electrically operated and mechanically latched in place.

NOTE: If a 4-pole neutral switching is required, the neutral will be on the furthest connection from the A-phase when looking at the switch.



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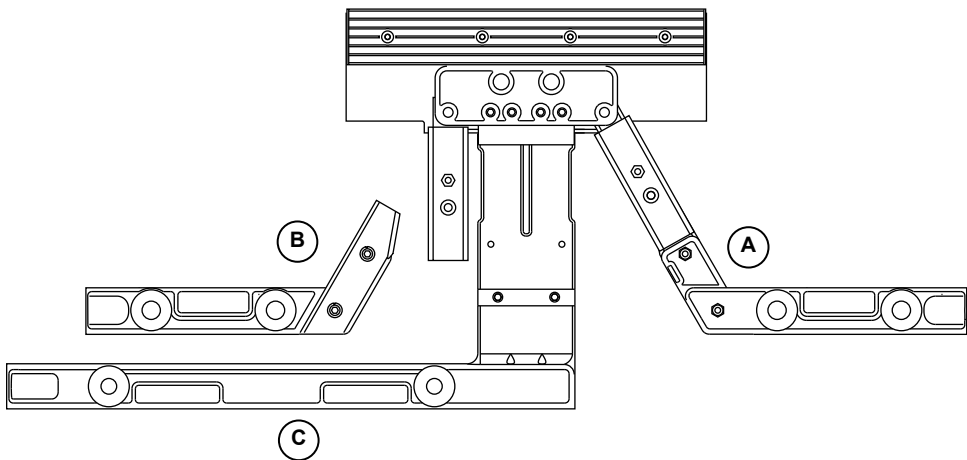
Figure 1-1. Typical 3-pole TX Transfer Mechanism

A	Charging Motors and Reduction Gears
B	TX Mechanism
C	Main Contacts
D	Limit Switches

Transfer Mechanism and Operation

Main Contacts at Source 1 (Utility)

The illustration shows the load terminals connected to the Source 1 (utility) terminals. In this position, the indicator will display “S1.” See [Figure 1-2](#).



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Figure 1-2. Main Contact at Source 1 (Utility)

A	Source 1 Position “ON”
B	Source 2 Position “OFF”
C	Load

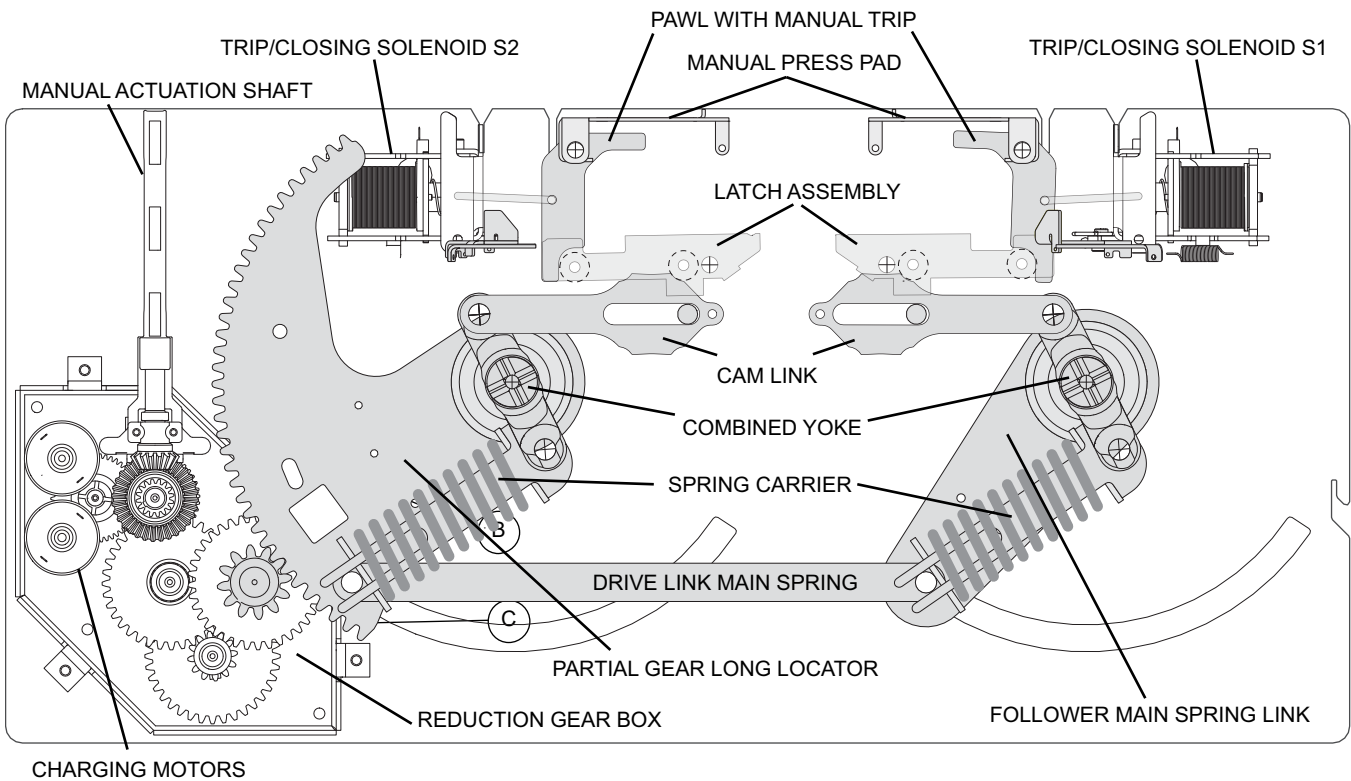
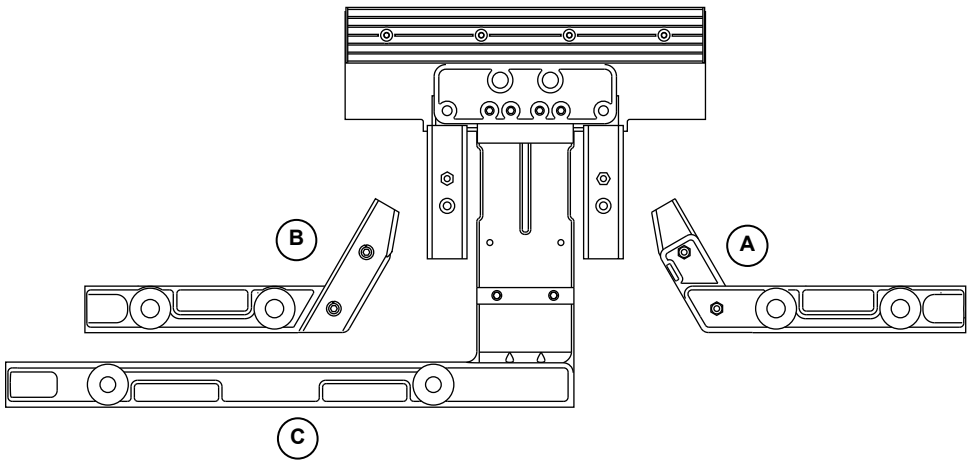


Figure 1-3. Contactor Drive Components, Main Contacts at Source 1 (Utility)

NOTE: Internal components are not separately serviceable. Items shown are for illustration purposes only.

Main Contacts at Neutral (Optional TDN)

Load terminals are disconnected from both power supply terminals. In this position, the indicator will display “0.” See [Figure 1-4](#).



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Figure 1-4. Main Contacts at Neutral

A	Source 1 Position “OFF”
B	Source 2 Position “OFF”
C	Load

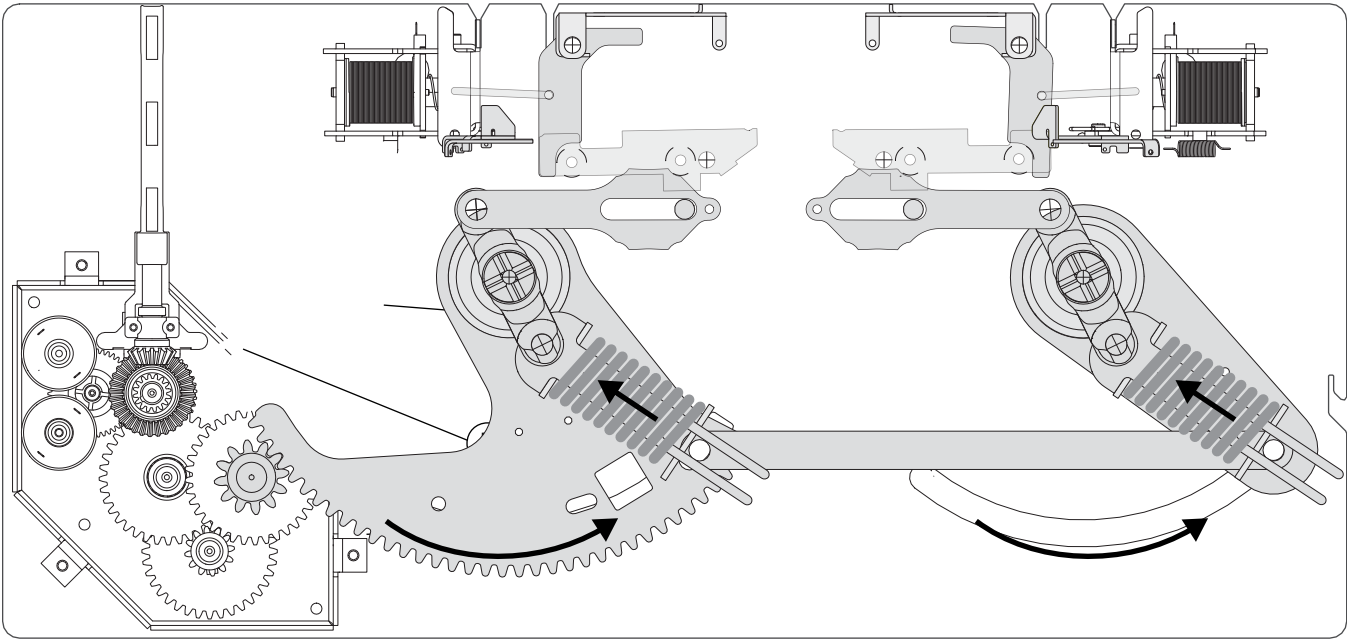
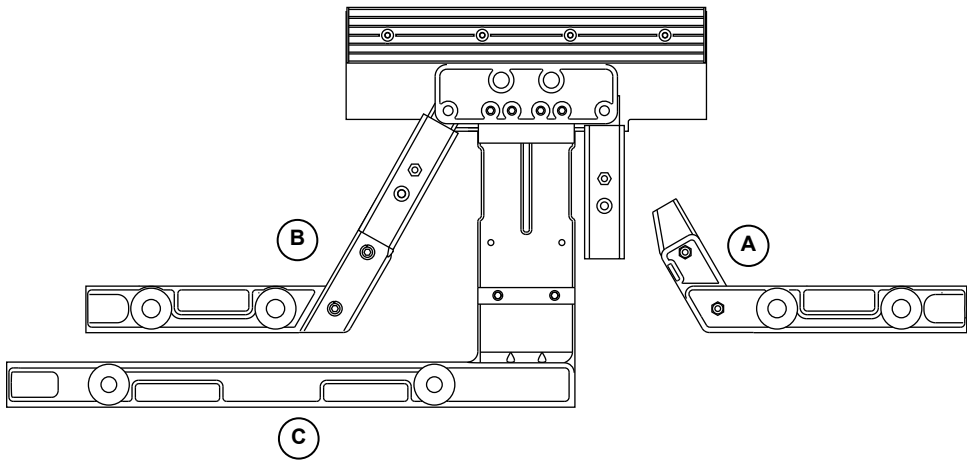


Figure 1-5. Contactor Drive Components – Springs Charged

NOTE: Internal components are not separately serviceable. Items shown are for illustration purposes only.

Main Contacts at Source 2 (Emergency)

Load terminals are connected to the Source 2 (emergency) power supply. In this position, the indicator will display “S2.” See [Figure 1-6](#).



000000

Figure 1-6. Main Contacts at Source 2 (Emergency)

A	Source 1 Position “OFF”
B	Source 2 Position “ON”
C	Load

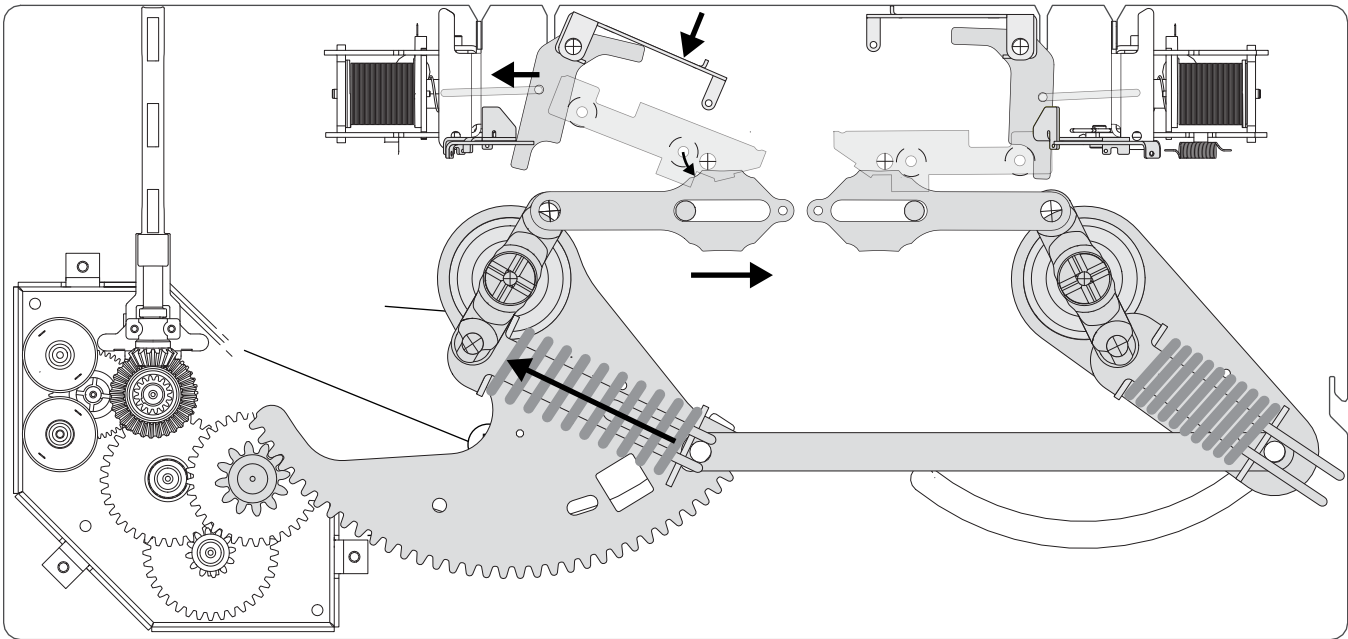


Figure 1-7. Contactor Drive Components, Main Contacts at Source 2 (Emergency)

NOTE: Internal components are not separately serviceable. Items shown are for illustration purposes only.

Transfer Mechanism

The transfer mechanism houses the main, current carrying contacts, along with other mechanical and electrical components, required for operating the switch. The main contacts are electrically operated and mechanically latched in place.

Power for the operating actuators is taken from the source of supply that the Customer Load is being transferred to. Therefore, transfer to any power source cannot occur unless that power source is available to the switch.

Main contacts are actuated by an over center spring linkage charged by a high speed motor. A pair of electric motors charge both Source 1 and Source 2 Main Charge springs at once. If only one spring is depleted, it will only charge that spring after operation. The rotating shaft assembly contains contacts that transfer the load between Source 1 and Source 2. These contacts ensure all phases are actuated together and maintain safe electrical clearances. The use of a duel set of movable contacts allows for some applications with closed transition operation.

Transfer Mechanism Operation

The primary driver of the mechanism is a spring, which is initially charged by a high speed DC motor pair using a proportional control system allowing consistent operation regardless of system voltage. These motors are used to charge both main operator springs. The charged spring is released by either manual or electrical DC pulse (via solenoid). A solenoid is used to release the latch from either the latched (OFF) position or latched (ON) position, completing the transfer operation.

The main charge spring of the mechanism allows consistent opening or closing operation. To manually change the Source Position, first the Main Spring must be charged. The Source state (Open or Closed) and the Spring Charge/Discharged Indicator is on the contactor (E). Once the Spring is charged, pressing either Source 1 or Source 2 Operation button (D) will change that Source State.

NOTE: For open transition switches, both Sources must be Open prior to Closing a Source due to mechanical interlock.

Manual Operation



⚠ DANGER

Electrocution. Do not manually transfer under load. Disconnect transfer switch from all power sources prior to manual transfer. Failure to do so will result in death or serious injury, and equipment damage.

(000132)

⚠ WARNING

Sudden start-up. Always set the switch control to MAINTENANCE MODE before working on equipment. Failure to do so could result in death or serious injury.

(000683)



⚠ WARNING

Hearing loss. Hearing protection is recommended when transferring load. Failure to wear hearing protection could result in permanent hearing loss.

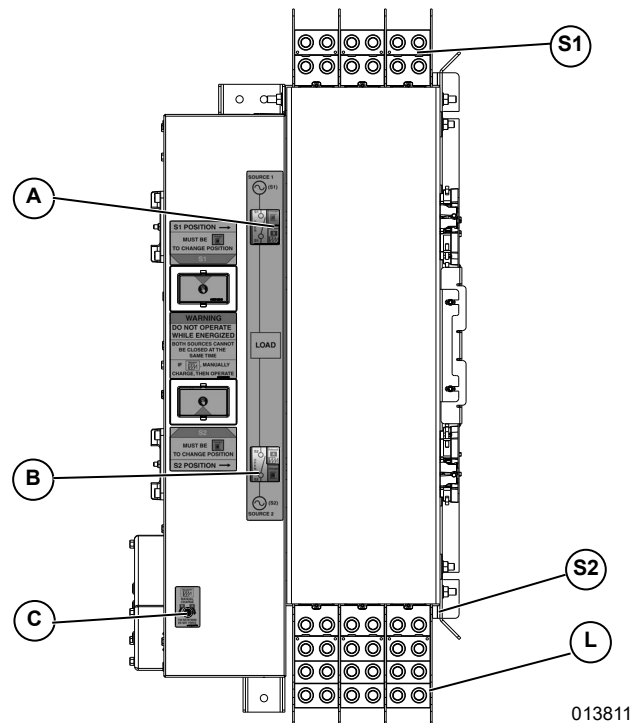
(000766)

Preparing for Manual Operation

IMPORTANT NOTE: During automatic operation, the mechanism is always charged to enable a fast transition between sources. When approaching the switch, assume the springs are charged. Check both windows to verify each spring state before taking any action on the mechanism.

A manual switch handle is shipped with the transfer switch. Manual operation must be checked before the transfer switch is operated electrically. To check manual operation, proceed as follows:

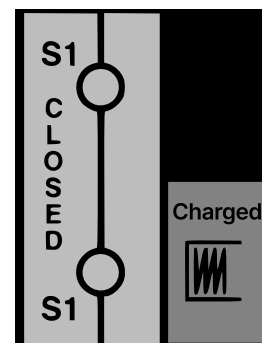
1. In the transfer switch enclosure, set the Maintenance Mode setting on the TX Switch Control to MAINTENANCE MODE. This prevents the generator from starting automatically as soon as the utility power source is turned OFF. If actuating prior to initial energization of control, isolate connector J3 of the customer interconnections to ensure generator start command is inhibited.
2. If so equipped, turn the generator's AUTO/OFF/MANUAL switch to OFF.
3. Turn OFF both normal and standby power supplies to the transfer switch, with whatever means provided, such as the main line circuit breaker(s).
4. See [Figure 3-8](#). Note position of transfer mechanism main contacts by observing the Source 1 (A) and Source 2 (B) windows to see:
 - If the source opened or closed.
 - If the spring charged or discharged.
 - What direction to rotate the manual handle (C) to charge the spring.



013811

Figure 3-8. Transfer Switch Mechanism

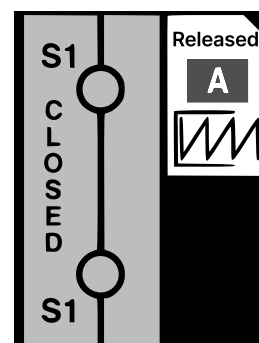
NOTE: The mechanism has two springs, one for each source. There is a gear box which charges or discharges both springs at the same time. Each source has a button to open or close the source.



014887

Figure 3-9. Source 1 Closed, Spring Charged

See [Figure 3-9](#). Source 1 is closed, and the Source 1 spring is charged. Press the manual press button to open Source 1.



014888

Figure 3-10. Source 1 Closed, Spring Released

See [Figure 3-10](#). Source 1 is closed, and the Source 1 spring is released. Source 1 spring will need to be charged to open Source 1. Charge the spring by rotating in the clockwise (A) direction.

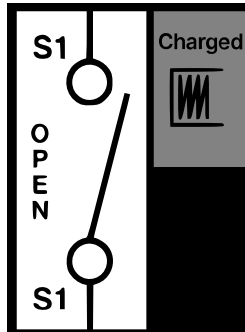


Figure 3-11. Source 1 Open, Spring Charged

014889

See [Figure 3-11](#). Source 1 is open, and the spring for Source 1 is charged. Press the manual press button to close Source 1.

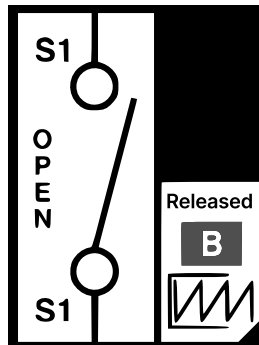


Figure 3-12. Source 1 Open, Spring Released

014890

See [Figure 3-12](#). Source 1 is open, and the Source 1 spring is released. Source 1 spring will need to be charged to close Source 1. Charge the spring by rotating in the counterclockwise (B) direction.

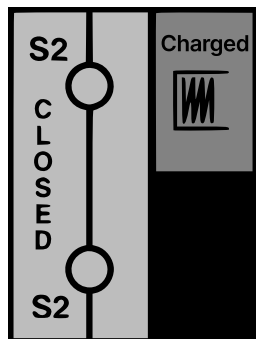


Figure 3-13. Source 2 Closed, Spring Charged

014891

See [Figure 3-13](#). Source 2 is closed, and the Source 2 spring is charged. Press the manual press button to open Source 2.

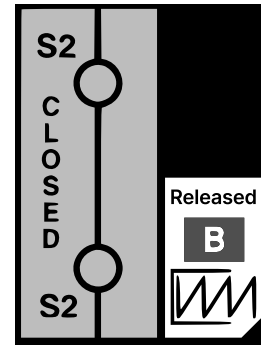


Figure 3-14. Source 2 Closed, Spring Released

014892

See [Figure 3-14](#). Source 2 is closed, and the Source 2 spring is released. Source 2 spring will need to be charged to open Source 2. Charge the spring by rotating in the counterclockwise (B) direction.

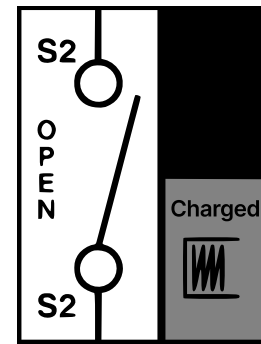


Figure 3-15. Source 2 Open, Spring Charged

014893

See [Figure 3-15](#). Source 2 is open, and the Source 2 spring is charged. Press the manual press button to close Source 2.

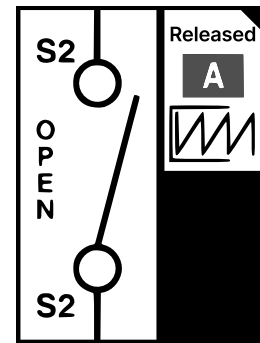
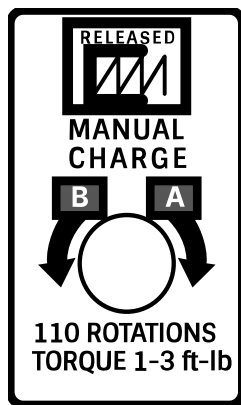


Figure 3-16. Source 2 Open, Spring Released

014894

See [Figure 3-16](#). Source 2 is open, and the Source 2 spring is released. Press the manual press paddle to close Source 2. Charge the spring by rotating in the clockwise (A) direction.



014895

Figure 3-17. Manual Charge Decal

See [Figure 3-17](#). This is the manual charge decal. Place the manual handle inside the hole and press all the way down. Check the indicator window to verify if the springs need to be charged and determine which direction the handle should be turned.

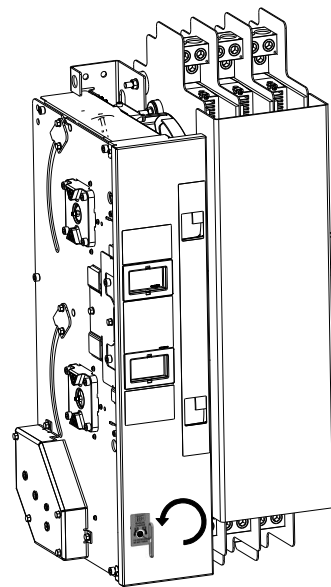
IMPORTANT NOTE: Do not use a drill or impact driver to charge mechanism.

Performing Manual Operation

The contactor has three positions. Connected to Source 1, connected to Source 2, or connected to no source. The closed-transition switches have a fourth state, connected to both Source 1 and Source 2.

When changing which source is connected to load, proceed as follows:

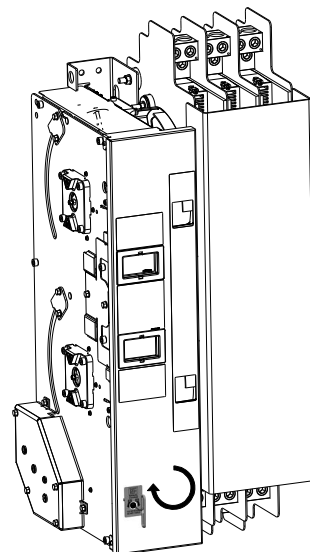
1. Verify no power is on the contactor.
2. See [Figure 3-9](#) through [Figure 3-16](#). Check source windows to see if mechanism needs to be charged.
3. See [Figure 3-18](#) and [Figure 3-19](#). Insert manual handle into the lower receptacle to charge a discharged main spring prior to manual change of switch state. Turn manual handle 110 times in the direction according to Source window if charging is needed.



013812

Figure 3-18. Rotate Manual Operator Counter-Clockwise

NOTE: Rotate handle 110 turns counterclockwise to Charge Main Spring to Open S2 or Close S1 when the main spring indicator shows discharged on the appropriate source.



013813

Figure 3-19. Rotate Manual Operator Clockwise

NOTE: Rotate handle 110 turns clockwise to Open S1 or Close S2 when the main spring indicator shows discharged on the appropriate source.

4. Remove manual handle.
5. See [Figure 3-8](#). Press “S1” or “S2” manual button to open one source.
6. See [Figure 3-8](#). Press “S1” or “S2” manual button to close the other source.

NOTE: Remove manual operating handle after use and return it to its storage location.

TX Controller

Configuring the Controller for Nominal Voltage

- The transfer switch controller can operate with multiple voltages.
- See [Figure 3-20](#). To configure the controller, remove the interlock plate (A) to expose the J24 (B) and J6 (C) connectors.

NOTE: If the voltage applied to the controller is 480 VAC and the harness connector is placed into J5, the controller screen will display “Please move J5 to J6”. This will display whenever the line-to-line voltage exceeds 350 VAC and the harness connector is incorrectly placed into J5.

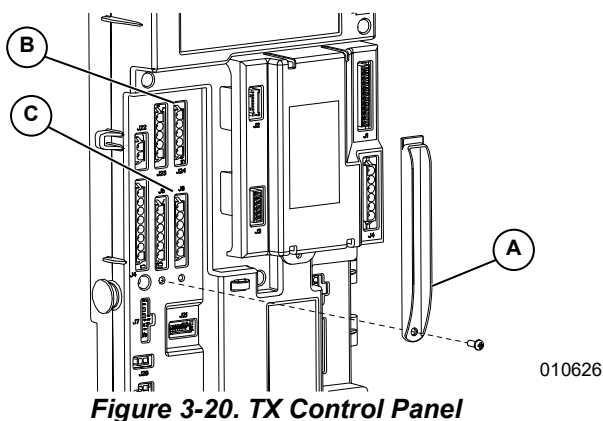


Figure 3-20. TX Control Panel

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Section 4 100–600 Amp TX Transfer Switch Troubleshooting

Fault Types

When a fault is presented, the Alarm LED on the front panel is illuminated and the Fault Indicator Relay is closed. For all faults, except for Source Miswire and Source Voltage High, the ATS control will continue to attempt all functions while in the Fault Indication State. The system needs to be inspected for proper functionality when this occurs.

Faults are displayed on the Warnings screen. Fault can be acknowledgeable (ACKABLE) or unacknowledgeable (UNACKABLE). When a fault is acknowledged (ACKD), it may stay on the screen, but the fault light will turn off and the relay opens.

If multiple faults exist, all faults will need to be ACKD in order to clear the fault indication. Faults are removed from the screen if fault conditions are removed.

Fault types include:

- Acknowledged (ACKD)
 - The fault is present, the Alarm LED is not lit, and the relay is open.
- Acknowledgeable (ACKABLE)
 - Fault is present and fault is asserted until user acknowledges fault. Once acknowledged, the Alarm LED turns off and the relay opens.
- Unacknowledgeable (UNACKABLE)
 - Fault is displayed and cannot be acknowledged by the user.

NOTE: See specific 600 amp contactor diagrams and information for troubleshooting and operation.

List of Faults

Fault	Description	Fault Types
ATS ERROR State	Indicates the ATS failed to transfer within the set amount of retries. The ATS must be inspected and reset from the ATS menu.	Acknowledgeable
Exercise Fail	If secondary source is interrupted for more than 60 seconds during exercise, exercise stops and this fault is displayed.	Acknowledgeable
Secondary Source Failure	If the generator is called and does not present valid voltage for more than 30 seconds, this warning is presented. It is cleared upon detection of good secondary source voltage.	Acknowledgeable
Crank Failure	Indicates the engine failed at start. Voltage was not detected within tolerance after 120 seconds.	Acknowledgeable
IO Board Failure	Indicates the I/O Board is disconnected. This warning only appears if an I/O board is required to drive the contactor and is not detected.	Unacknowledgeable
Contactor Operation Failure	Contactor feedback indicates the previous operation did not succeed on the last attempt.	Acknowledgeable
Contactor Position Failure	Contactor position feedback is not correct for what the automatic operation dictates.	Unacknowledgeable
Supply Voltage High	Indicates that 480V is applied to the Control Power Circuit. The Control may be damaged with extended operation at 480V. Refer to Configuring Controller for Nominal Voltage to remedy.	Unacknowledgeable

S1 Miswire	This is the warning structure to indicate if S1 is wired with incorrect phase rotation. The warning comes up when Phase B is detected to be in the angle that C should be in.	Unacknowledgeable
S2 Miswire	This is the warning structure to indicate if S2 is wired with incorrect phase rotation. The warning comes up when Phase B is detected to be in the angle that C should be in.	Unacknowledgeable
Permissive Fault	This fault appears if the control is programmed for a permissive input signal and the signal does not appear after programmed time (11 seconds by default).	Acknowledgeable
Gen Shunt Fault	This appears if the control senses the paralleling relay has tripped. Acknowledge the fault to reset the relay. The Shunt trip on the generator can then be reset. Contact an IASD for maintenance.	Acknowledgeable
RA or H-100 Comm Loss	This warning appears when either the ATS Remote Annunciator (RA) loses communications for at least 30 seconds or the H-100 loses communications for at least 20 seconds when HTS Emulation mode is enabled.	Acknowledgeable

Time Delay Neutral

When the switch is held in time delay neutral the solenoid plunger catches and holds the switch in the neutral position. Once the neutral timer has expired the control board commands the solenoid to be energized to release the switch to the selected position. Applied voltage should be approximately 40–60 VDC.

IMPORTANT NOTE: The tests in this section are optimized for use with a Fluke 117 digital multimeter (or equivalent) using appropriate probe adapters.

Place test leads on Wire 870 and Wire 871 at the control board J2 connection. Set meter to read DC voltage manual range maximum DC reading and record Min/Max.

Typical coil resistance is approximately 25 Ohms.

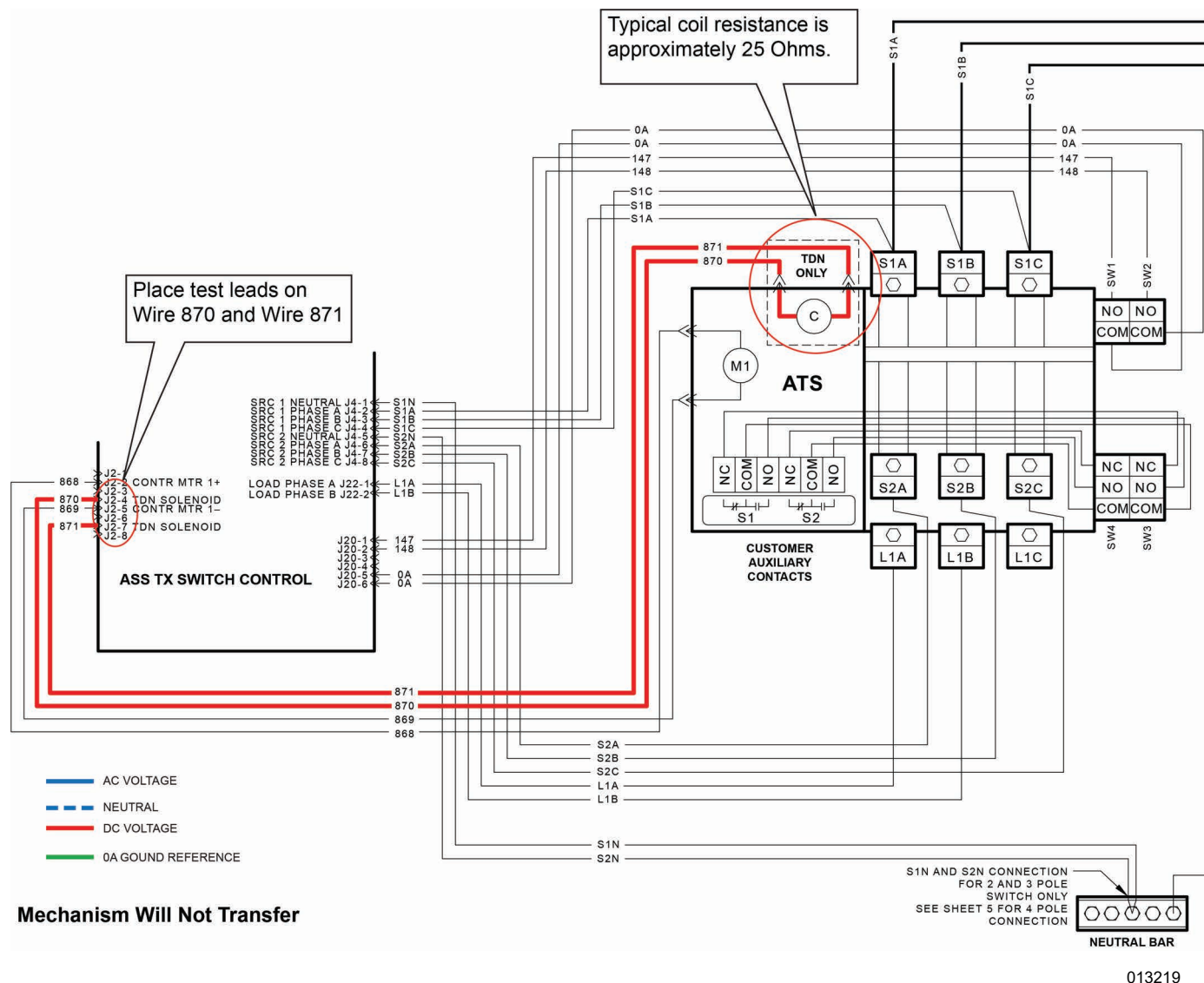


Figure 4-21.

Drive Motor Operation/Diagnostic

The transfer switch mechanism is driven to transfer by a DC motor located inside the mechanism assembly. The switch will transfer from one source to the other by either maintaining or reversing the polarity of the DC voltage delivered to the motor via the control board. The DC voltage being delivered to the motor is rectified line-to-line AC source voltage (300 milliseconds at approximately 120–190 VDC).

IMPORTANT NOTE: The tests in this section are optimized for use with a Fluke 117 digital multimeter (or equivalent) using appropriate probe adapters.

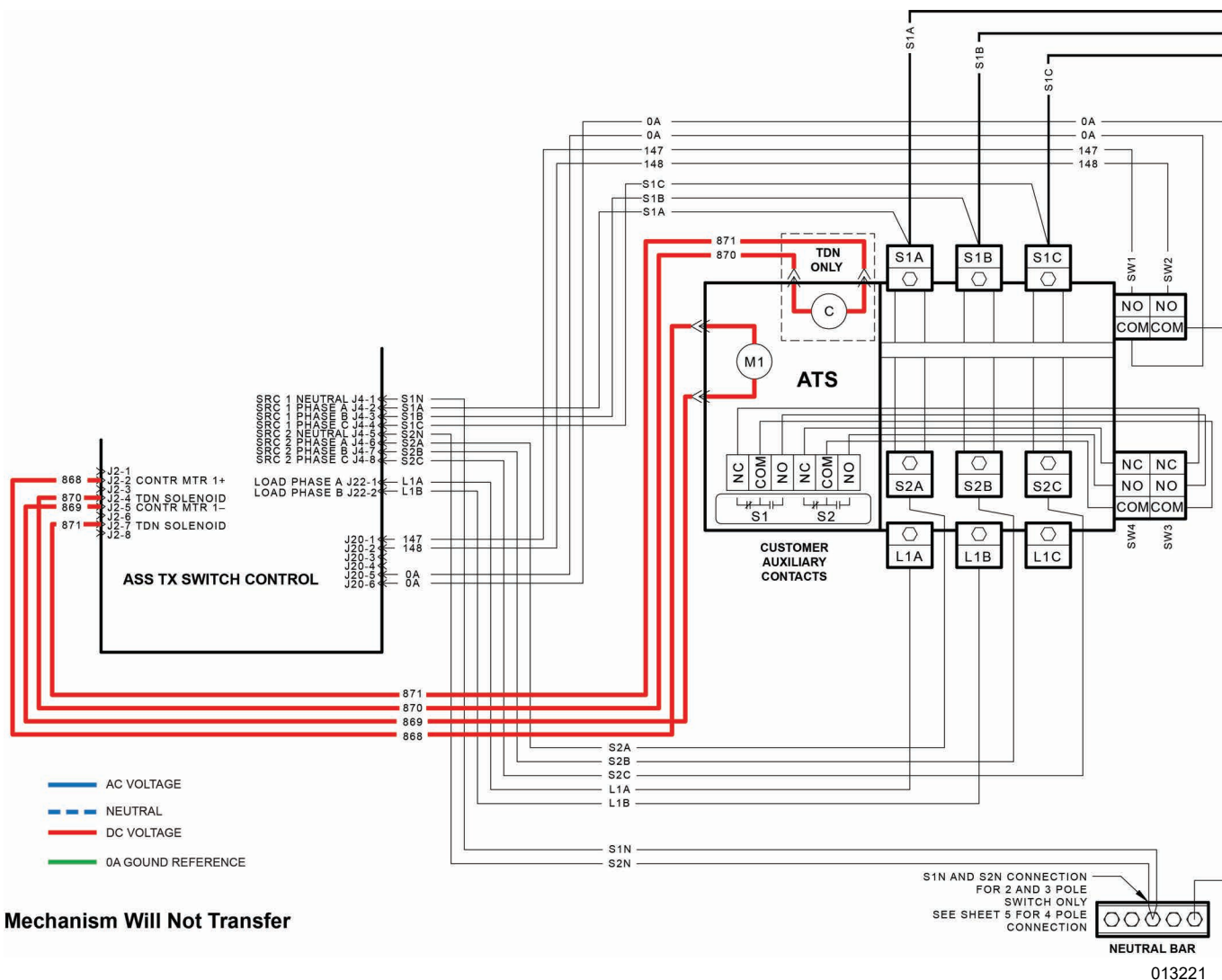
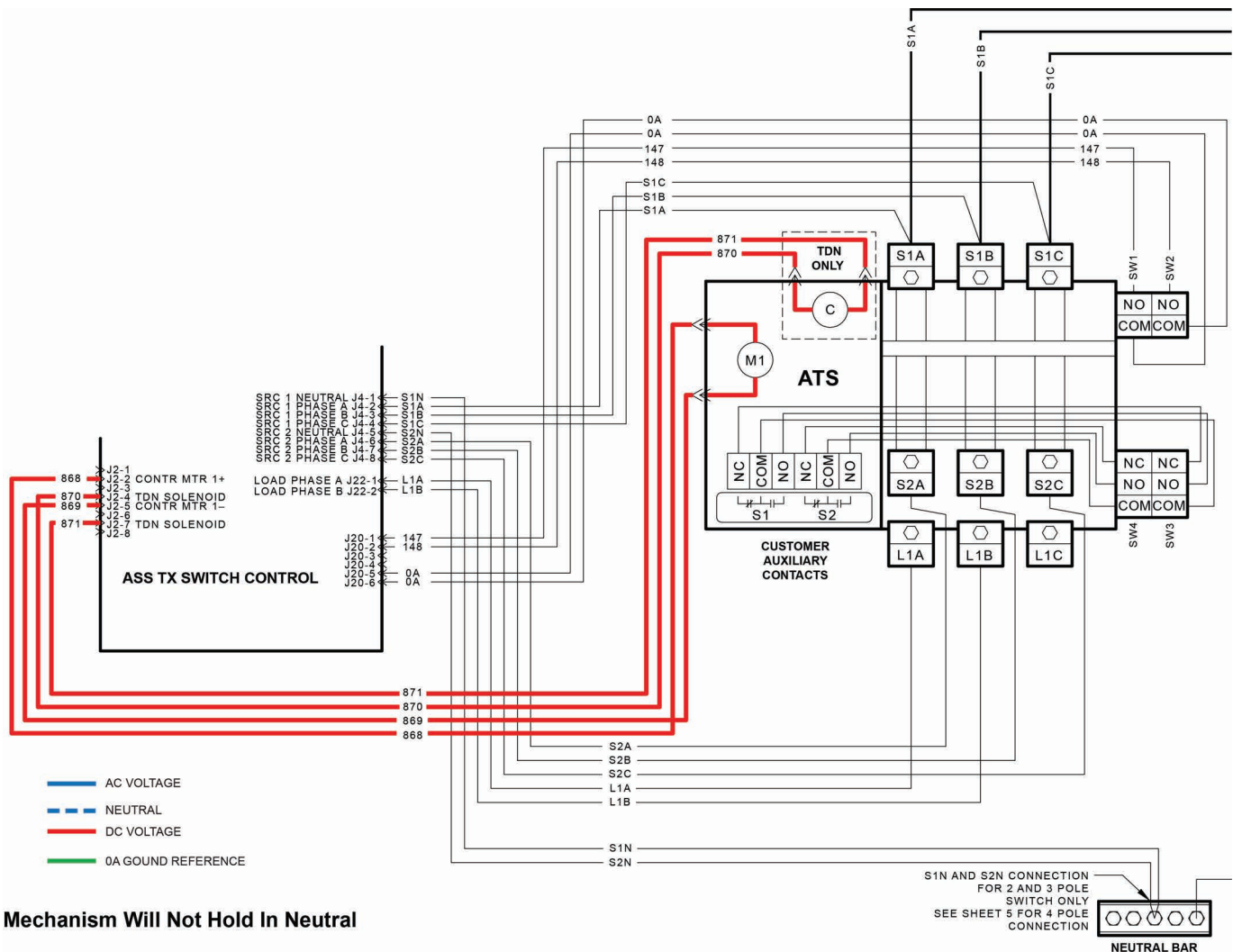


Figure 4-23.

Mechanism Will Not Hold In Neutral

The transfer switch can be purchased with a 3-position contact that will allow for Time Delay In Neutral (TDN). TDN is accomplished by physically holding the switch in the neutral position until a solenoid is energized by approximately 40–60 VDC from the control board. The solenoid releases the switch from the neutral position to the selected position. The solenoid is energized by DC voltage from the control board (approximately 40–60 VDC).

IMPORTANT NOTE: The tests in this section are optimized for use with a Fluke 117 digital multimeter (or equivalent) using appropriate probe adapters.



Mechanism Will Not Hold In Neutral

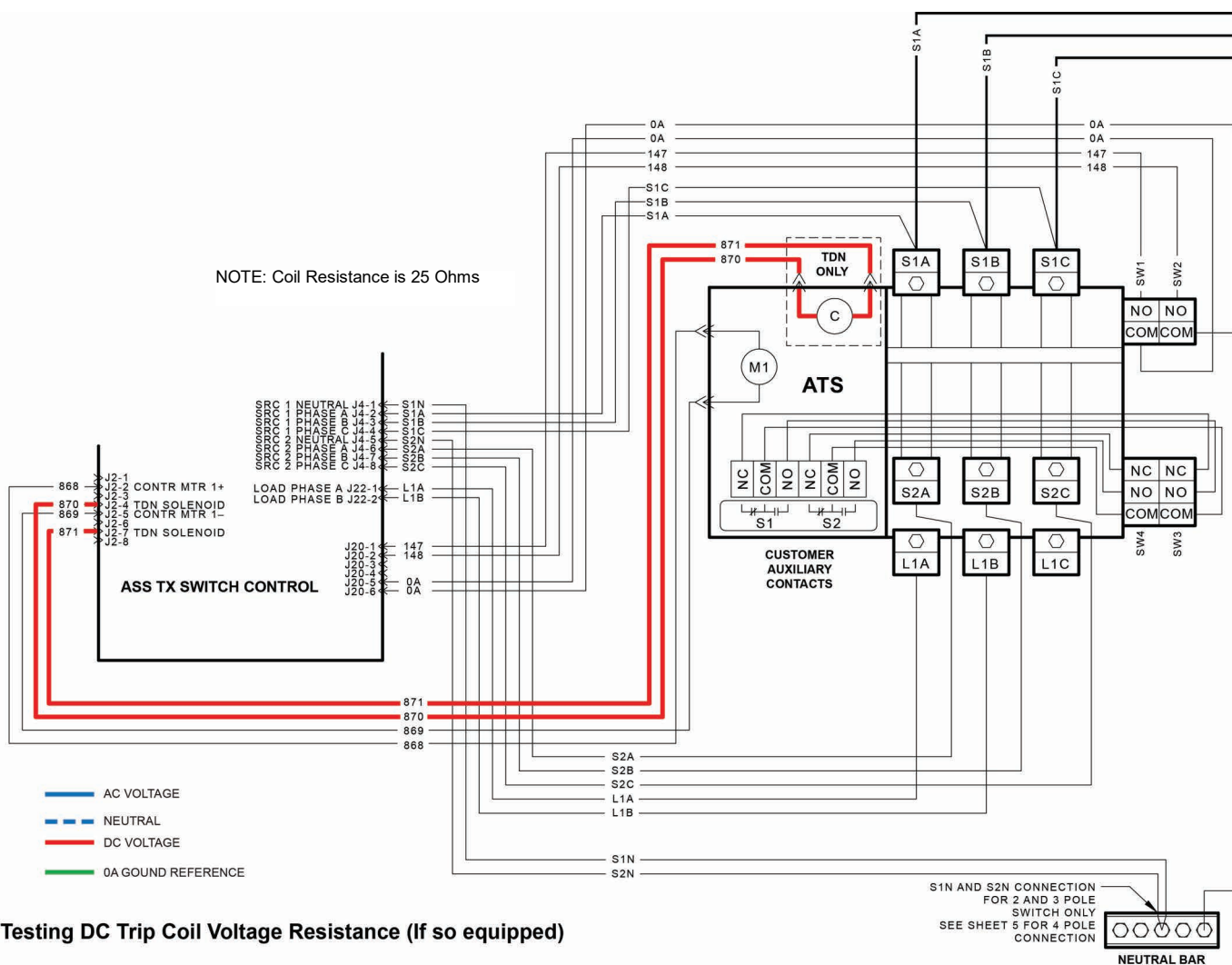
Figure 4-24.

Testing DC Trip Coil Voltage Resistance (If so equipped)

Time Delay In Neutral Control (If equipped)

The switch mechanism has a solenoid installed inside the contactor. The solenoid catches the mechanism in mid travel and holds it in the neutral position. After an internal timer has expired the control board will signal the solenoid to allow the switch to complete its movement to the next position.

IMPORTANT NOTE: The tests in this section are optimized for use with a Fluke 117 digital multimeter (or equivalent) using appropriate probe adapters.



Testing DC Trip Coil Voltage Resistance (If so equipped)

Figure 4-25.

013196

Testing Motor Drive DC Voltage/Resistance

The DC voltage motor drive is used to move the switch contacts from one position to the next. The controller sends a DC voltage to the transfer switch mechanism on Wires 868 and 869.

IMPORTANT NOTE: The tests in this section are optimized for use with a Fluke 117 digital multimeter (or equivalent) using appropriate probe adapters.

Set the meter to read DC volts. Use the Range Hi/Lo to set range to “Manual 6.00” and record “Min Max.”

NOTE: Using the record MIN MAX function of the meter will capture peak closed circuit voltage and polarity being used to energize the DC drive motor.

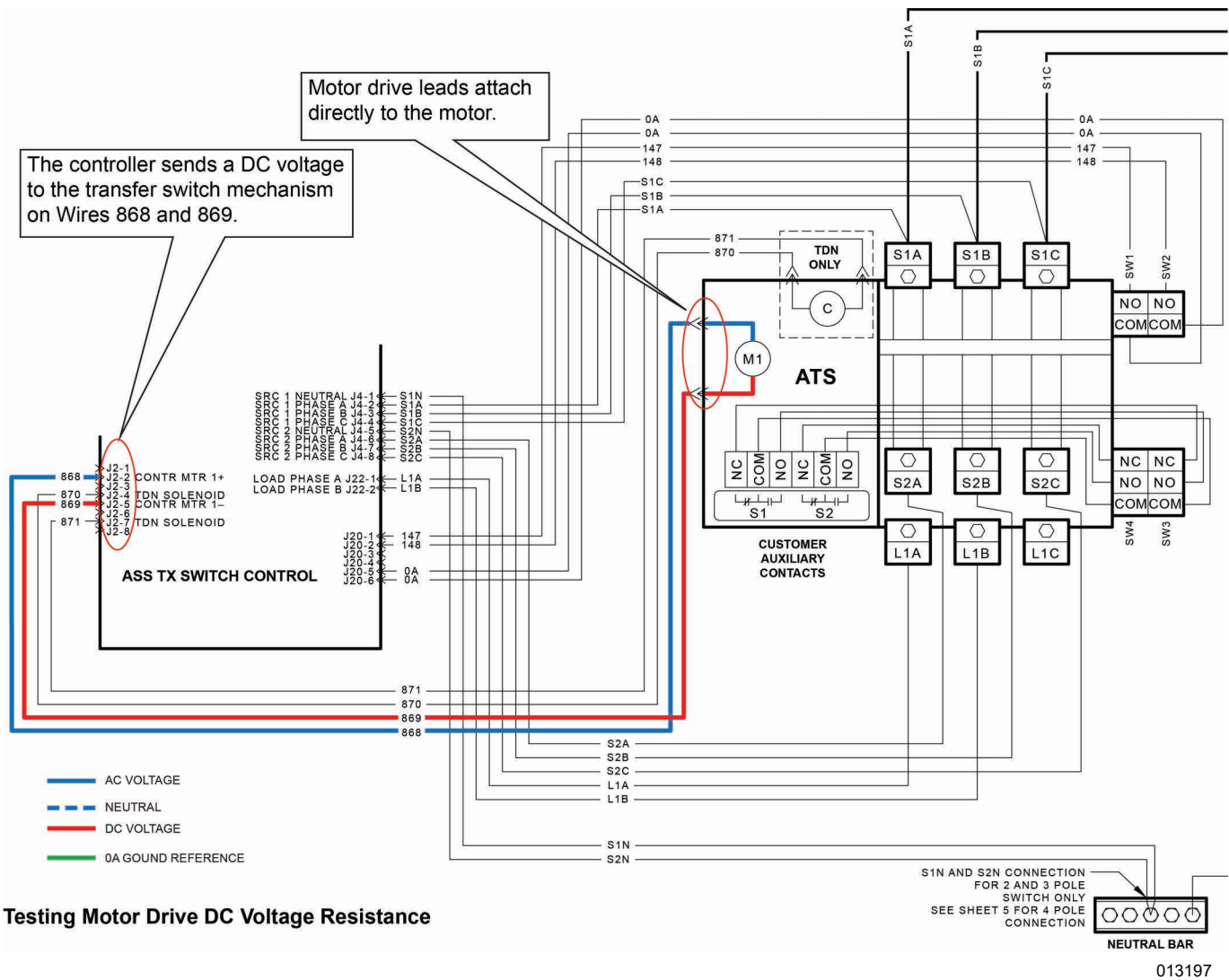


Figure 4-26.

Testing Generator Sensing Voltage

If the controller is not measuring source 2 voltage (typically standby voltage) start by measuring the line-to-line voltage at S2A, S2B and S2C at the contactor mechanism. Measure both the line-to-line and line to the Neutral Bar. This voltage should be measured at the SRC 2 connector, pins J4-6 to J4-8 with J4-5 being the Neutral.

IMPORTANT NOTE: The tests in this section are optimized for use with a Fluke 117 digital multimeter (or equivalent) using appropriate probe adapters.

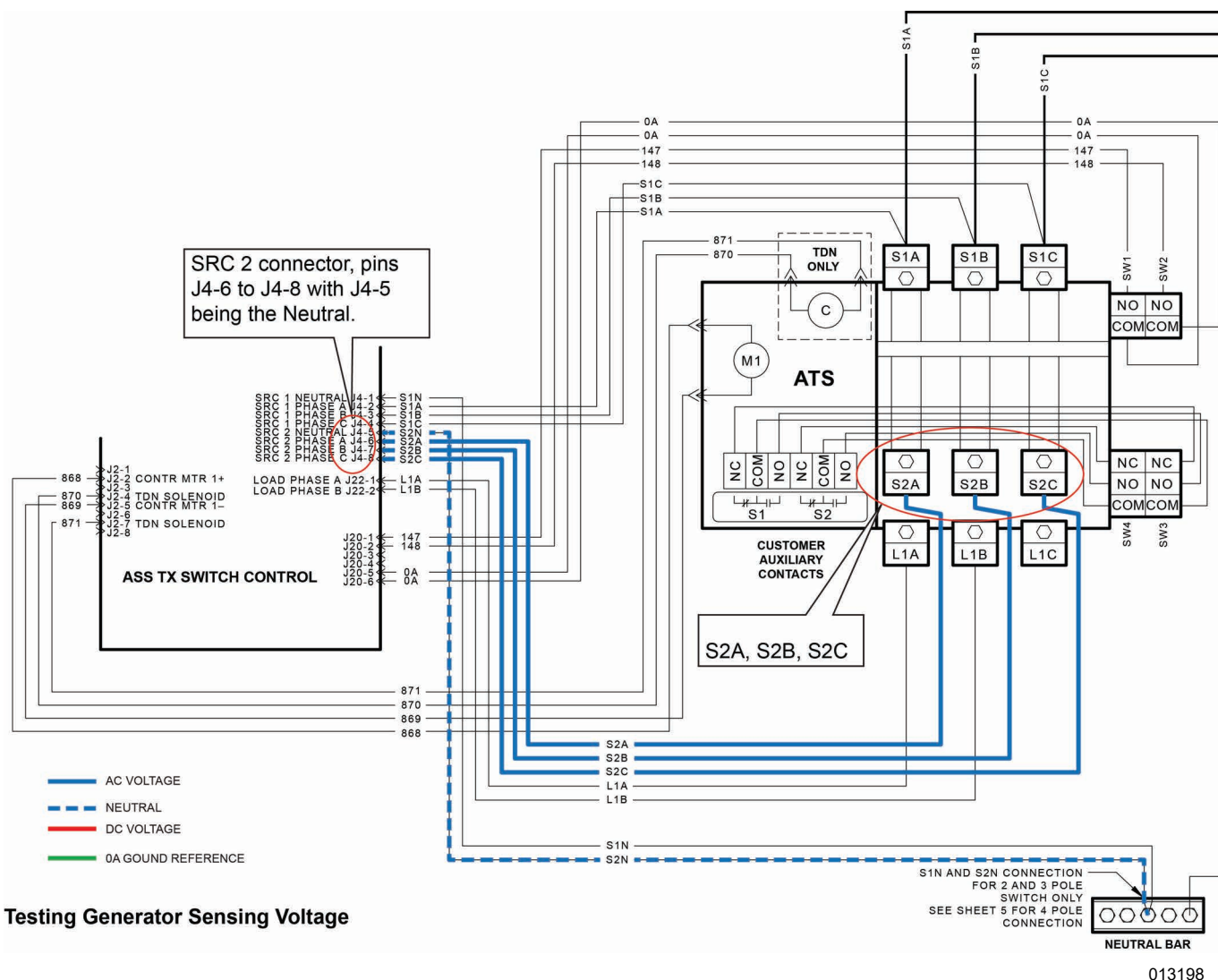


Figure 4-27.

Testing Utility Position Indication

If the controller is not displaying the correct transfer switch position indication then test the position indication circuit. The circuit consists of Wire 0A (an isolated internal ground wire) and Wires 147 and 148. When the transfer switch is in the S1 position 3.3 VDC should be measured between Wires 148 and 0A, and 0 volts between Wires 147 and 0A. When the transfer switch is in the S2 position 3.3 VDC should be measured between Wires 147 and 0A, and 0 volts between Wires 148 and 0A.

IMPORTANT NOTE: The tests in this section are optimized for use with a Fluke 117 digital multimeter (or equivalent) using appropriate probe adapters.

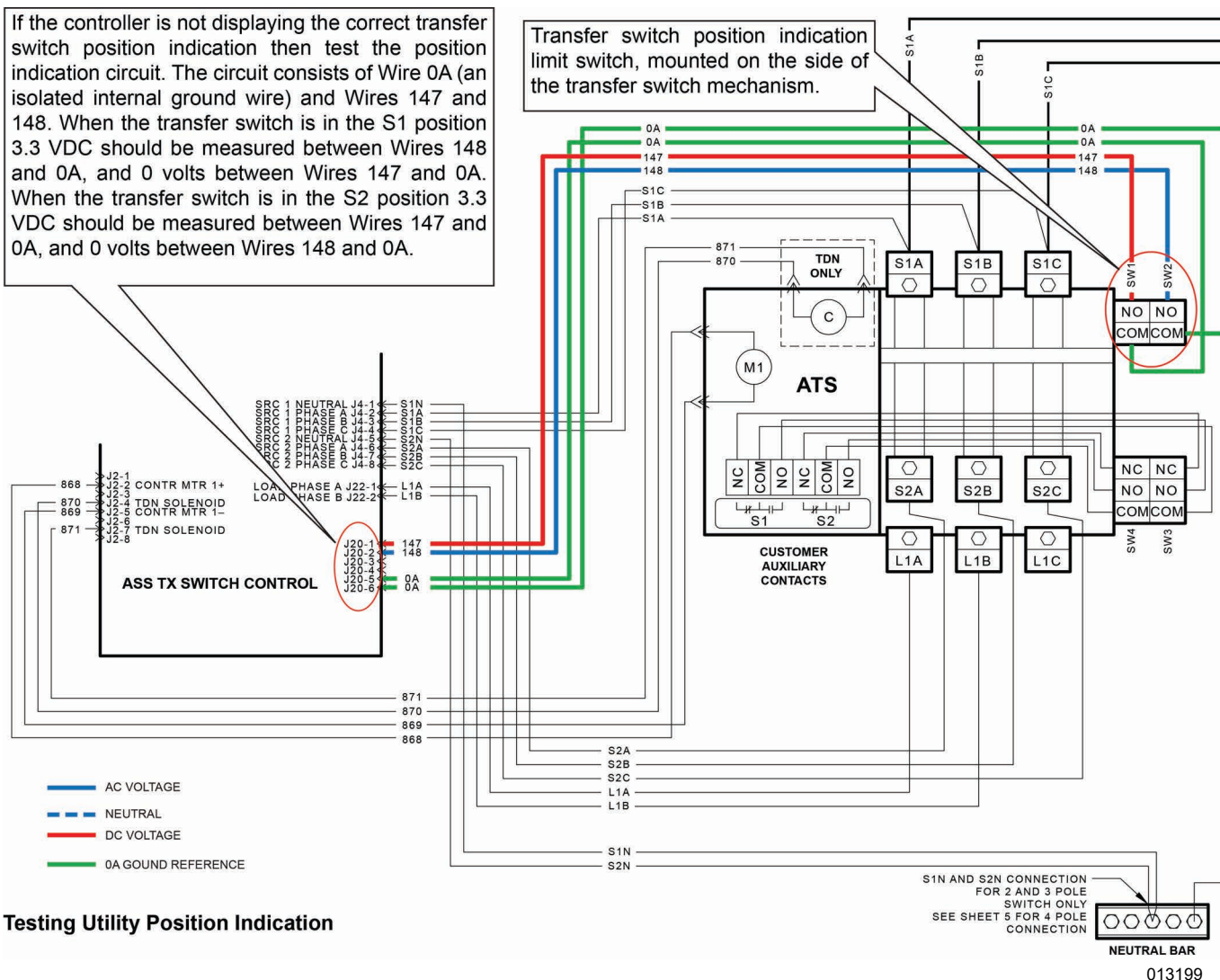
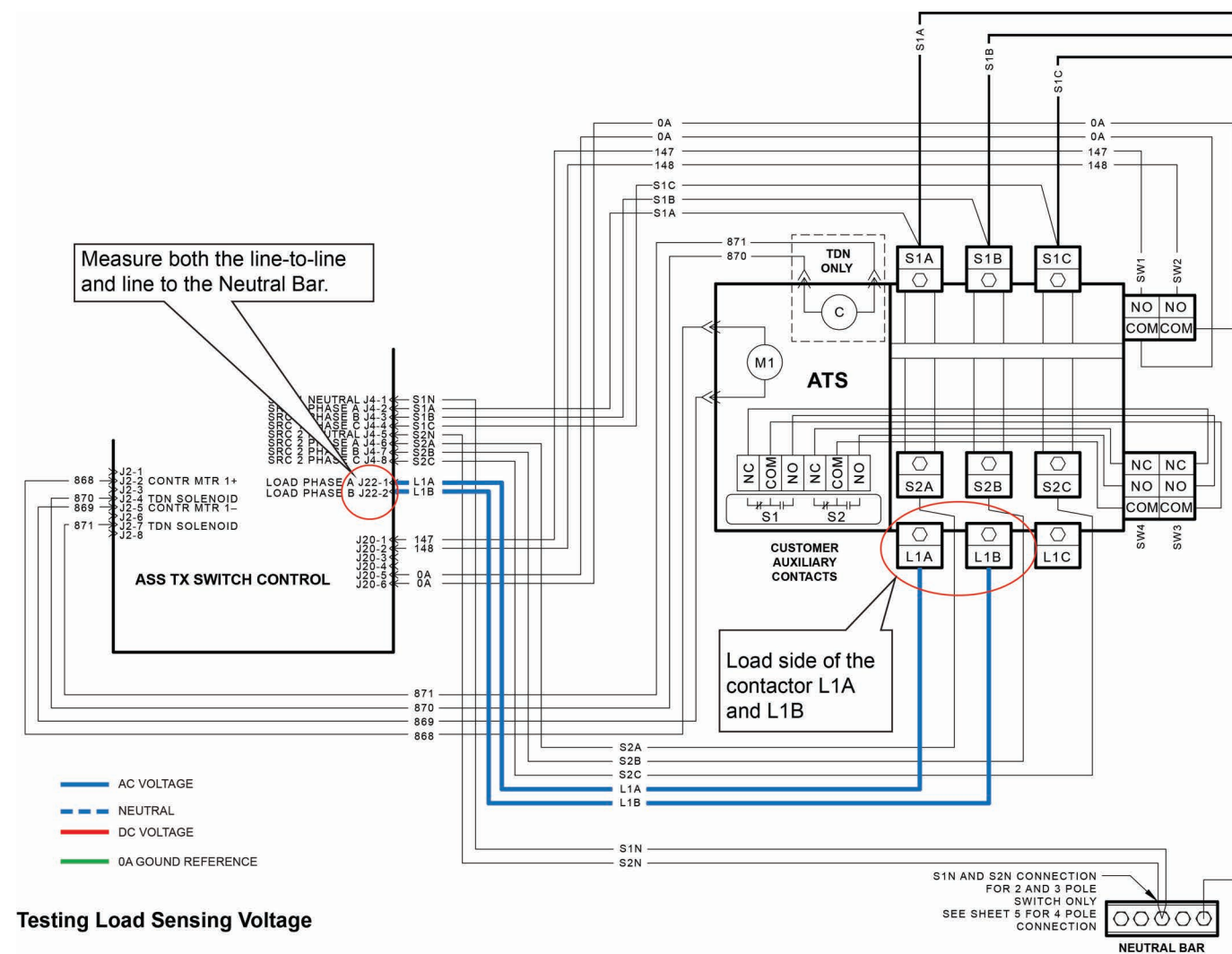


Figure 4-28.

Testing Heater Power Supply Source

The anti-condensate heater is powered from the load terminals of the contactor, through the controller J22 connector. If the controller heater is not functional, start by measuring line-to-line voltage at L1A and L1B. Line-to-line voltage should then be measured at the J22-1 and J22-2 connector. If power is available at this connector, continue to verify power is available before the heater relay and that the relay is being activated by the control according to the heater schematic.

IMPORTANT NOTE: The tests in this section are optimized for use with a Fluke 117 digital multimeter (or equivalent) using appropriate probe adapters.



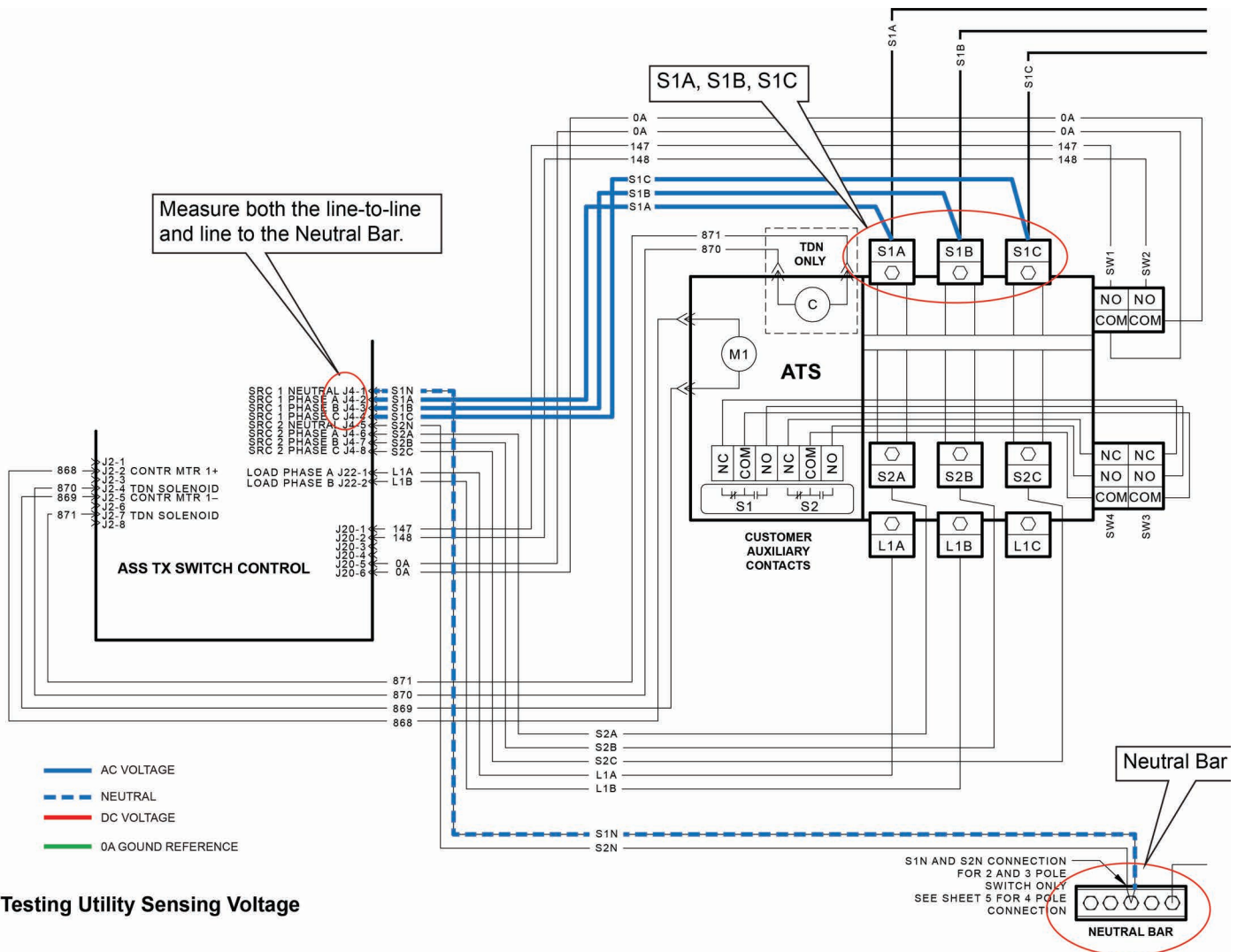
013200

Figure 4-29.

Testing Utility Sensing Voltage

If the controller is not measuring source 1 voltage (typically Utility Voltage) start by measuring the line-to-line voltage at S1A, S1B and S1C. Measure both the line-to-line and line to the Neutral Bar. This Voltage should then be measured at the SRC 1 connector J4-2 to J4-4 with J4-1 being the neutral.

IMPORTANT NOTE: The tests in this section are optimized for use with a Fluke 117 digital multimeter (or equivalent) using appropriate probe adapters.



013201

Figure 4-30.

Section 5 TX TTS Transfer Switch Troubleshooting

W-Type Transfer Switch – No Transfer to Standby



⚠ DANGER

Electrocution. High voltage is present at transfer switch and terminals. Contact with live terminals will result in death or serious injury.

(000129)

1. Remove all power sources from switch.
2. Set a digital multimeter (DMM) to measure resistance (Ω).
3. Isolate Wire E2.
4. Check resistance of Wire E2 from terminal B2 to main lug E2.
5. If wire is open, replace wire and re-test switch.

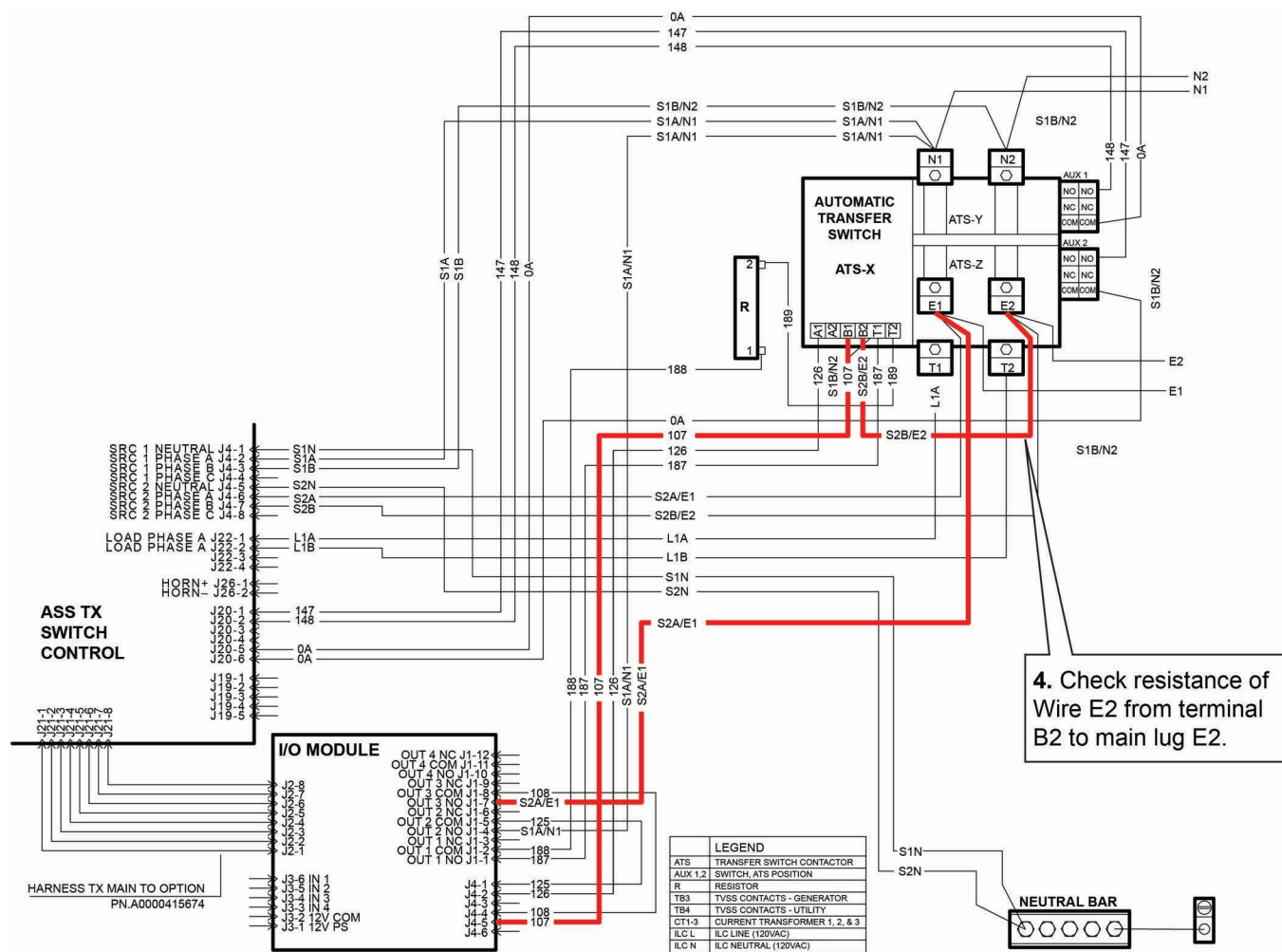


Figure 1-31.

013190



(000129)

1. Check for AC generator voltage at B1 and B2 on the transfer switch mechanism terminal strip.
 - If generator voltage is present at B1 and B2, the problem is internal to the mechanism. Contact service for further action.

- If generator voltage is NOT present at B1 and B2, troubleshoot the wiring and relay contacts.
- Place the black test lead on main lug E2 of the transfer switch.

3. Set DMM to measure AC volts.
4. Use the red test lead to measure for generator voltage at B1, Wire 107.
 - If generator voltage is not present proceed to next step (do not remove the wire).

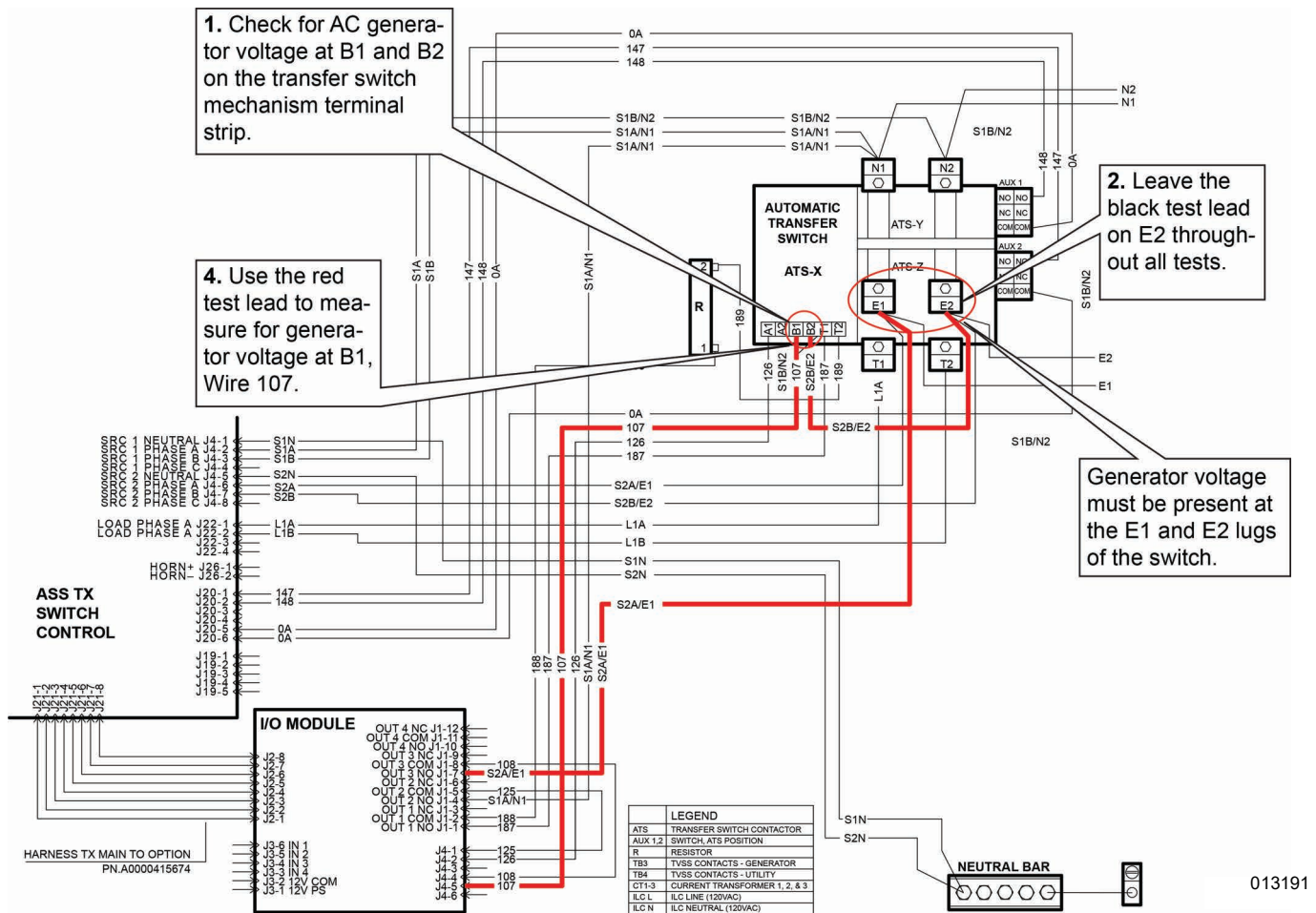


Figure 1-32.

W-Type Transfer Switch – No Transfer to Standby



⚠ DANGER

Electrocution. High voltage is present at transfer switch and terminals. Contact with live terminals will result in death or serious injury.

(000129)

1. Place the black test lead on main lug E2 of the transfer switch.

NOTE: Leave the black test lead on main lug E2 throughout remaining steps.

2. Measure for voltage at J4-5 Wire 107 on the I/O board.
 - If generator voltage is measured this indicates Wire 107 is open, test and replace Wire 107.

- If generator voltage is not measured proceed to next step.
3. Measure for voltage at J1-7 Wire S2A/E1 on the I/O board (do not remove the wire).
 - If generator voltage is measured this indicates a problem in the I/O board normally open contacts.
 - If generator voltage is not measured proceed to next step.
4. Measure for voltage at S2A/E1 main lug of the transfer switch (do not remove the wire).
 - If generator voltage is measured then Wire S2A/E1 is open, test and replace Wire E1.
 - If generator voltage is not measured this indicates the problem is not with the switch but with the generator supply voltage. Refer to the diagnostic manual for the generator.

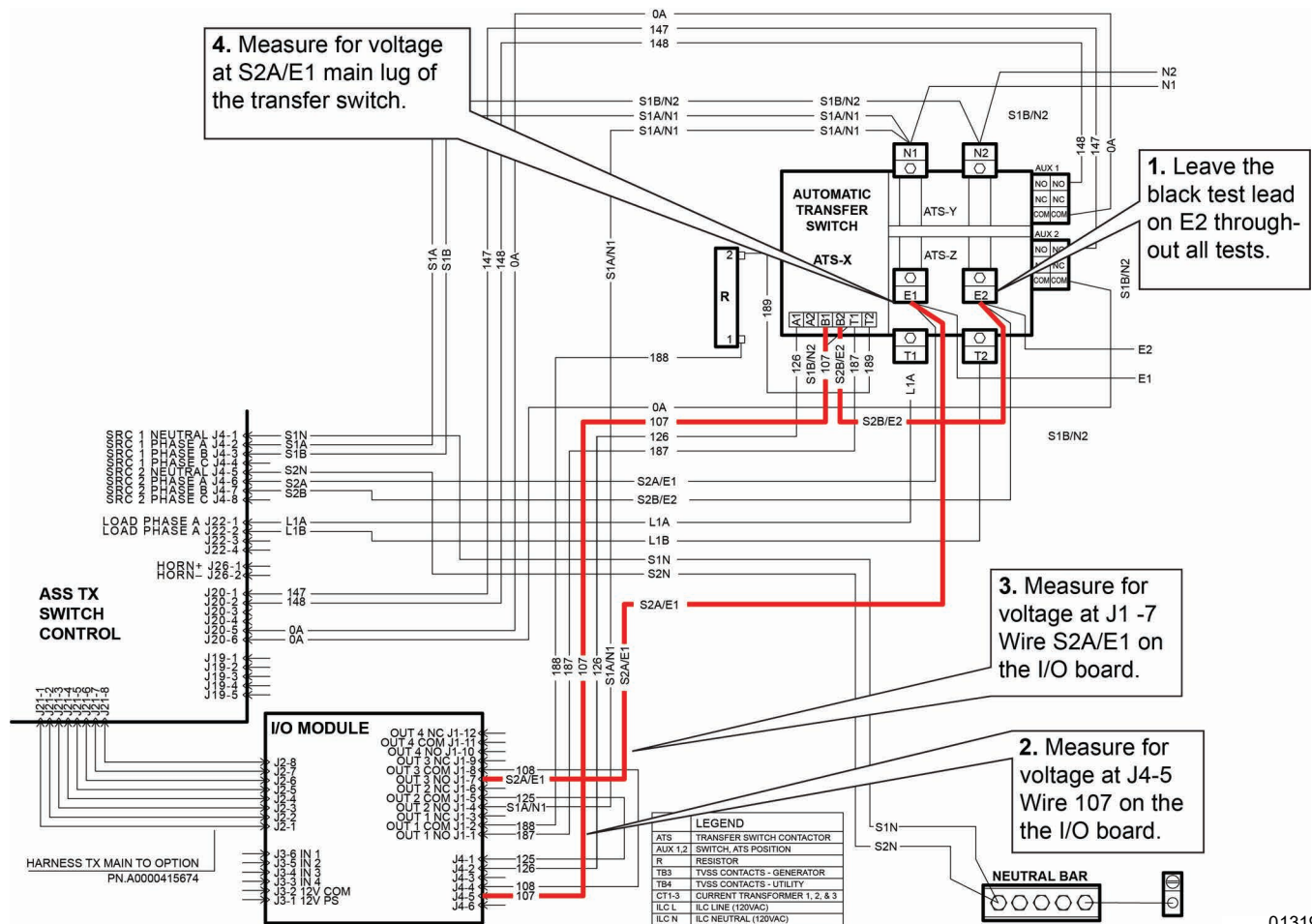


Figure 1-33.

013192

W-Type Transfer Switch – No Transfer to Utility



⚠ DANGER

Electrocution. High voltage is present at transfer switch and terminals. Contact with live terminals will result in death or serious injury.

(000129)

1. Remove all power sources from switch.
2. Set a digital multimeter (DMM) to measure resistance (Ω).
3. Isolate Wire S1B/N2.
4. Check resistance of Wire S1B/N2 from terminal A2 to main lug N2.
5. If wire is open, replace wire and re-test switch.

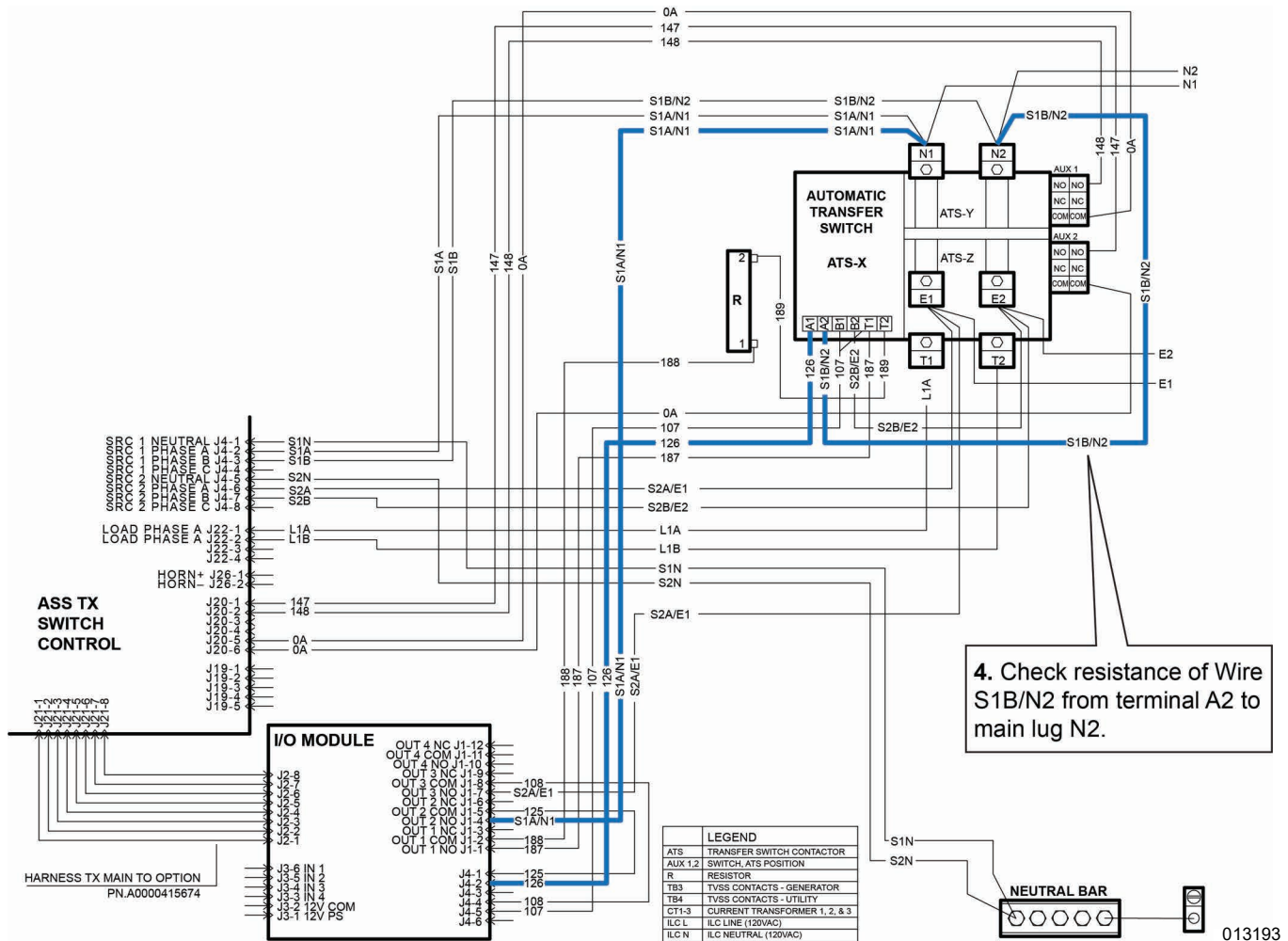


Figure 1-34.

W-Type Transfer Switch – No Transfer to Utility



⚠ DANGER

Electrocution. High voltage is present at transfer switch and terminals. Contact with live terminals will result in death or serious injury.

(000129)

NOTE: Utility voltage must be present at the N1 and N2 lugs of the switch (I/O board must be de-energized).

1. Check for AC utility voltage across A1 and A2 on the transfer switch mechanism terminal strip.
 - If utility voltage is present at A1 and A2, the problem is internal to the mechanism. Contact service for further action.

- If utility voltage is NOT present at A1 and A2, troubleshoot the wiring and relay contacts.
- 2. Place the black test lead on main lug N2 of the transfer switch.

NOTE: Leave the black test lead on main lug N2 throughout remaining steps.

3. Set DMM to measure AC volts.
4. Use the red test lead to measure for generator voltage at A1, Wire 126.
- If generator voltage is not present proceed to next step (do not remove the wire).

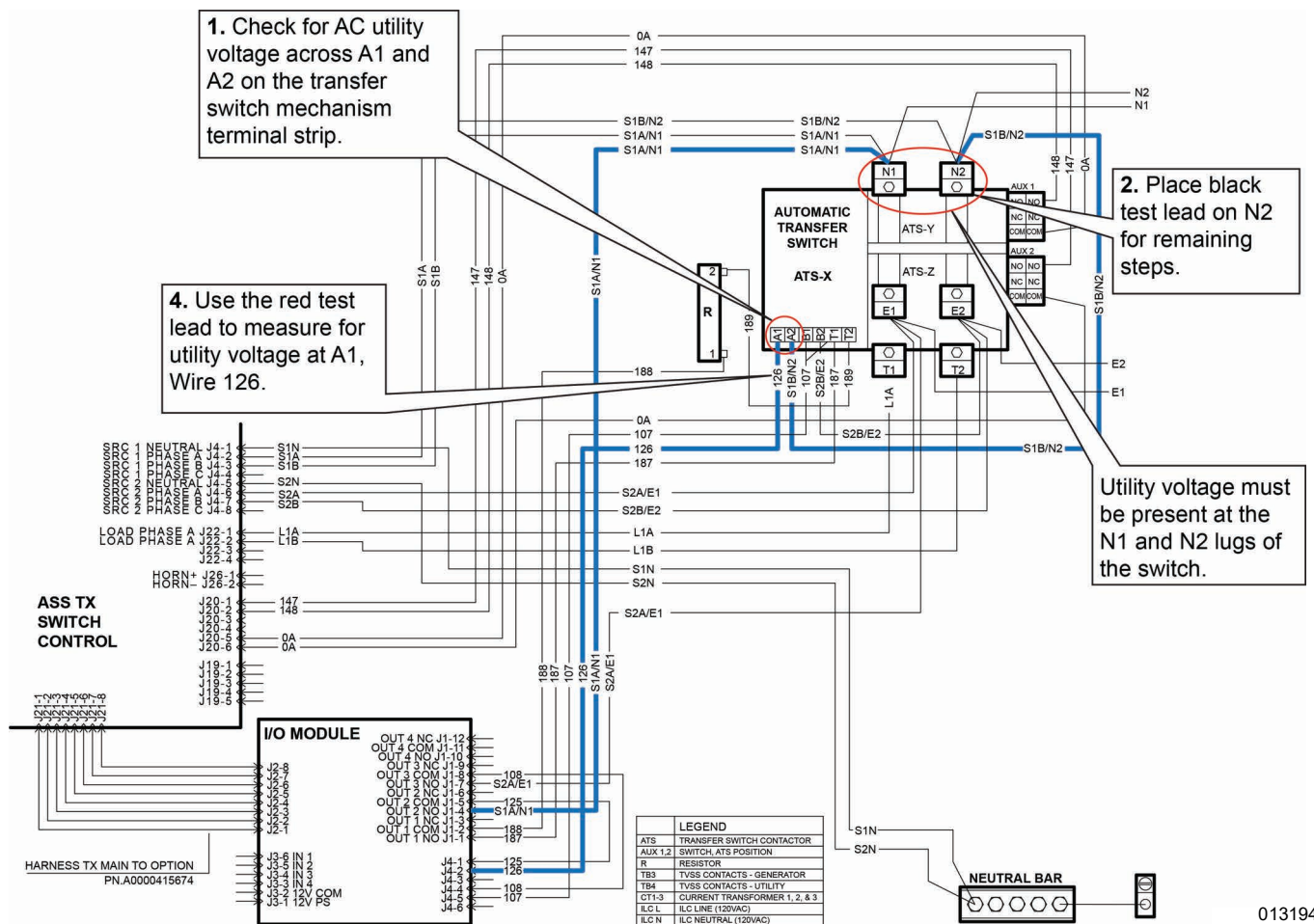


Figure 1-35.

013194

W-Type Transfer Switch – No Transfer to Utility



DANGER

Electrocution. High voltage is present at transfer switch and terminals. Contact with live terminals will result in death or serious injury.

(000129)

1. Place the black test lead on main lug N2 of the transfer switch.

NOTE: Leave the black test lead on main lug N2 throughout remaining steps.

NOTE: Relay must be de-energized for next step.

2. Measure for utility voltage at J4-2 Wire 126 on the I/O board (do not remove the wire).
 - If utility voltage is measured this indicates Wire 126 is open. Test and replace Wire 126.

- If utility voltage is not measured proceed to next step.
3. Measure for voltage at J1-4 Wire S1A/N1 on the I/O board (do not remove the wire).
 - If utility voltage is measured this indicates a problem in the I/O board normally closed contacts.
 - If utility voltage is not measured proceed to next step.
 4. Measure for voltage at main lug N1 of the transfer switch (do not remove the wire).
 - If utility voltage is measured then Wire S1A/N1 is open. Test and replace Wire S1A/N1.
 5. Measure for voltage at J1-4 Wire S1A/N1 on the I/O board.
 - If utility voltage is not measured this indicates the problem is not with the switch but with the utility supply voltage.

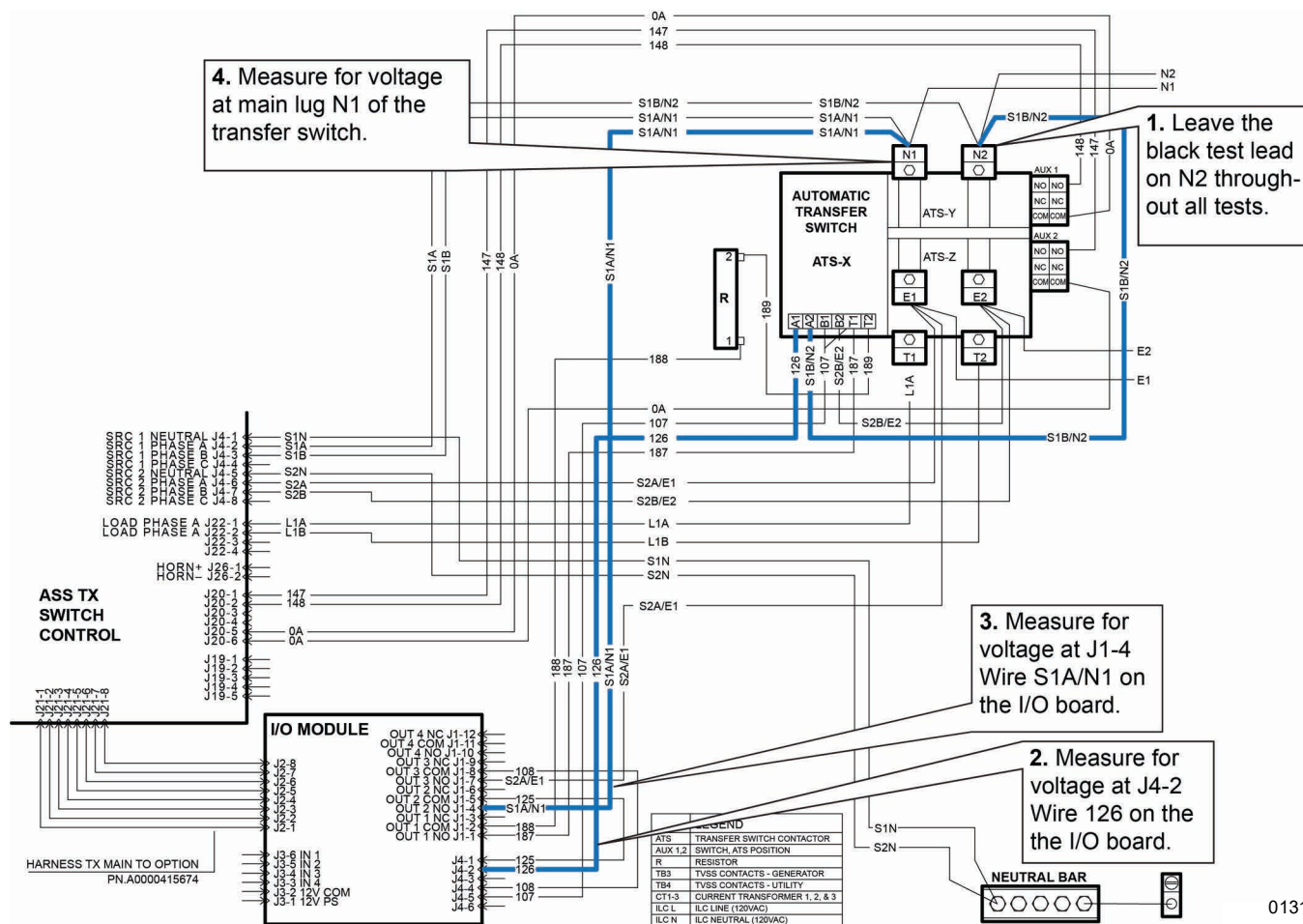


Figure 1-36.

013195

Section 6 800–1600 Amp TX Transfer Switch Troubleshooting

Before Troubleshooting

Verify switch configuration and on-site voltage are correctly matched.

MIN MAX AVG Recording Mode



⚠ DANGER

Electrocution. High voltage is present at transfer switch and terminals. Contact with live terminals will result in death or serious injury.

(000129)

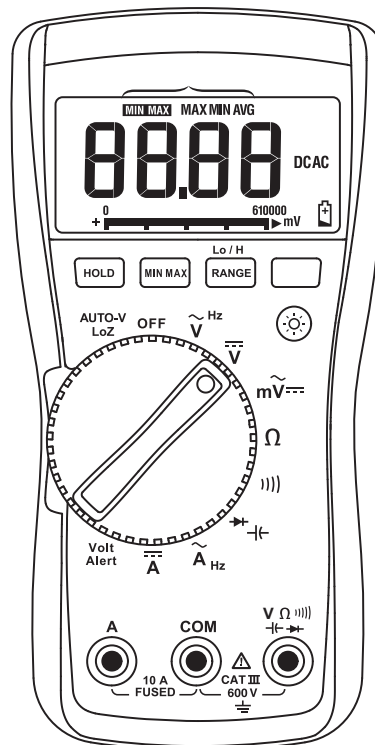


Figure 1-37.

The MIN MAX AVG recording mode captures the minimum and maximum input values (ignoring overloads), and calculates a running average of all readings. When the digital multimeter (DMM) detects a new high or low, the DMM beeps.

NOTE: Autoranging and Battery Saver™ are disabled in MIN MAX AVG mode.

1. Set the meter to read DC volts.
2. Use the Range Hi/Lo to set range to “Manual 6.00”.
3. Select MinMax mode.

NOTE: Using the record MIN MAX function of the meter will capture peak closed circuit voltage and polarity being used to energize the DC drive motor.

NOTE: Procedure based on Fluke 117.

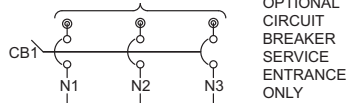
Switch Won't Trip To Neutral

Check diode function

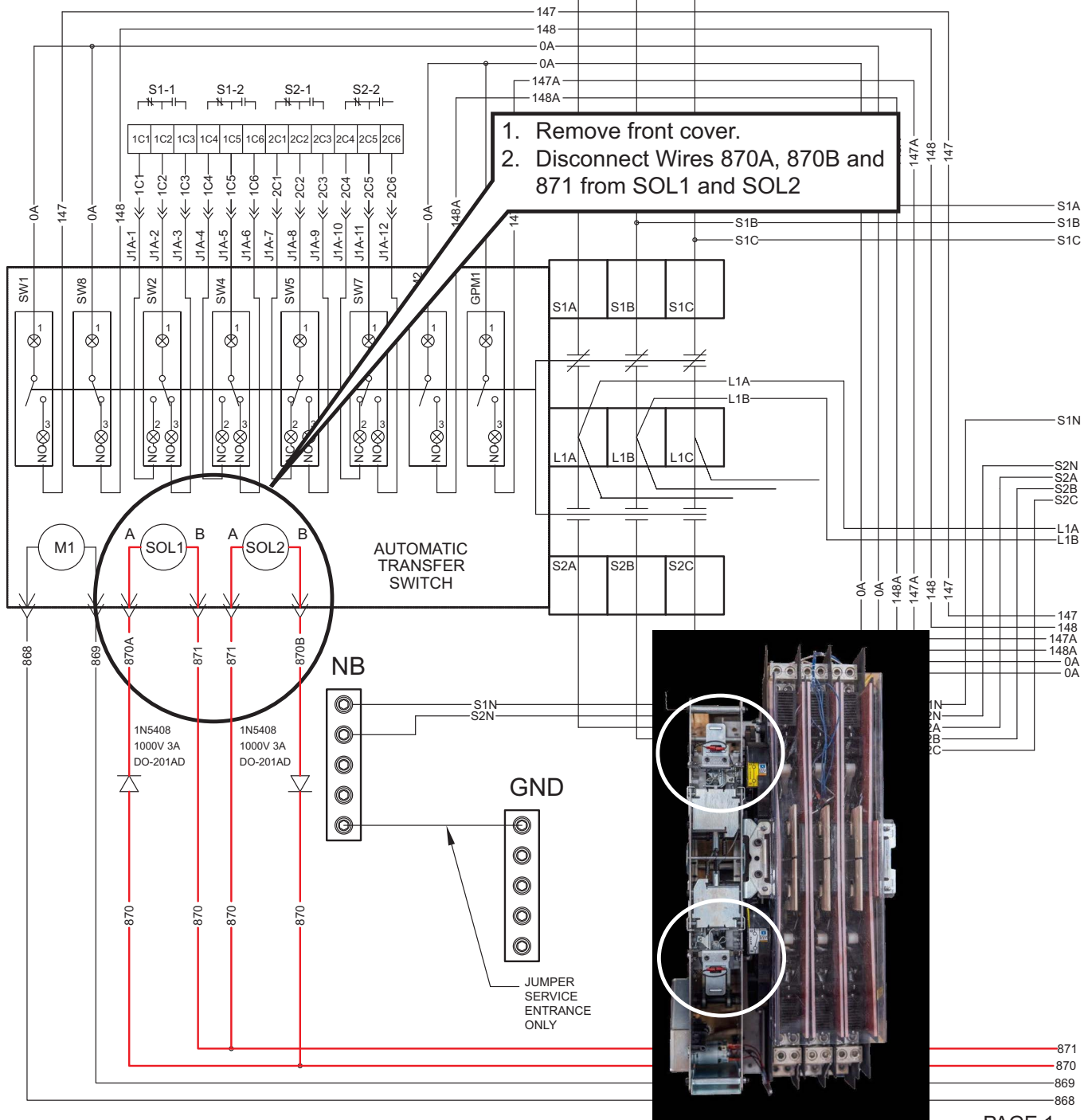
Wire 870A (at SOL1) to Wire 870
(harness end at TX controller)

Wire 870B (at SOL2) to Wire 870
(harness end at TX controller)

CUSTOMER CONNECTION
- UTILITY -



NOTE:
ALL CONTACTS SHOWN
WITH TRANSFER SWITCH
IN UTILITY POSITION.

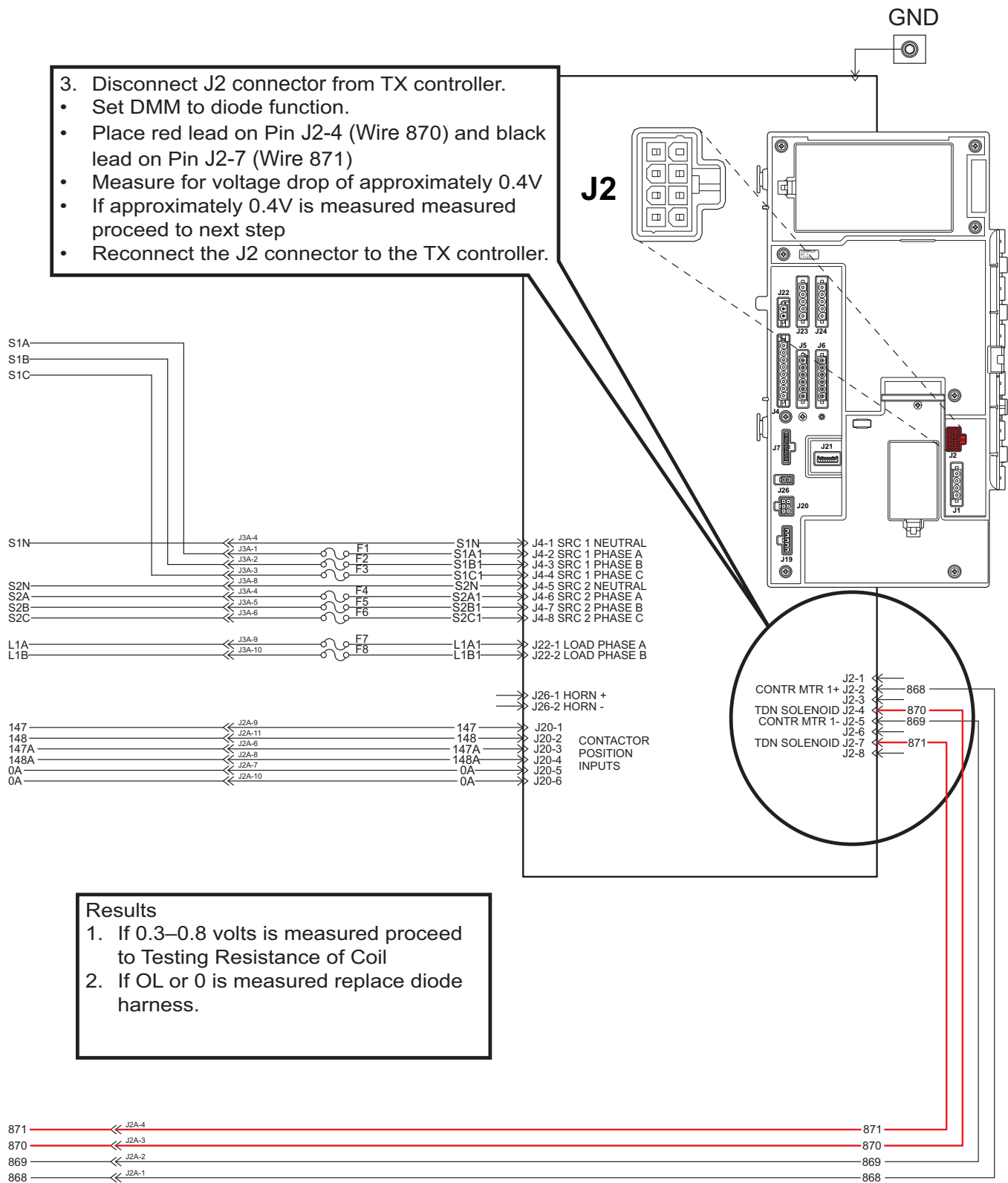


PAGE 1

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Figure 1-38.

Switch Won't Trip To Neutral



PAGE 2

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Figure 1-39.

Diagnostic Repair Manual



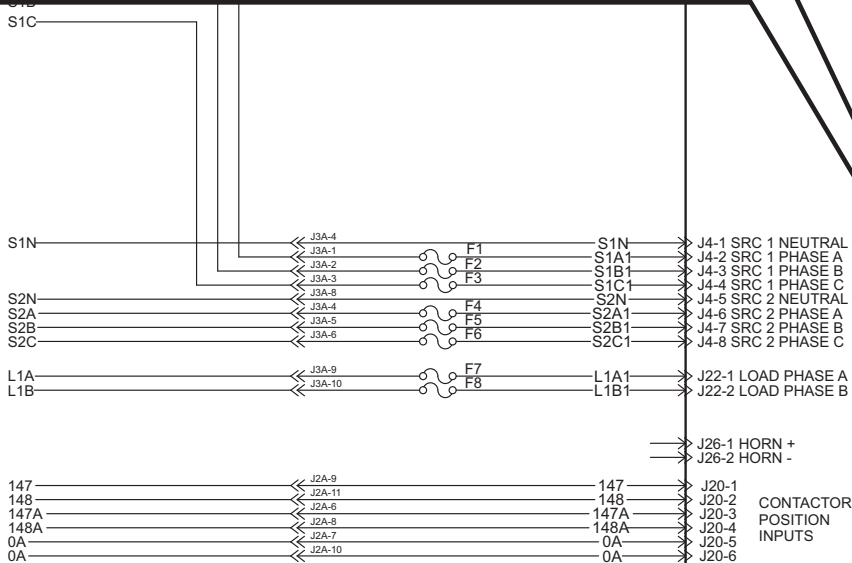
Figure 1-40.

Testing Motor Drive Voltage and Resistance

Testing Voltage

1. Set DMM to read DC volts.
 2. Use the Range Hi/Lo to set range to “Manual 6.00” and record “Min Max.”
 3. Back probe test leads on J2-2 (Wire 868) and J2-5 (Wire 869) at the control board J2 connection.
- 150 VDC (non-polarity specific) is delivered to the drive motors on Wires 868 and 869 while the controller is energizing the circuit. The drive motors will be audible while charging.

If 0 VDC or low VDC is measured from the control board, control board may be faulty. Before condemning control board validate that neutral connection is present at control board.

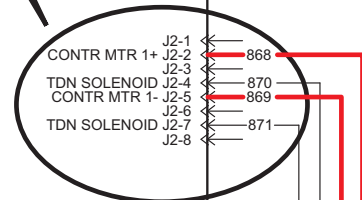
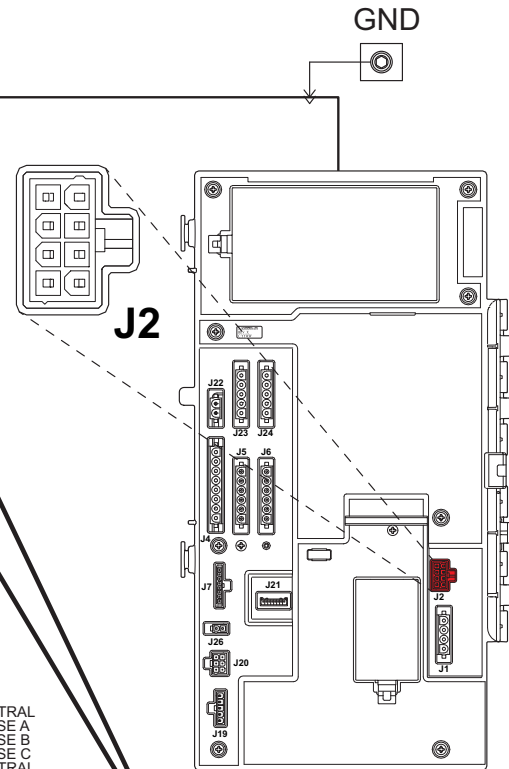
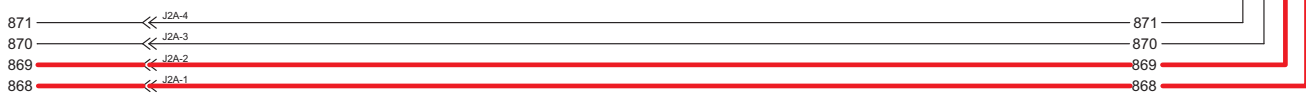


Testing Resistance

1. Remove J2 connector from controller.
2. Set DMM to read resistance.
3. Back-probe Pins J2-4 (Wire 870) and J2-7 (Wire 871).
4. A resistance of 12–20 Ohms should be measured.

Result

1. If on open (OL) is measured test the harness for continuity and each motor for resistance.



PAGE 2

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Figure 1-41.

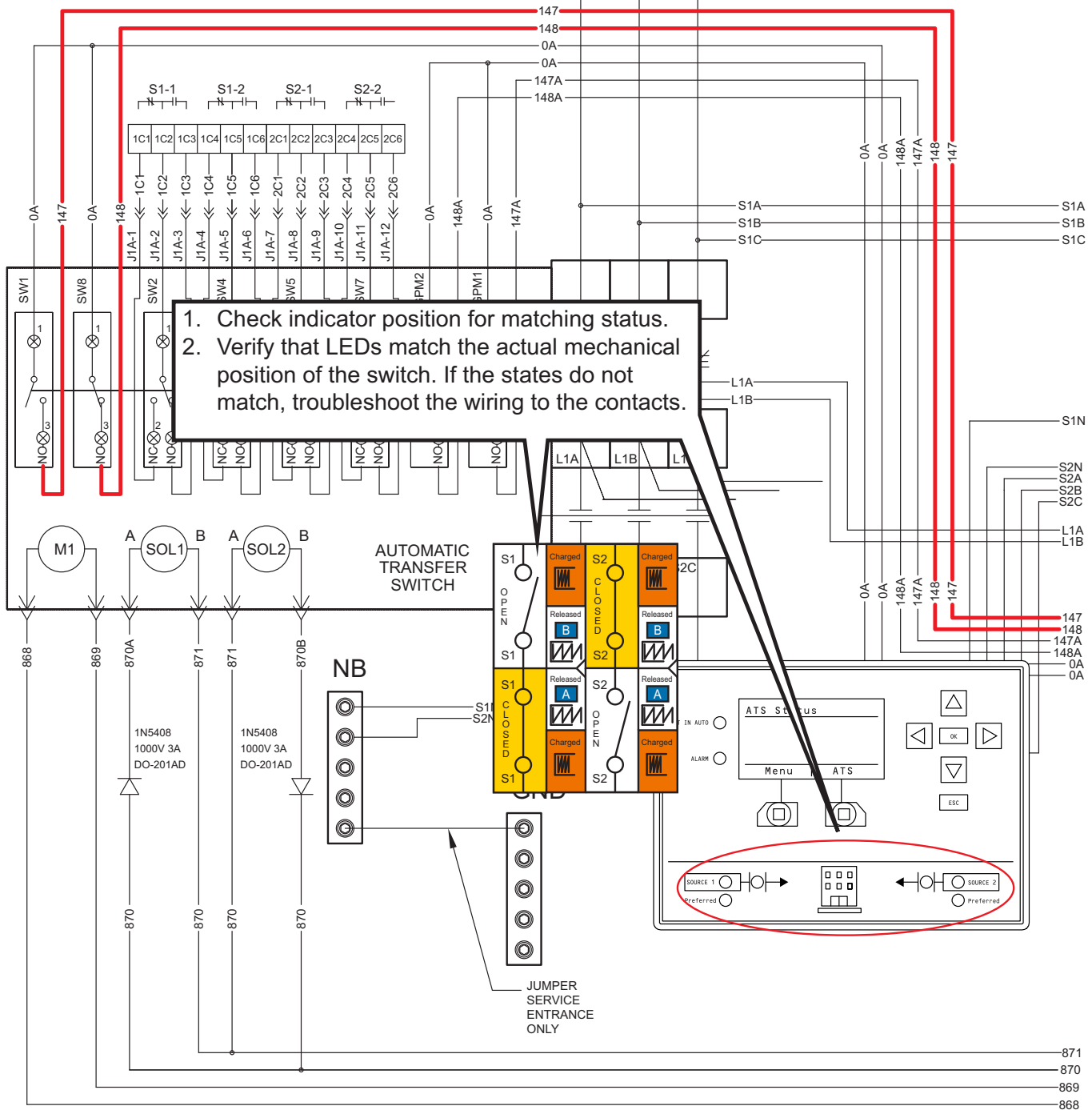
Validating Limit Switch Feedback

Check SW1 and SW8 at controller connector for continuity (Wire 147 and Wire 148).

CUSTOMER CONNECTION
- UTILITY -

OPTIONAL
CIRCUIT
BREAKER
SERVICE
ENTRANCE
ONLY

NOTE:
ALL CONTACTS SHOWN
WITH TRANSFER SWITCH
IN UTILITY POSITION.

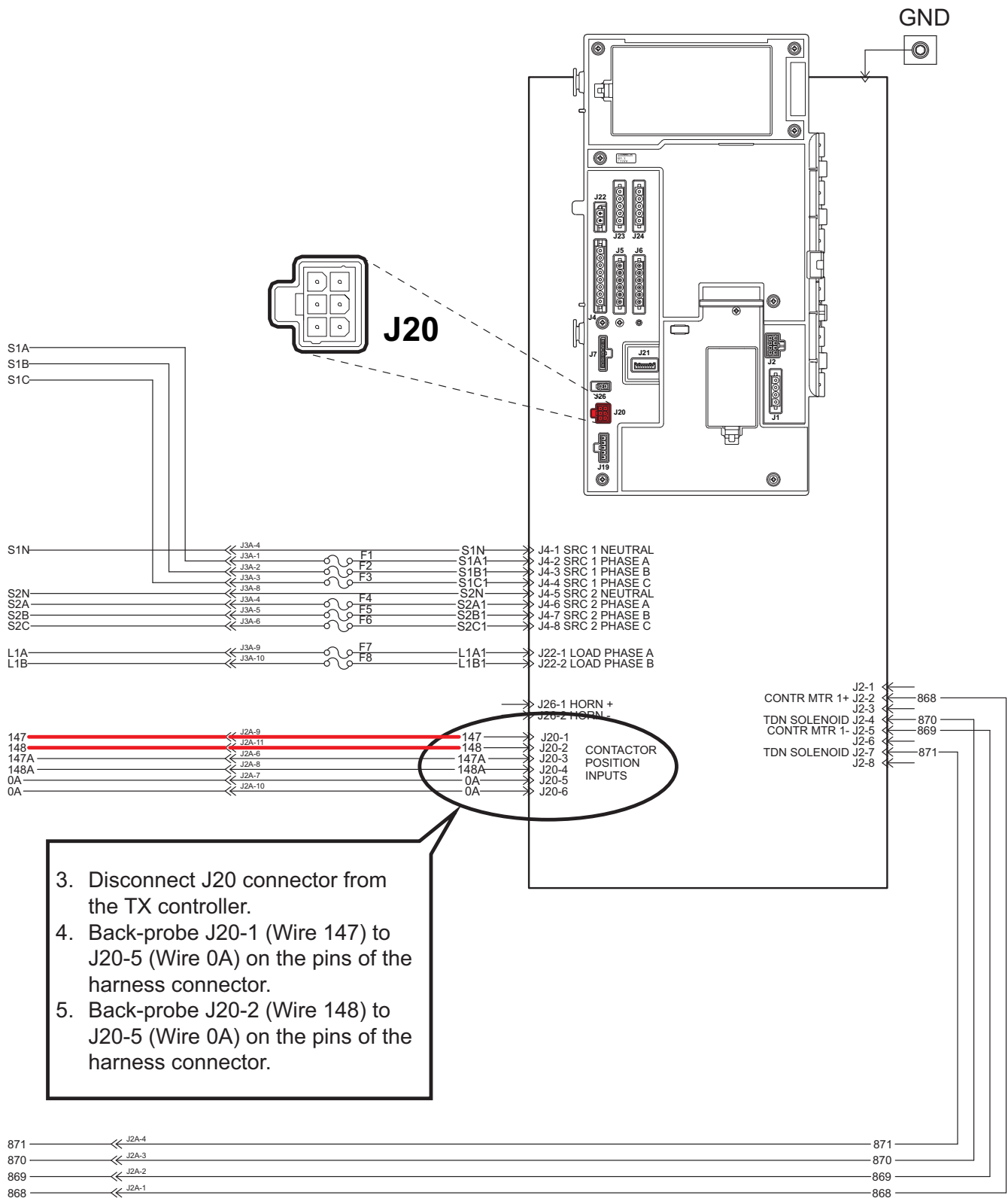


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Figure 1-42.

Validating Limit Switch Feedback

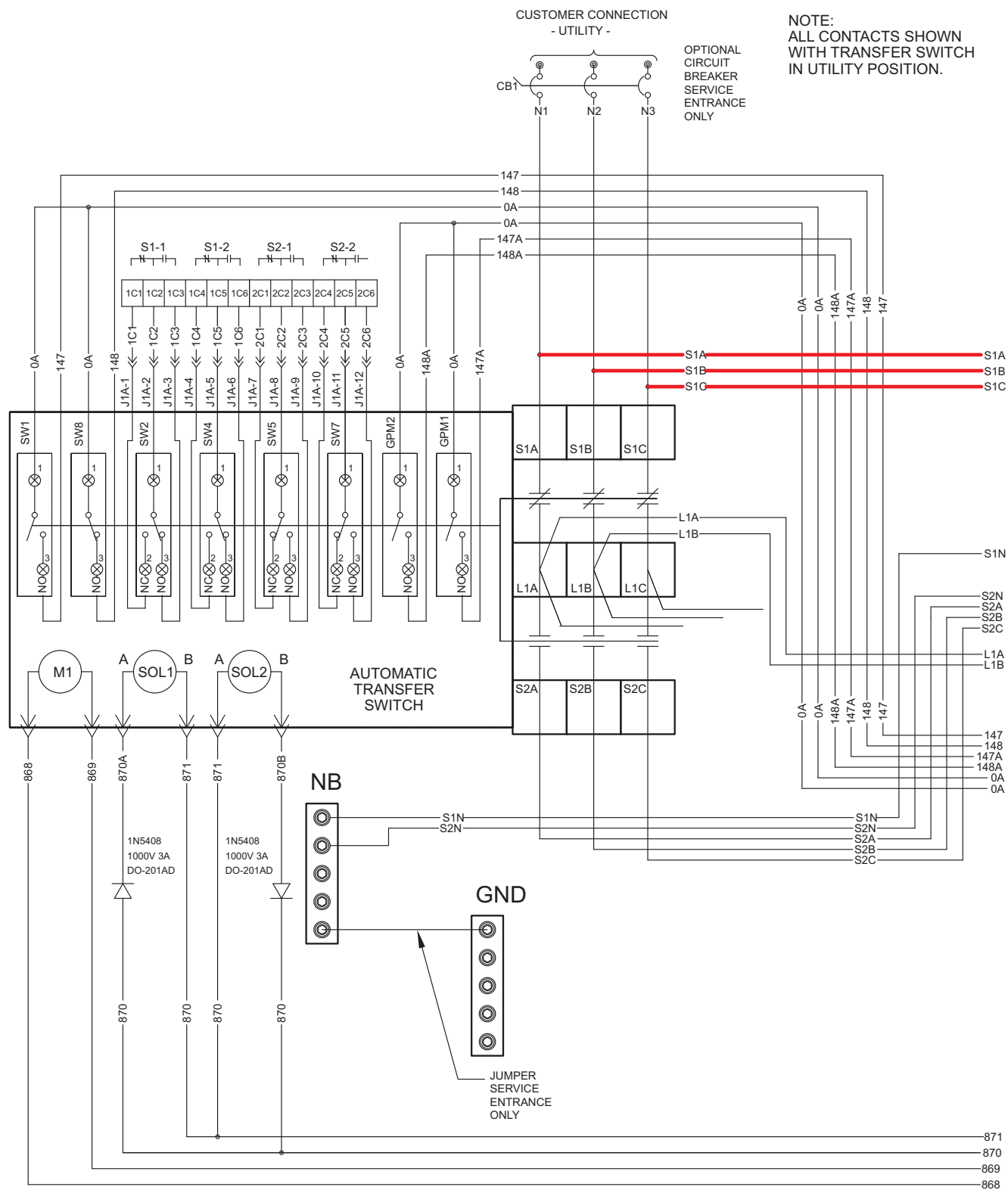


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Figure 1-43.

Testing Source Voltage



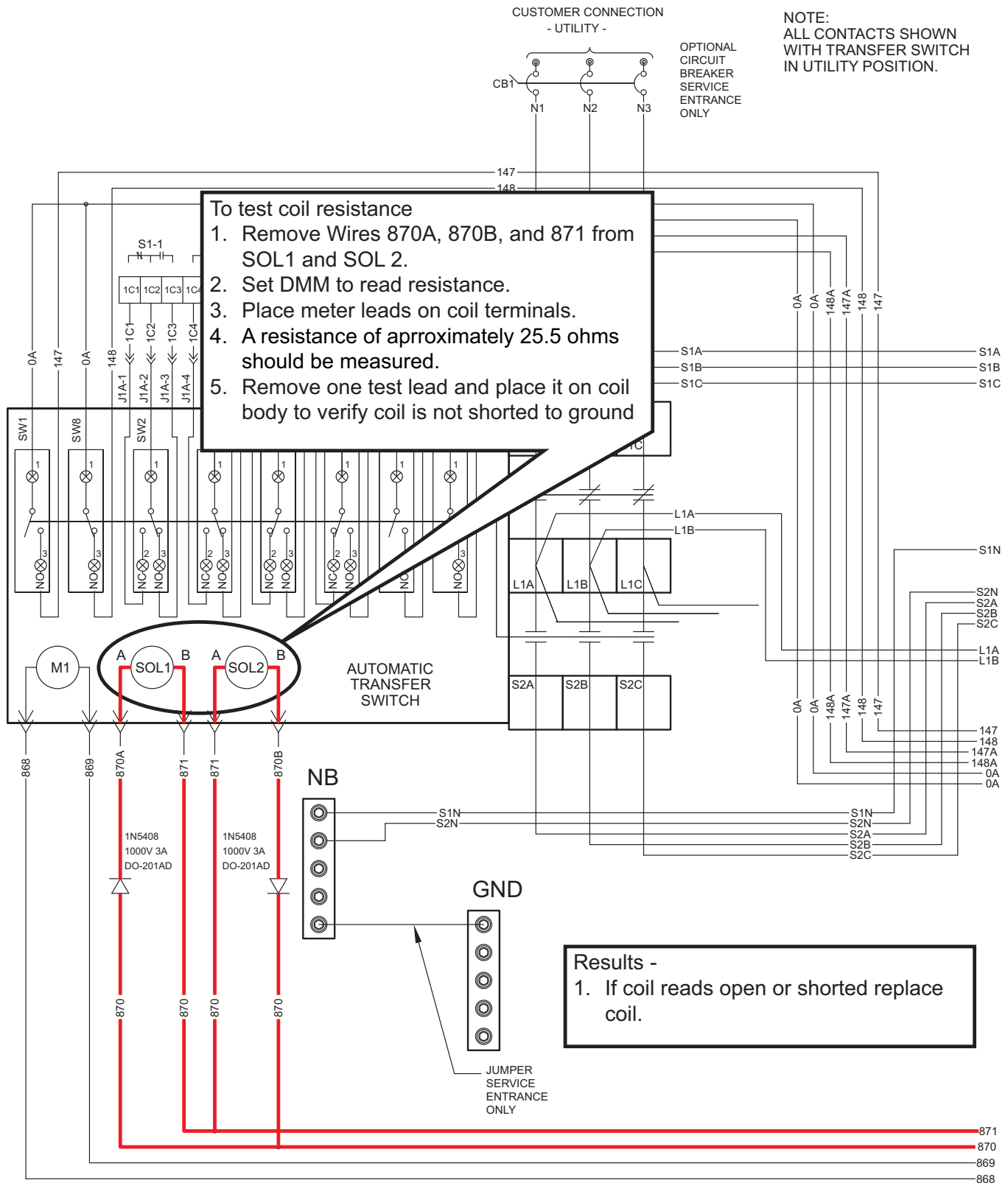
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Figure 1-44.

Testing Source Voltage

Testing Resistance of Coil

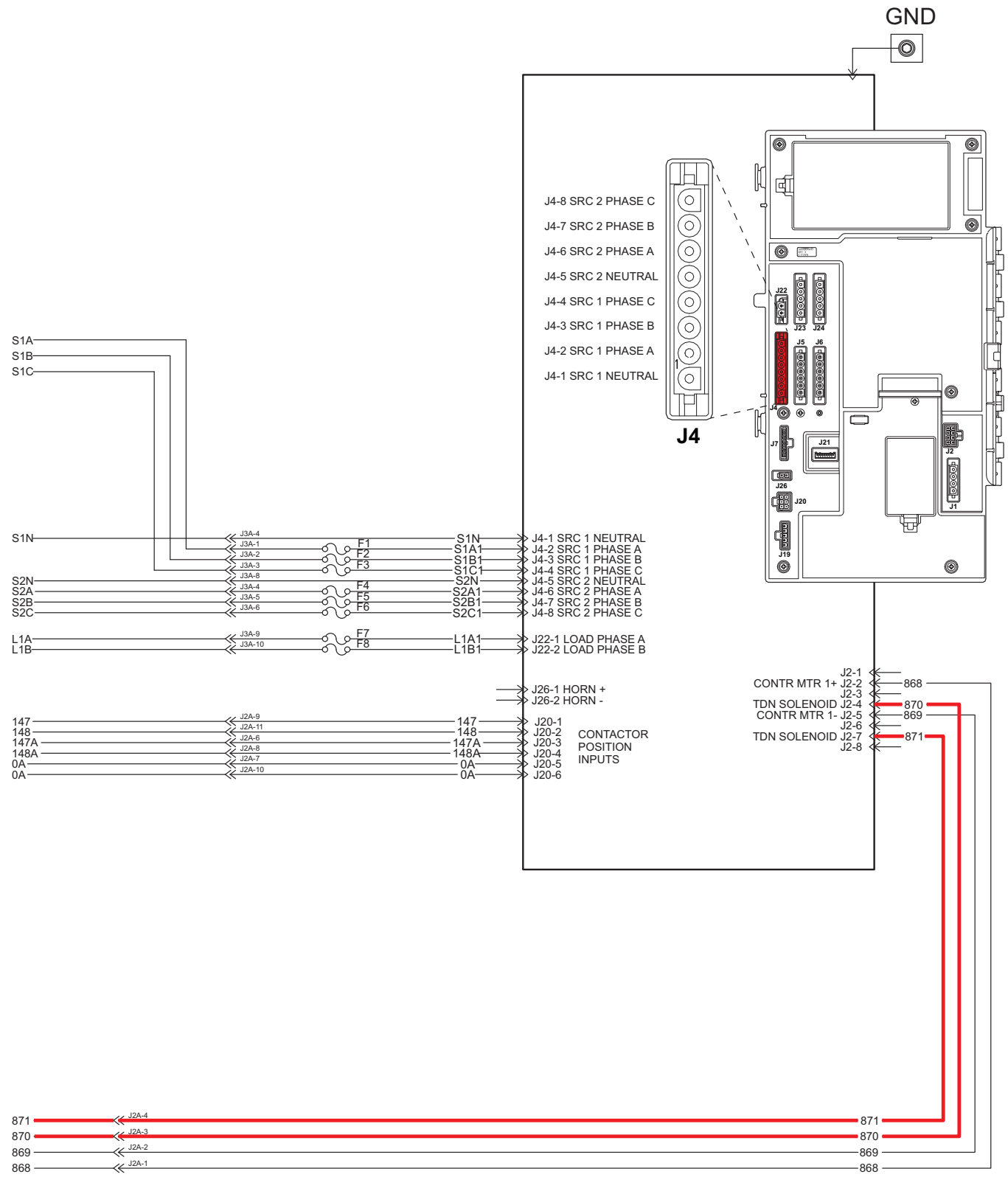


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Figure 1-46.

Testing Resistance of Coil



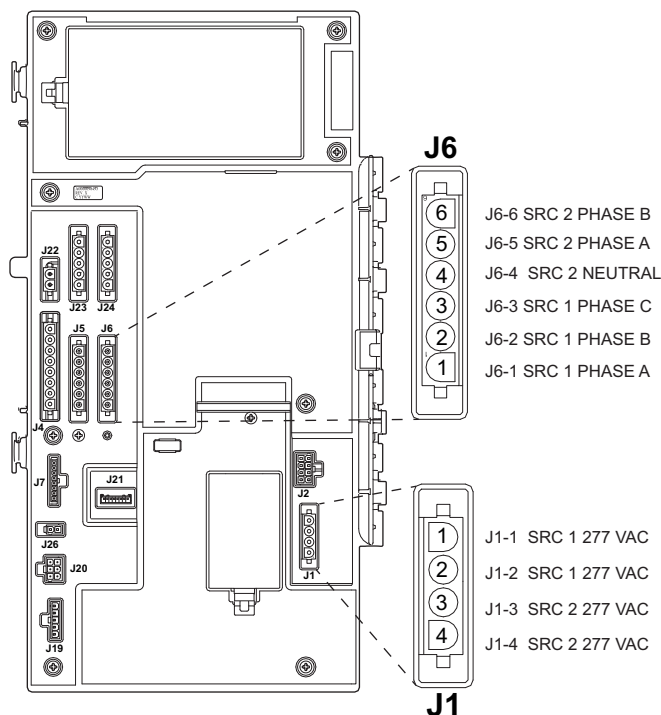
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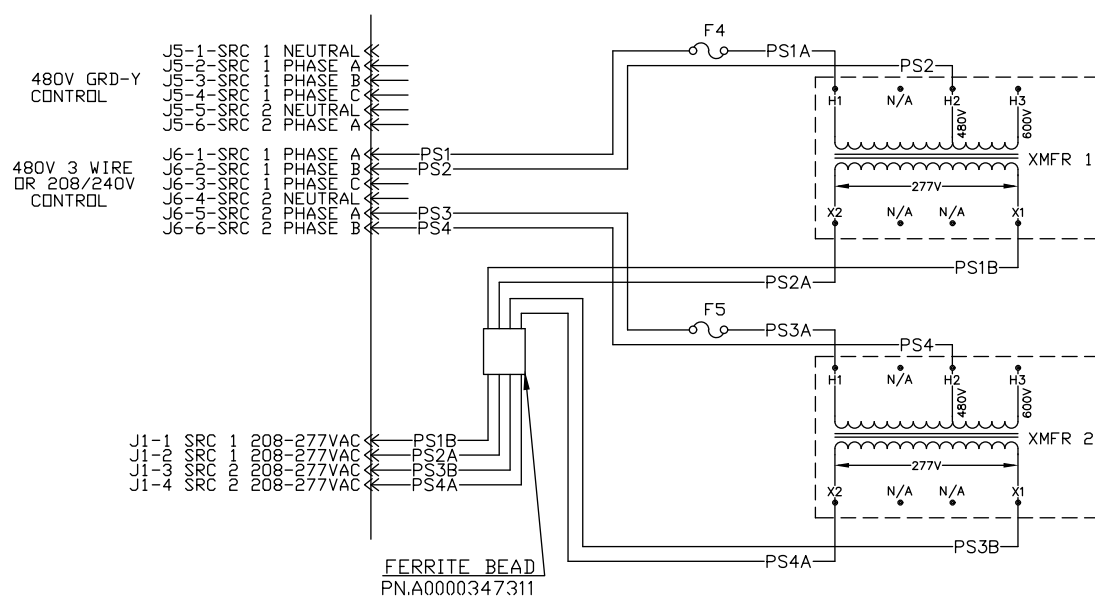
Figure 1-47.

Verify Presence Of Transformer Voltage 480/600V Switches

Transfer switches that are on a 480 Volt three wire connection (i.e. 480 volt delta) or a 600 volt wye, a transformer kit is installed between the J6 connector and the J1 connector. The transformer kit steps down voltage from the J6 connector (480 or 600 VAC) to approximately 290 VAC. This voltage is delivered to the J1 connector as motor drive supply voltage.



480V CONNECTION WITH TRANSFORMERS



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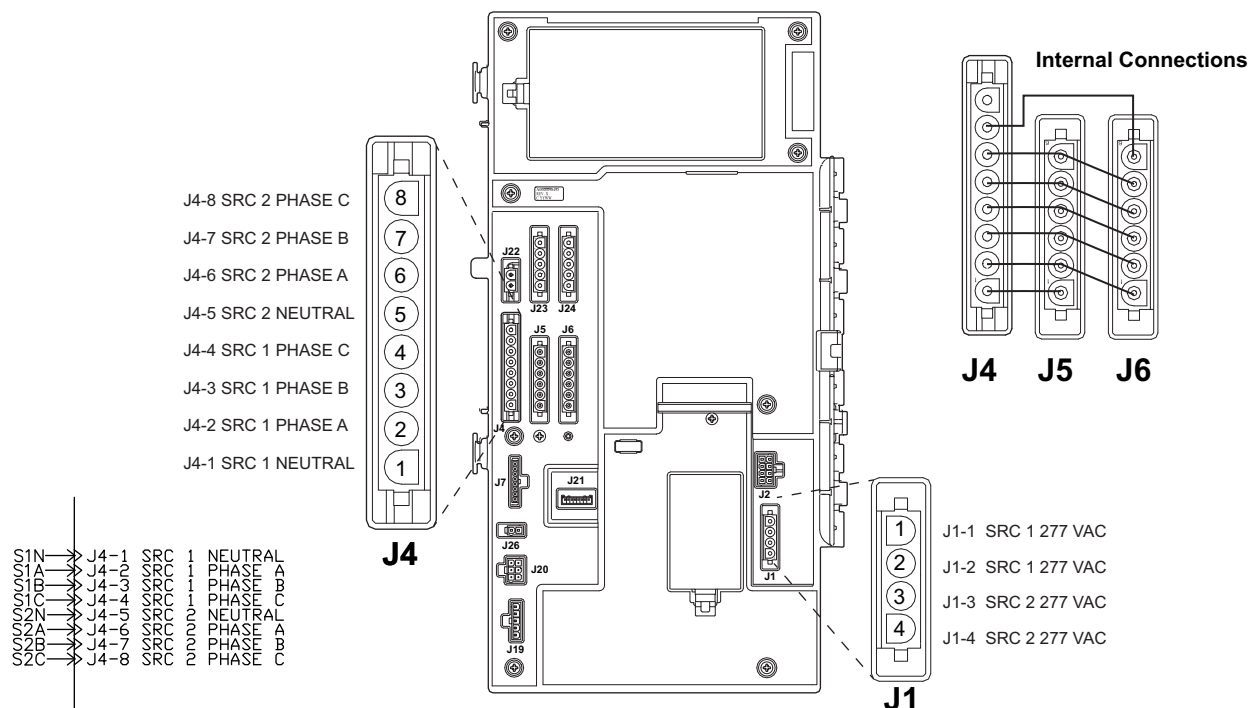
Figure 1-48.

Missing Neutral – Testing Source 1

Testing Source 1

Source 1 must be present to perform this test

1. Identify the J4 connector on the back of the TX Control Board.
2. Back-probe one test lead at J4 connector at pin J4-1
3. Back-probe other test lead at J4-2, J4-3 and J4-4. Approximately 277 VAC should be measured.
4. If 277 VAC is not measured, the problem is a non-existing neutral connection. Test Source 1 for proper neutral.



1. Identify the J1 connector on the back of the TX Control Board.
2. With Source 1 present, back-probe and measure pin J1-1 to pin J1-2 on a 480 Volt Wye system. Approx. 277 VAC should be present.

480V GRD-Y CONTROL

If 277 VAC is not measured with Source 1 present proceed to next step – Testing Source 1 at J5 Harness.

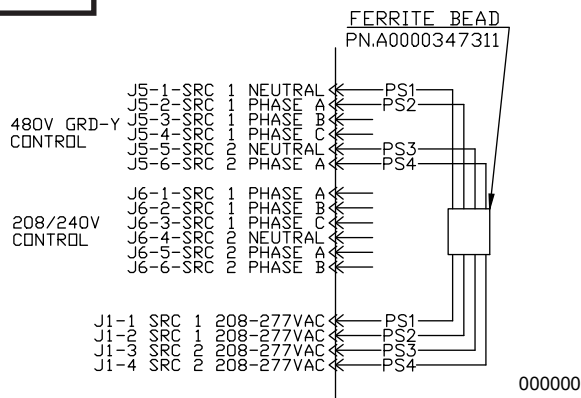
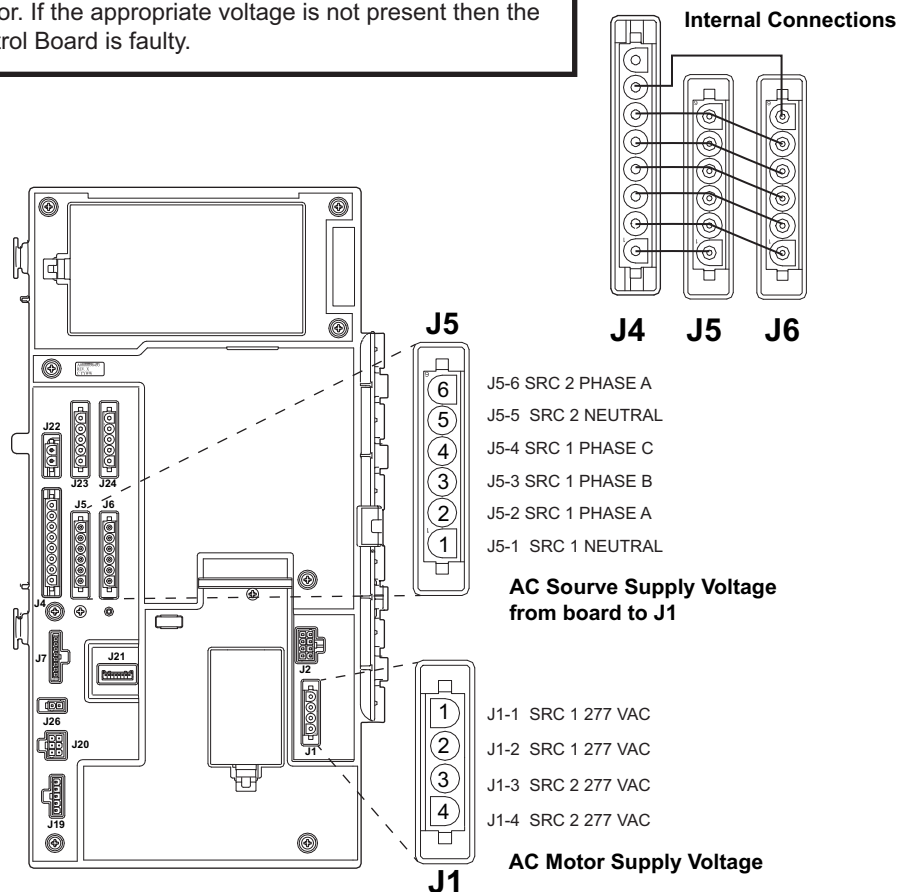


Figure 1-49.

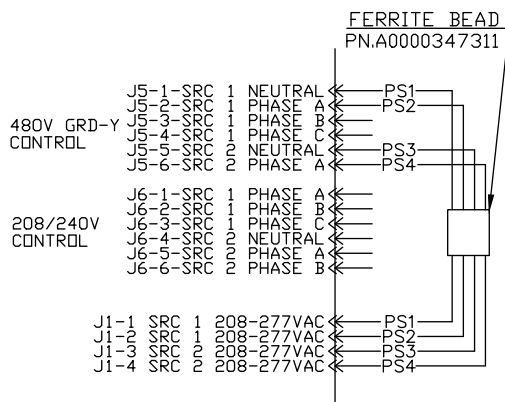
Missing Neutral – Testing Source 1 at J5 Harness

Testing Source 1 at J5 Harness

1. Perform a point-to-point continuity check. If the harness has continuity, then measure the output voltage of the J5 connector. If the appropriate voltage is not present then the TX Control Board is faulty.



480V GRD-Y CONTROL



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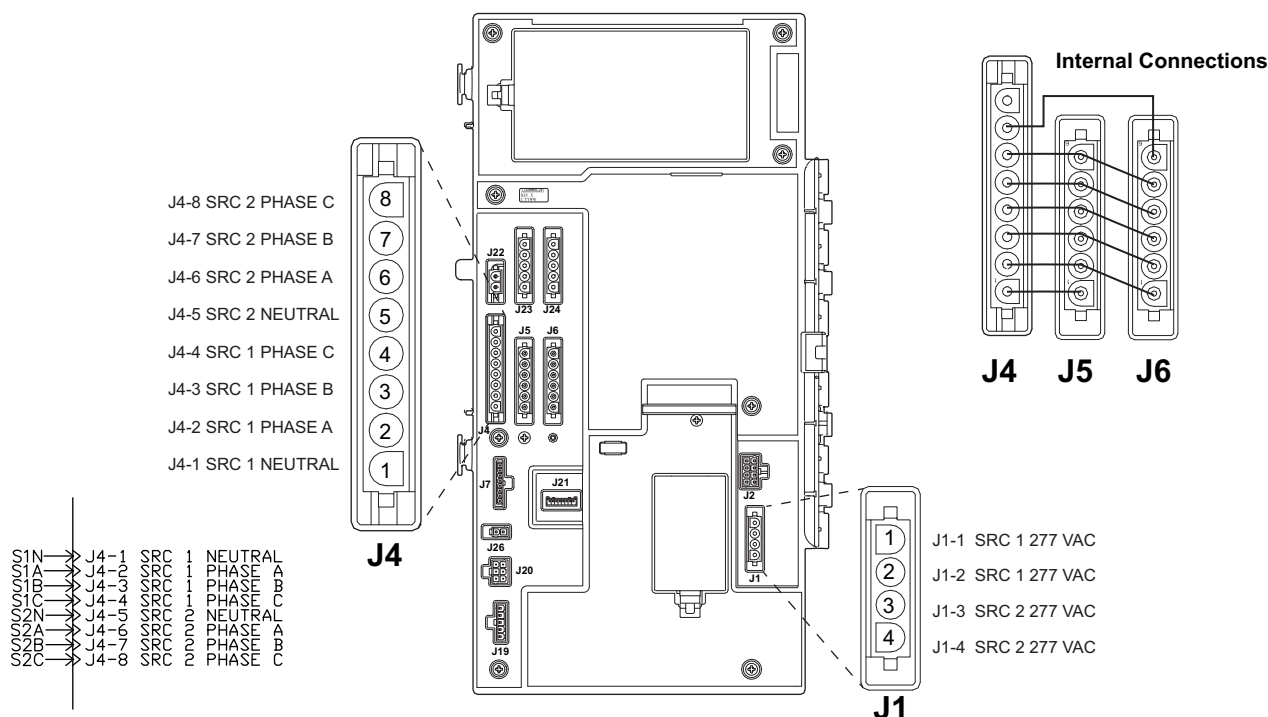
Figure 1-50.

Missing Neutral – Testing Source 2

Testing Source 2

Source 2 must be present to perform this test

1. Identify the J4 connector on the back of the TX Control Board.
2. Back-probe one test lead at J4 connector at pin J4-5
3. Back-probe other test lead at J4-6, J4-7 and J4-8. Approximately 277 VAC should be measured.
4. If 277 volts is not measured, then the problem is a non-existing neutral connection. Test Source 2 for proper neutral.



1. Identify the J1 connector on the back of the TX Control Board.
2. With Source 2 present, back-probe and measure pin J1-3 to pin J1-4 on a 480 Volt Wye system. Approx. 277 VAC should be present.

If 277 VAC is measured with Source 2 present proceed to next step – Testing Source 2 at J5 Harness.

480V GRD-Y CONTROL

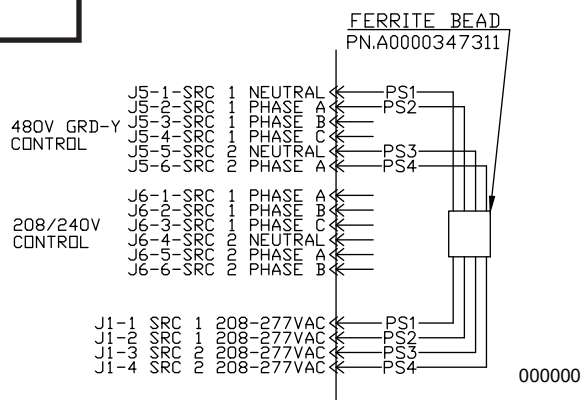
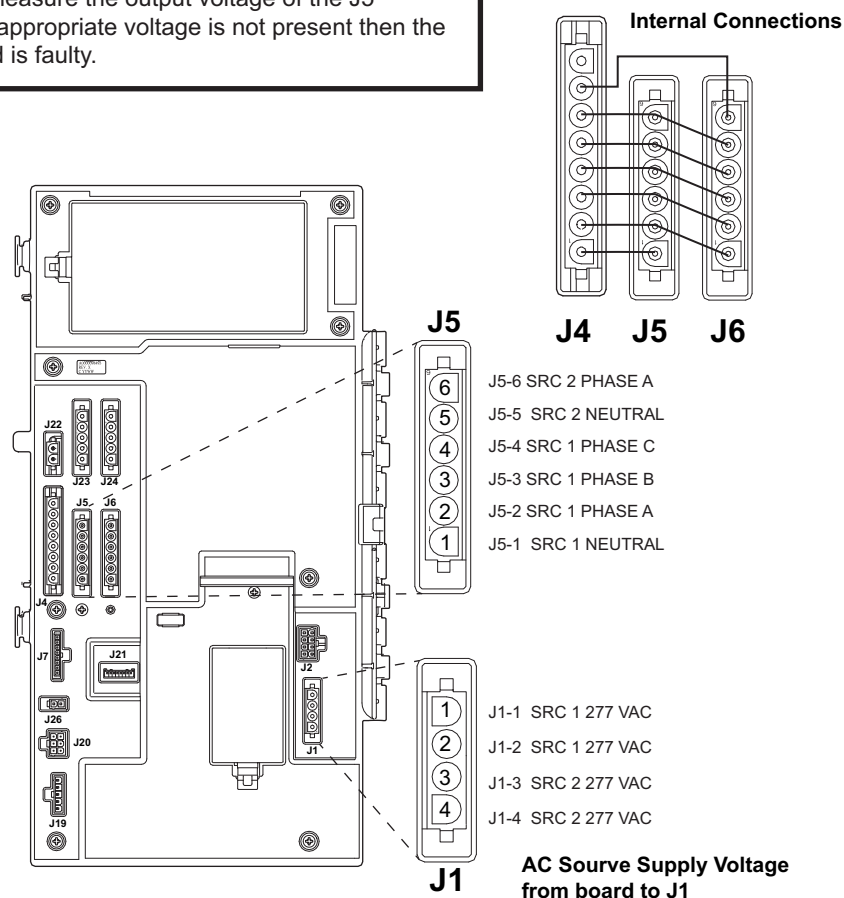


Figure 1-51.

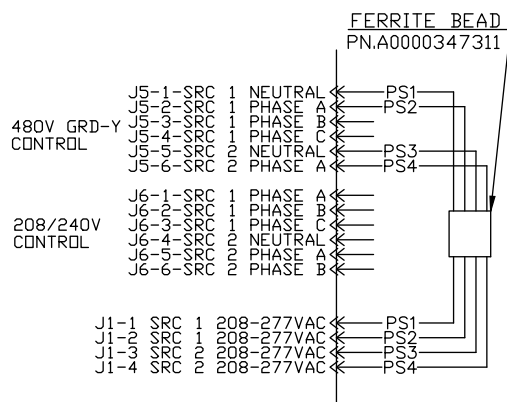
Missing Neutral – Testing Source 2 at J5 Harness

Testing Source 2 at J5 Harness

1. Perform a point-to-point continuity check. If the harness has continuity, then measure the output voltage of the J5 connector. If the appropriate voltage is not present then the TX Control Board is faulty.



480V GRD-Y CONTROL



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Figure 1-52.

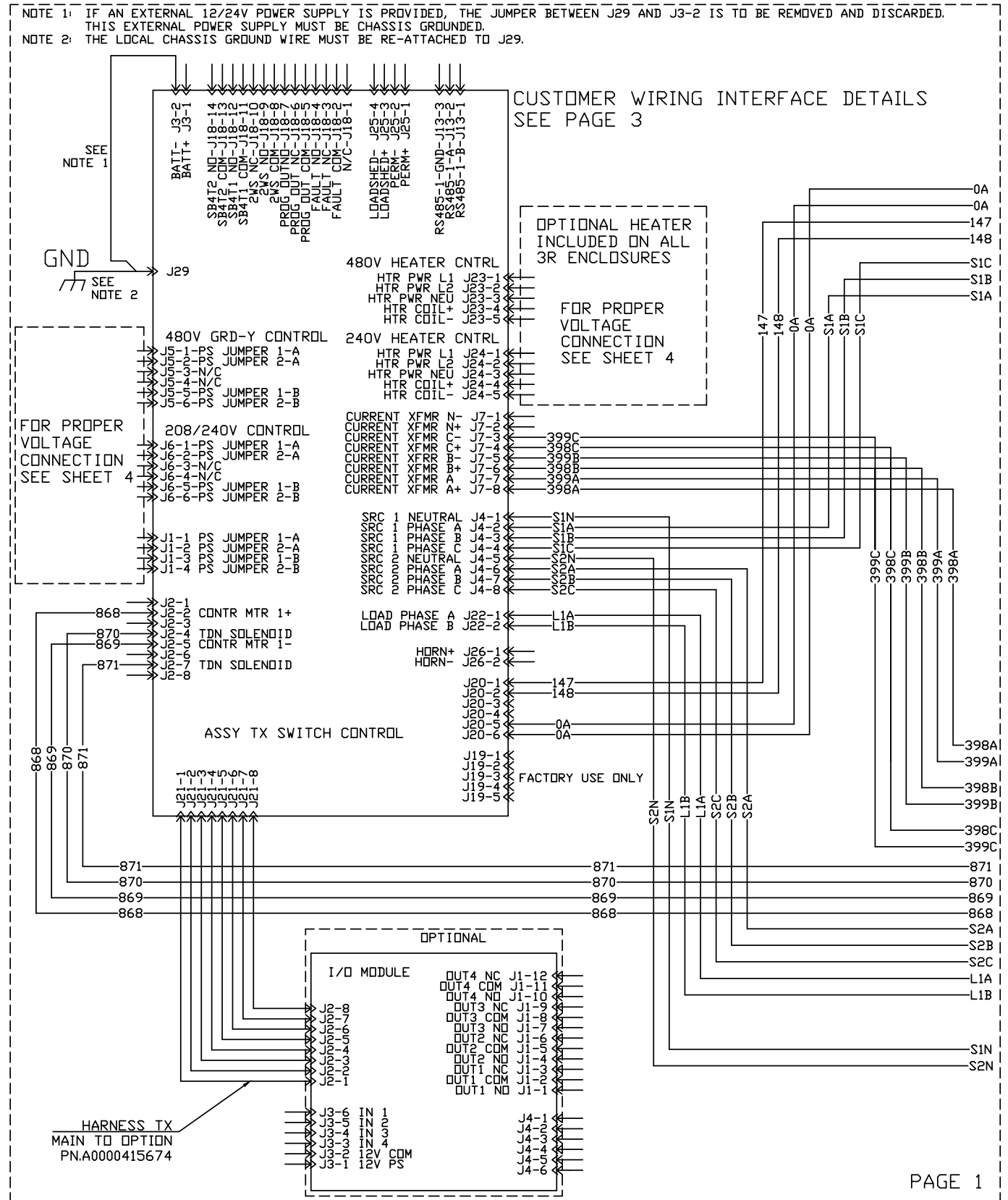
Section 7 Electrical Data

Introduction

The diagrams in this section are provided for general reference only. For unit specific diagrams refer to the Service and Support page at the manufacturer's website

A0000347534-F WD TX Control Transfer Switch (page 1 of 6)

GROUP G



REVISION: CN-0084280F

DATE: 4/18/24

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PAGE 2

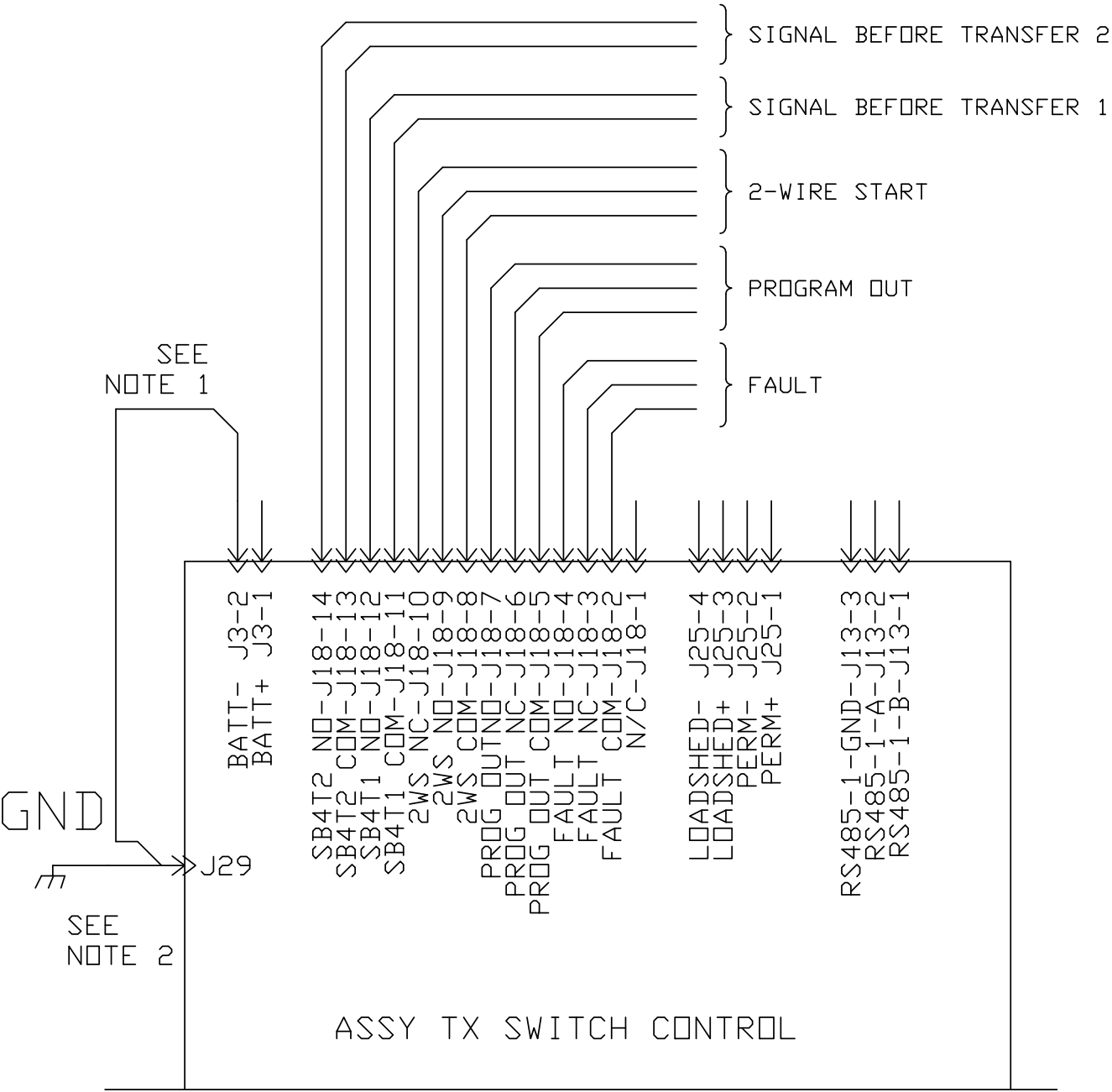
DRAWING #: A0000347534

PAGE 2 OF 6

A0000347534-F WD TX Control Transfer Switch (page 3 of 6)

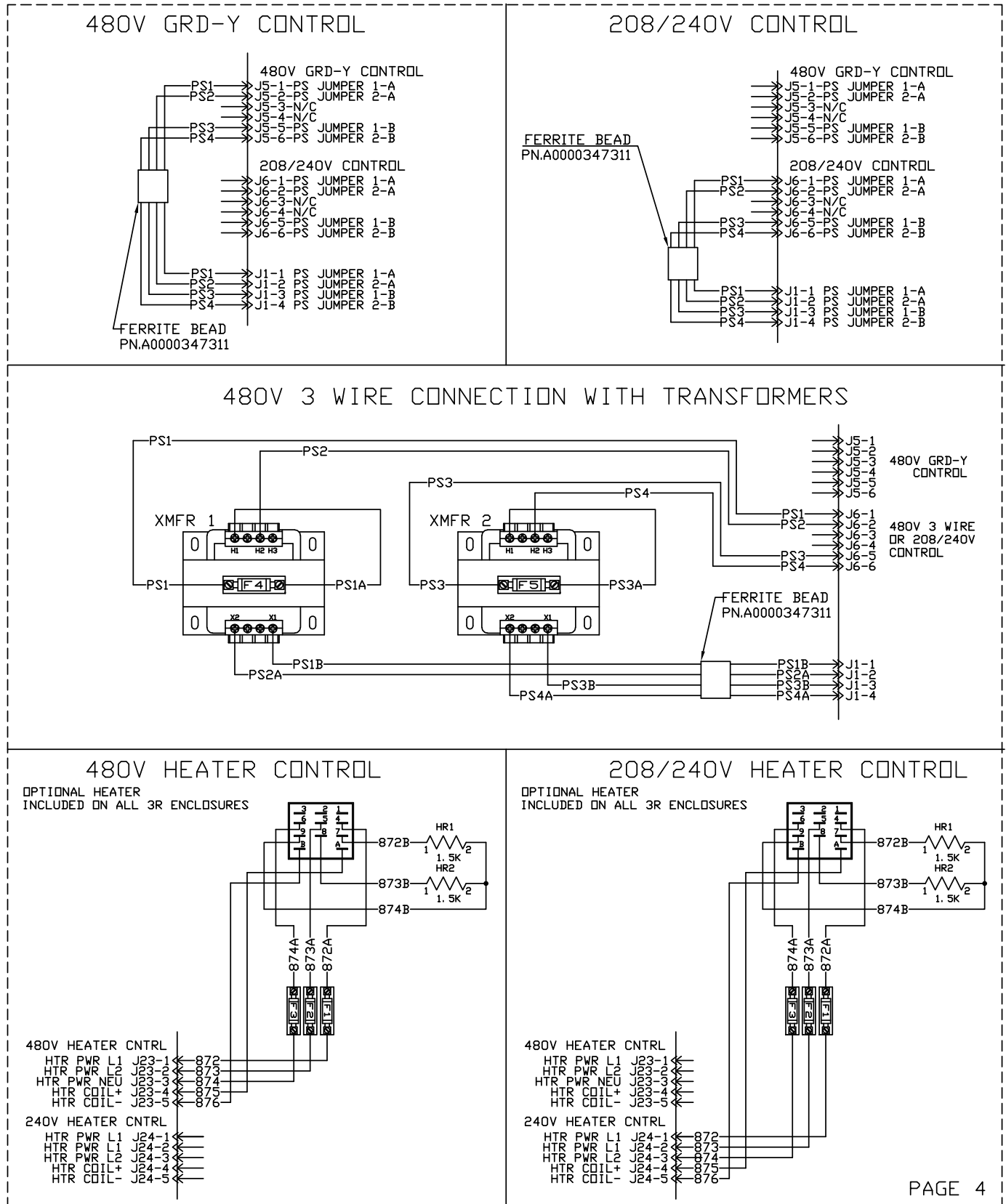
GROUP G

- NOTE 1: IF AN EXTERNAL 12/24V POWER SUPPLY IS PROVIDED, THE JUMPER BETWEEN J29 AND J3-2 IS TO BE REMOVED AND DISCARDED.
- NOTE 2: THE LOCAL CHASSIS GROUND WIRE MUST BE RE-ATTACHED TO J29.



A0000347534-F WD TX Control Transfer Switch (page 4 of 6)

GROUP G



PAGE 4

WIRING - DIAGRAM

TX CONTROL TRANSFER SW

DRAWING #: A0000347534

REVISION: CN-0084280F

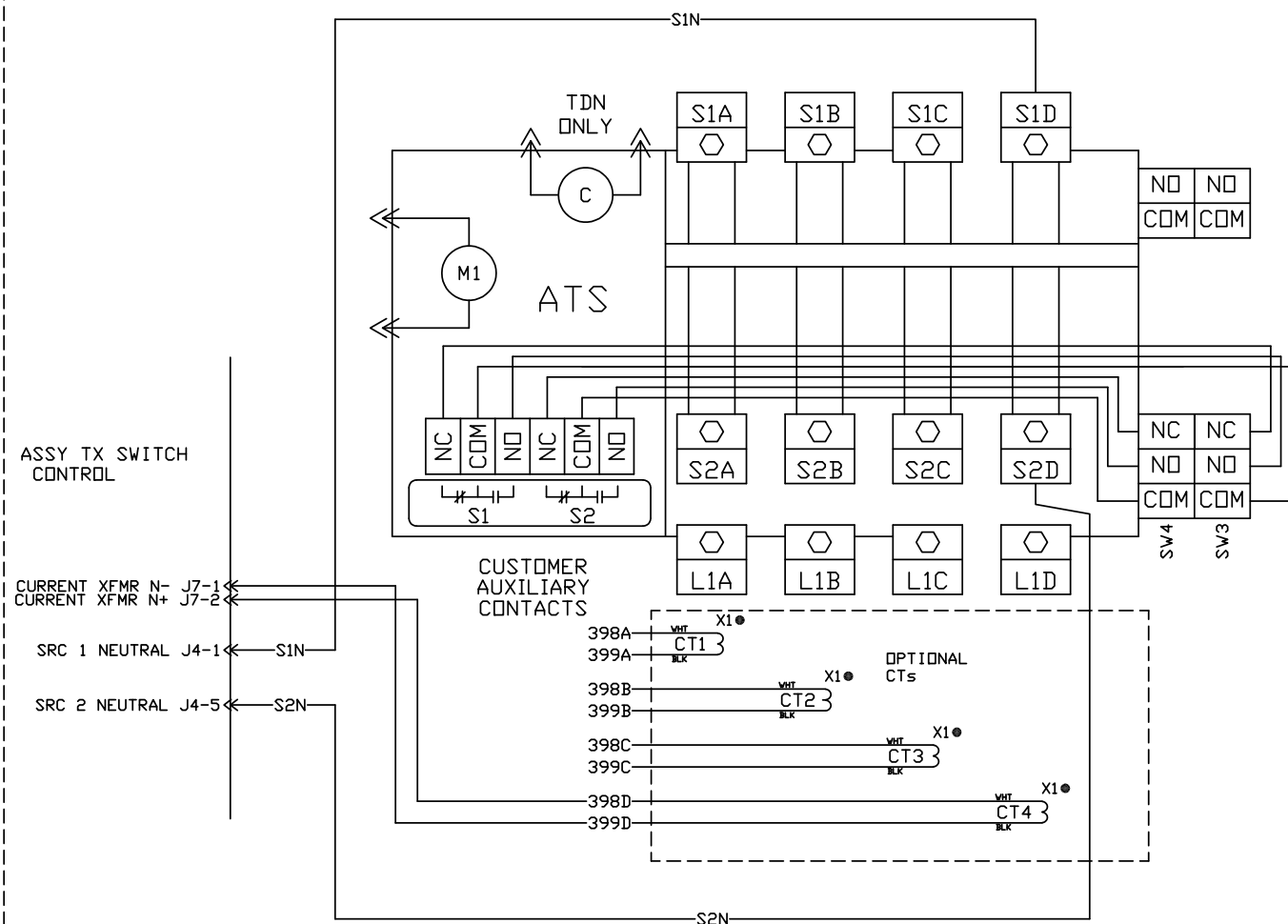
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A0000347534-F WD TX Control Transfer Switch (page 5 of 6)

GROUP G

4 POLE SWITCHED NEUTRAL CONNECTIONS



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A0000347534-F WD TX Control Transfer Switch (page 6 of 6)**GROUP G**

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REVISION: CN-0048815E
DATE: 10/7/21

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WIRING - DIAGRAM
TX CONTROL TRANSFER SW
DRAWING #: A0000347534

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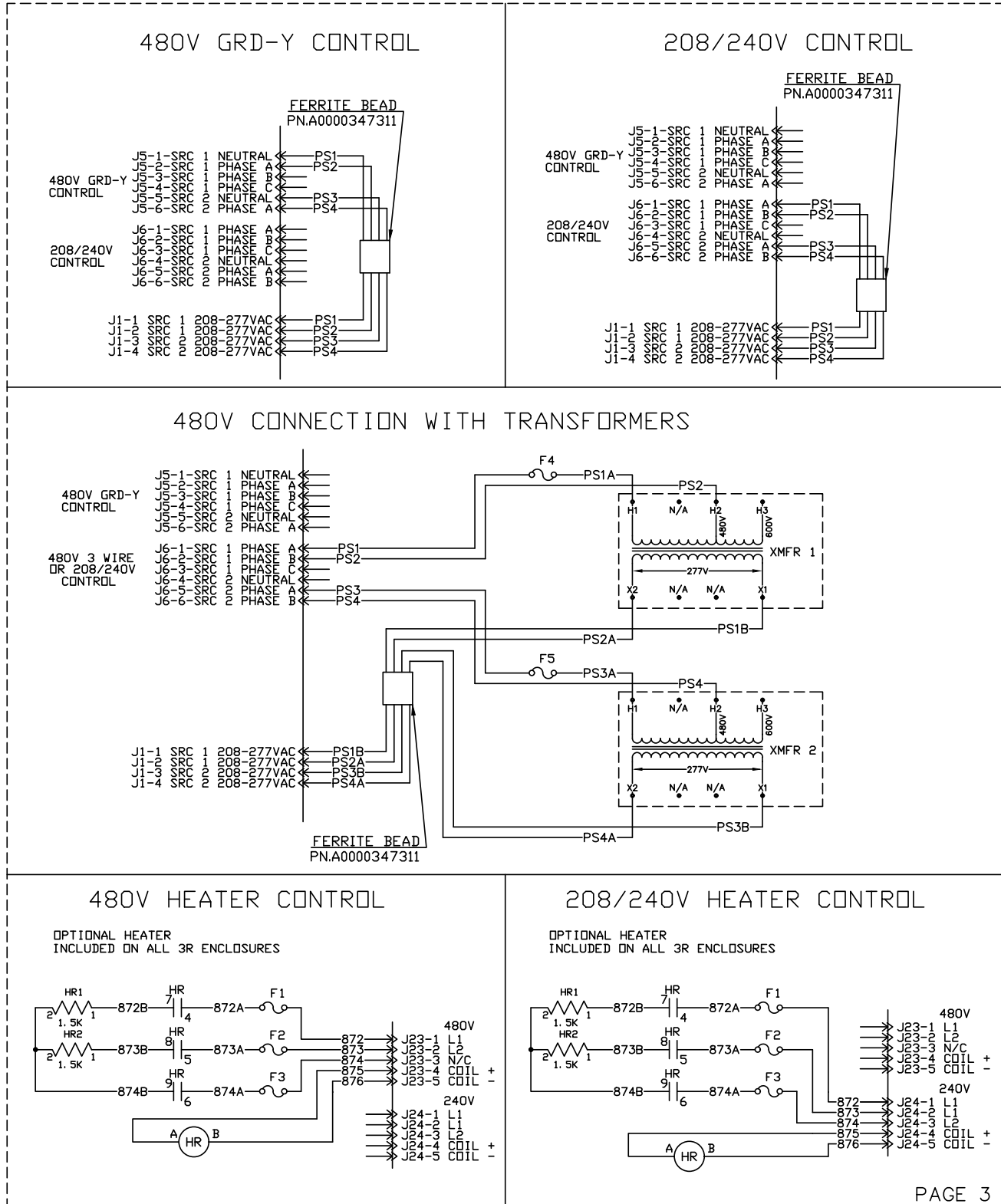
SCHEMATIC - DIAGRAM
TX CONTROL 600A TRANSFER SW
DRAWING #: A0000347536



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A0000347536-F SD TX Control 600A Transfer Switch (page 3 of 6)

GROUP G



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DATE: 5/1/24

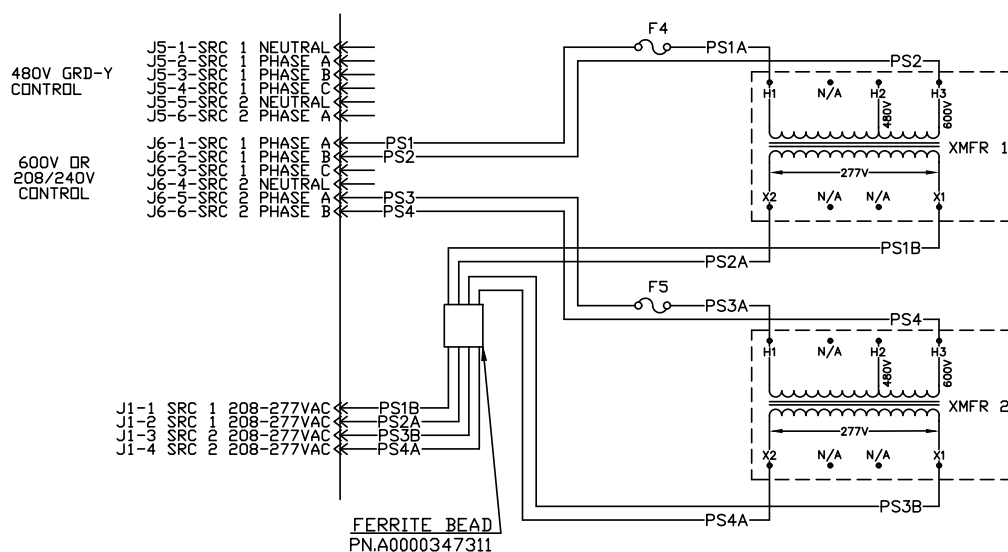
PAGE 3 OF 6

SCHEMATIC - DIAGRAM
TX CONTROL 600A TRANSFER SW
DRAWING #: A0000347536

A0000347536-F SD TX Control 600A Transfer Switch (page 4 of 6)

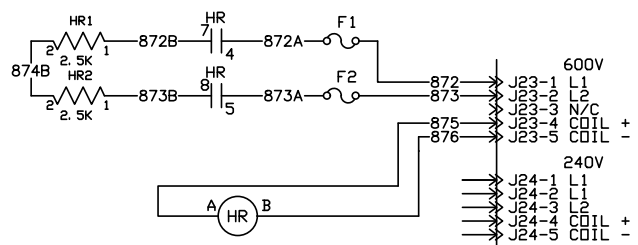
GROUP G

600V CONNECTION WITH TRANSFORMERS



600V HEATER CONTROL

OPTIONAL HEATER
INCLUDED ON ALL 3R ENCLOSURES



PAGE 4

REVISION: CN-0084844-F
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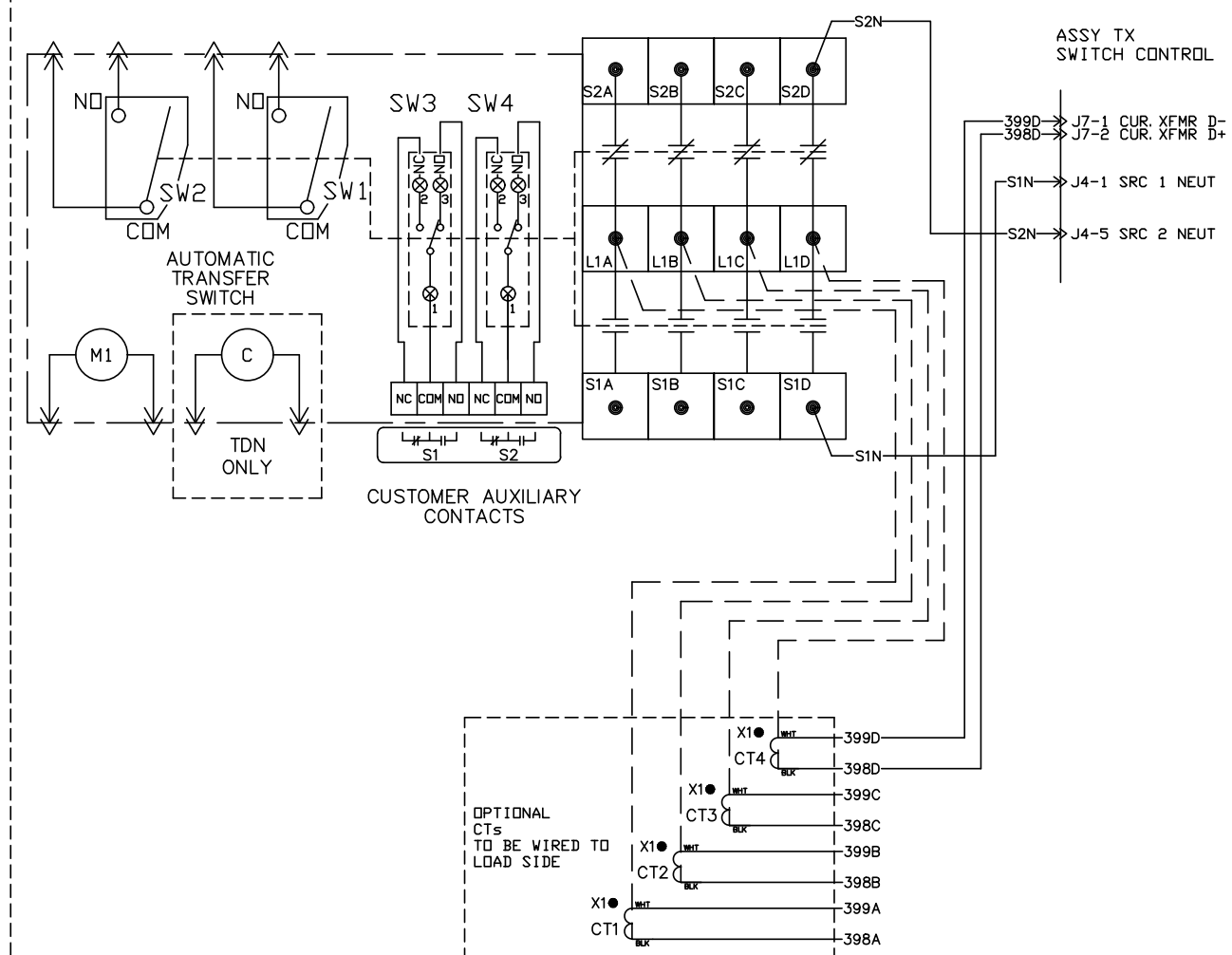
PAGE 4 OF 6

SCHEMATIC - DIAGRAM
TX CONTROL 600A TRANSFER SW
DRAWING #: A0000347536

A0000347536-F SD TX Control 600A Transfer Switch (page 5 of 6)

GROUP G

4 POLE SWITCHED NEUTRAL CONNECTIONS



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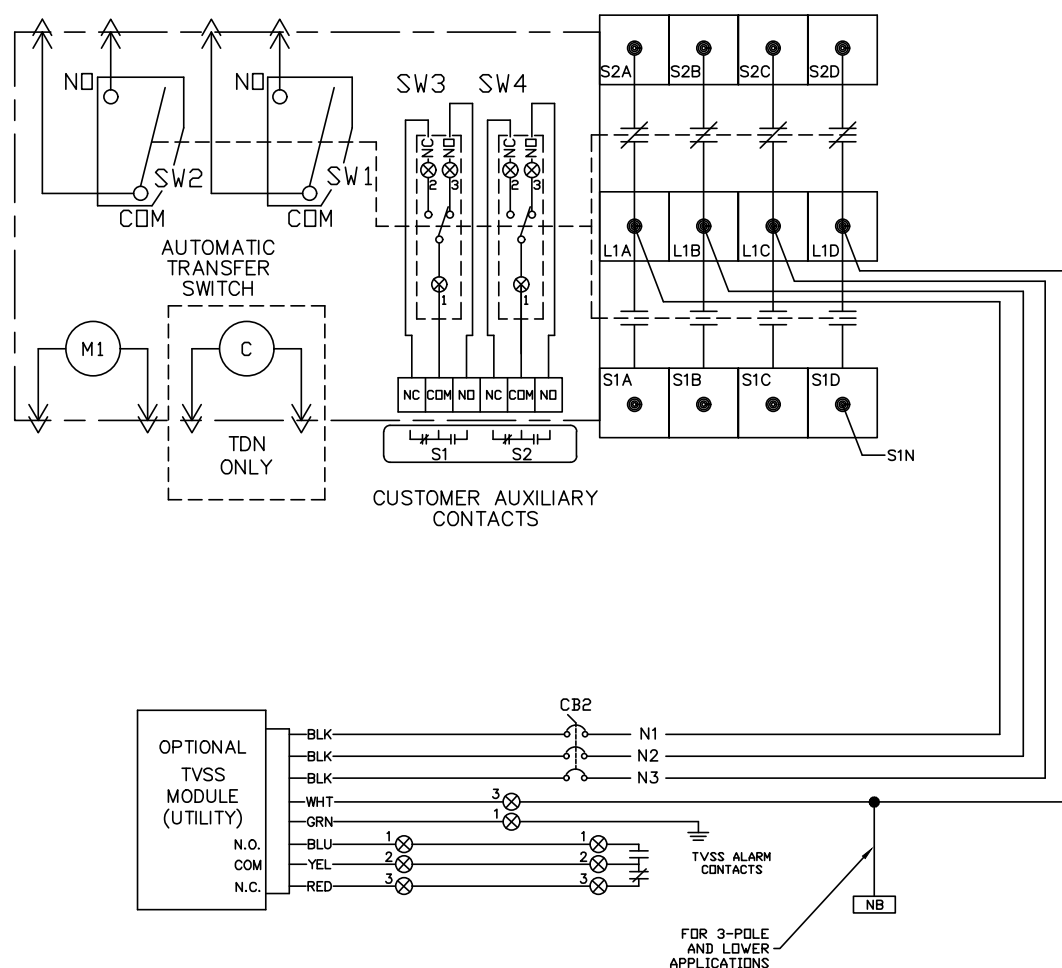
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SCHEMATIC - DIAGRAM
TX CONTROL 600A TRANSFER SW
DRAWING #: A0000347536

A0000347536-F SD TX Control 600A Transfer Switch (page 6 of 6)

GROUP G

3 & 4-POLE TVSS OPTION FOR TX TRANSFER SWITCH



PAGE 6

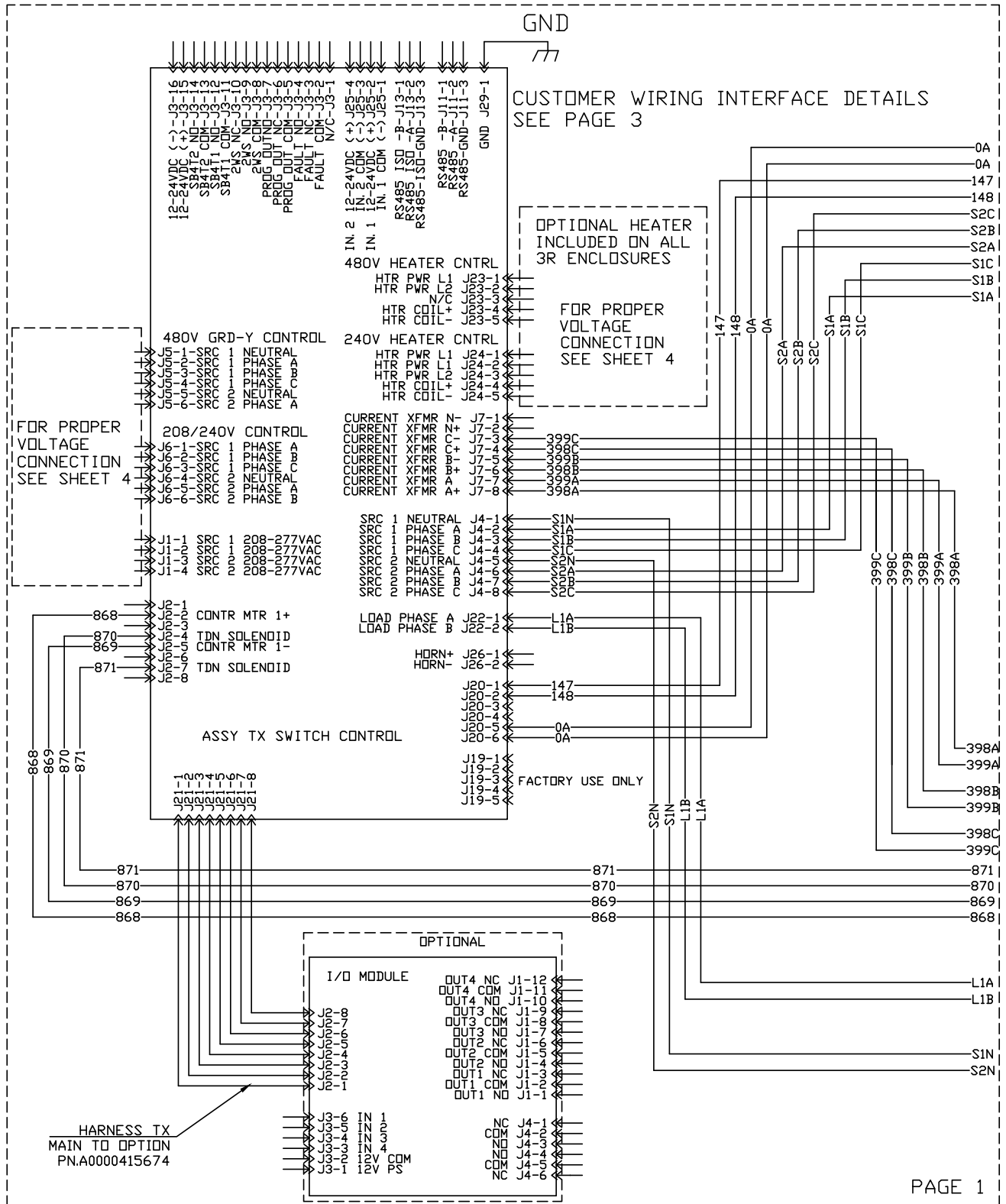
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SCHEMATIC - DIAGRAM
TX CONTROL 600A TRANSFER SW
DRAWING #: A0000347536

A0000349100-F SD TX Control 600A Transfer Switch (page 1 of 8)

GROUP G



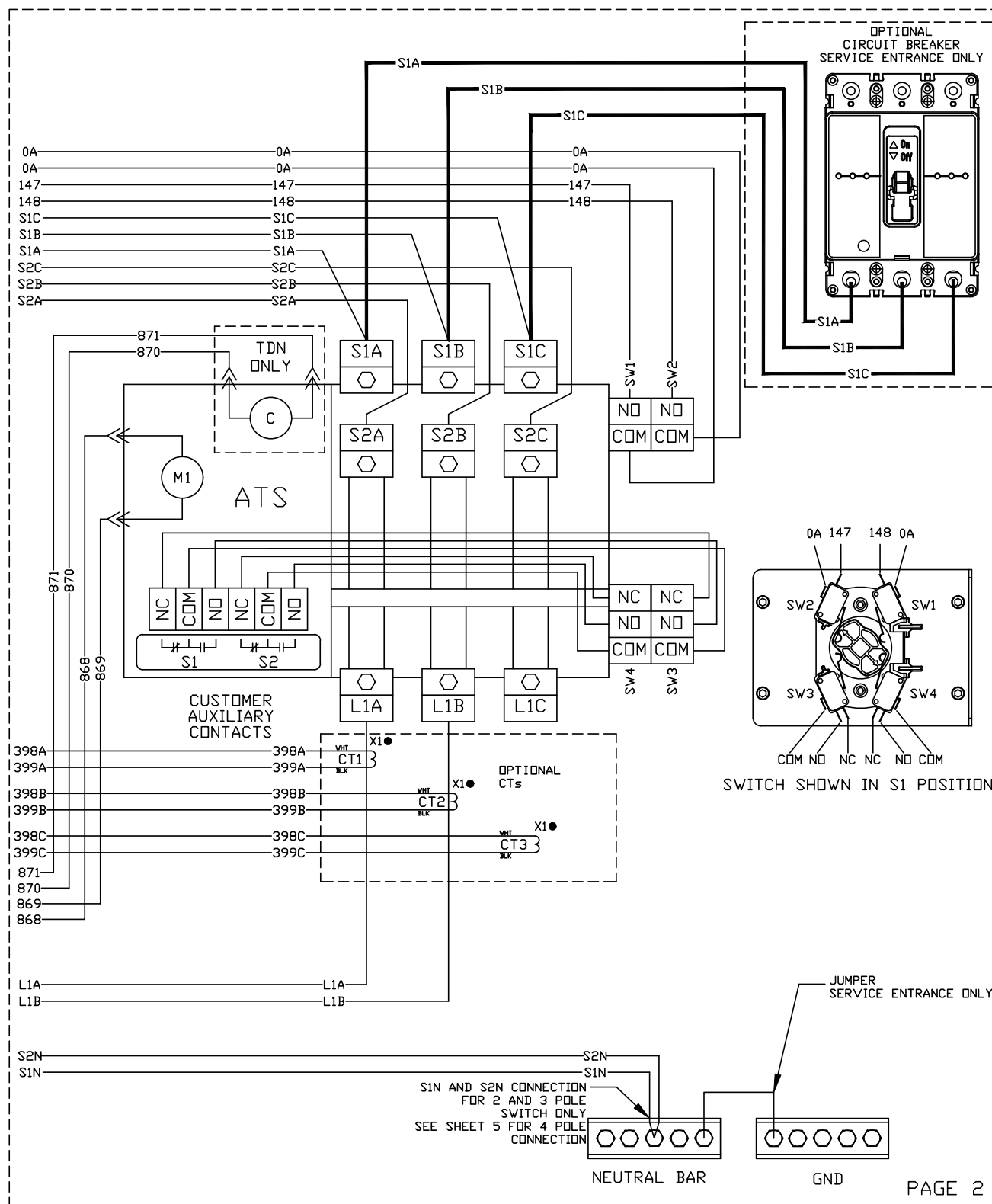
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A0000349100-F SD TX Control 600A Transfer Switch (page 2 of 8)

GROUP G



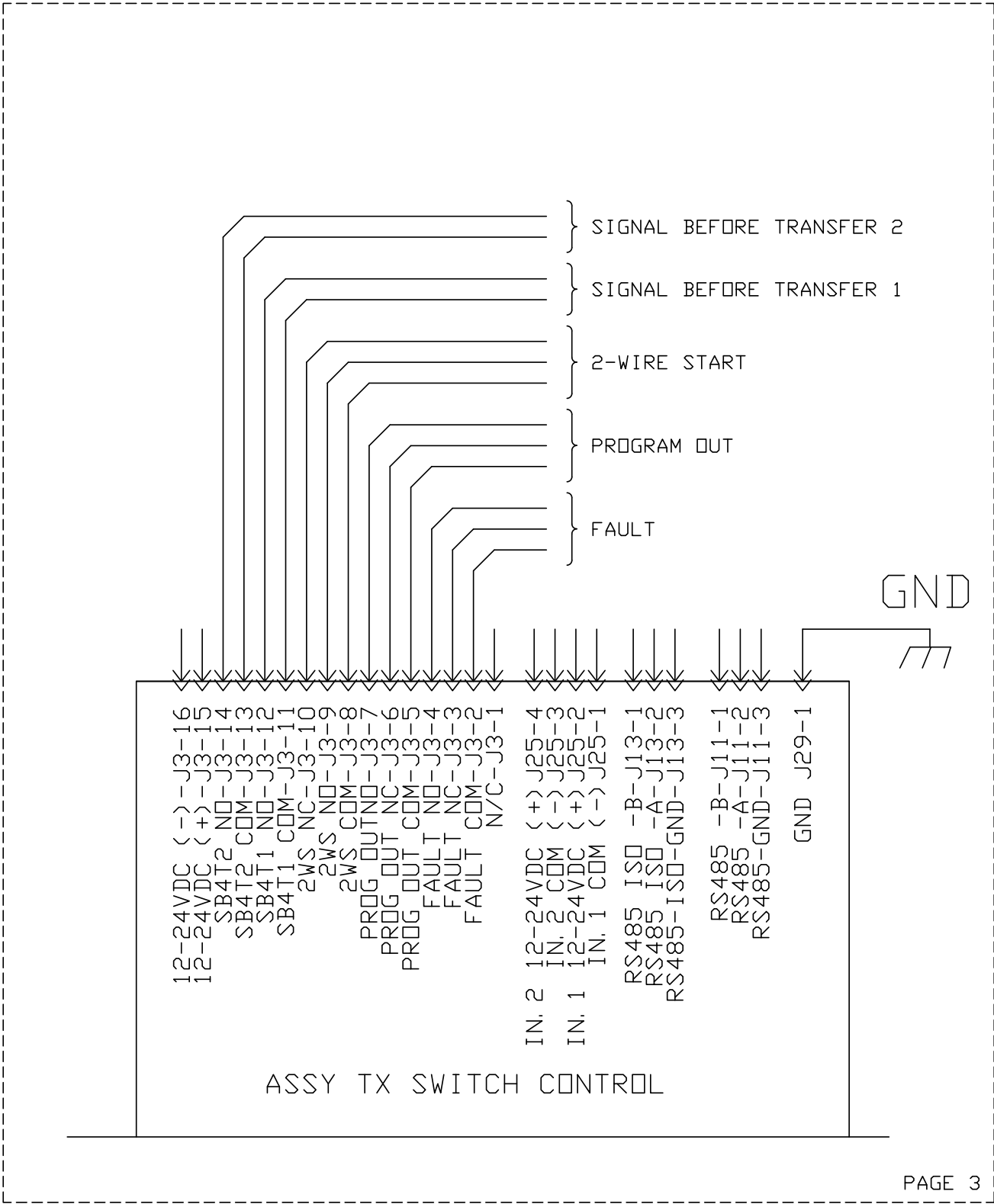
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TX CONTROL 600A TRANSFER SW
DRAWING #: A0000349100

A0000349100-F SD TX Control 600A Transfer Switch (page 3 of 8)

GROUP G



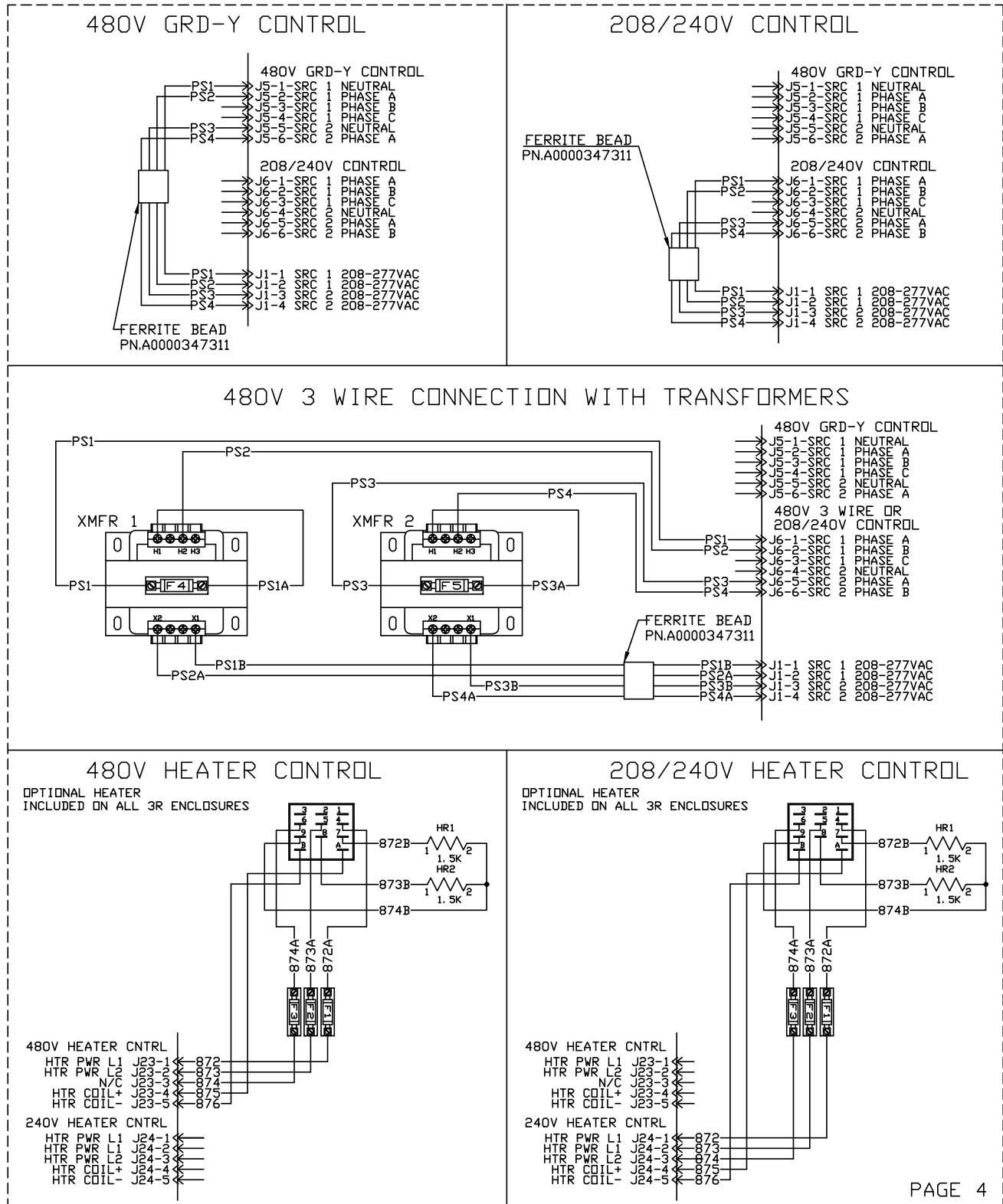
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TX CONTROL 600A TRANSFER SW
DRAWING #: A0000349100

A0000349100-F SD TX Control 600A Transfer Switch (page 4 of 8)

GROUP G



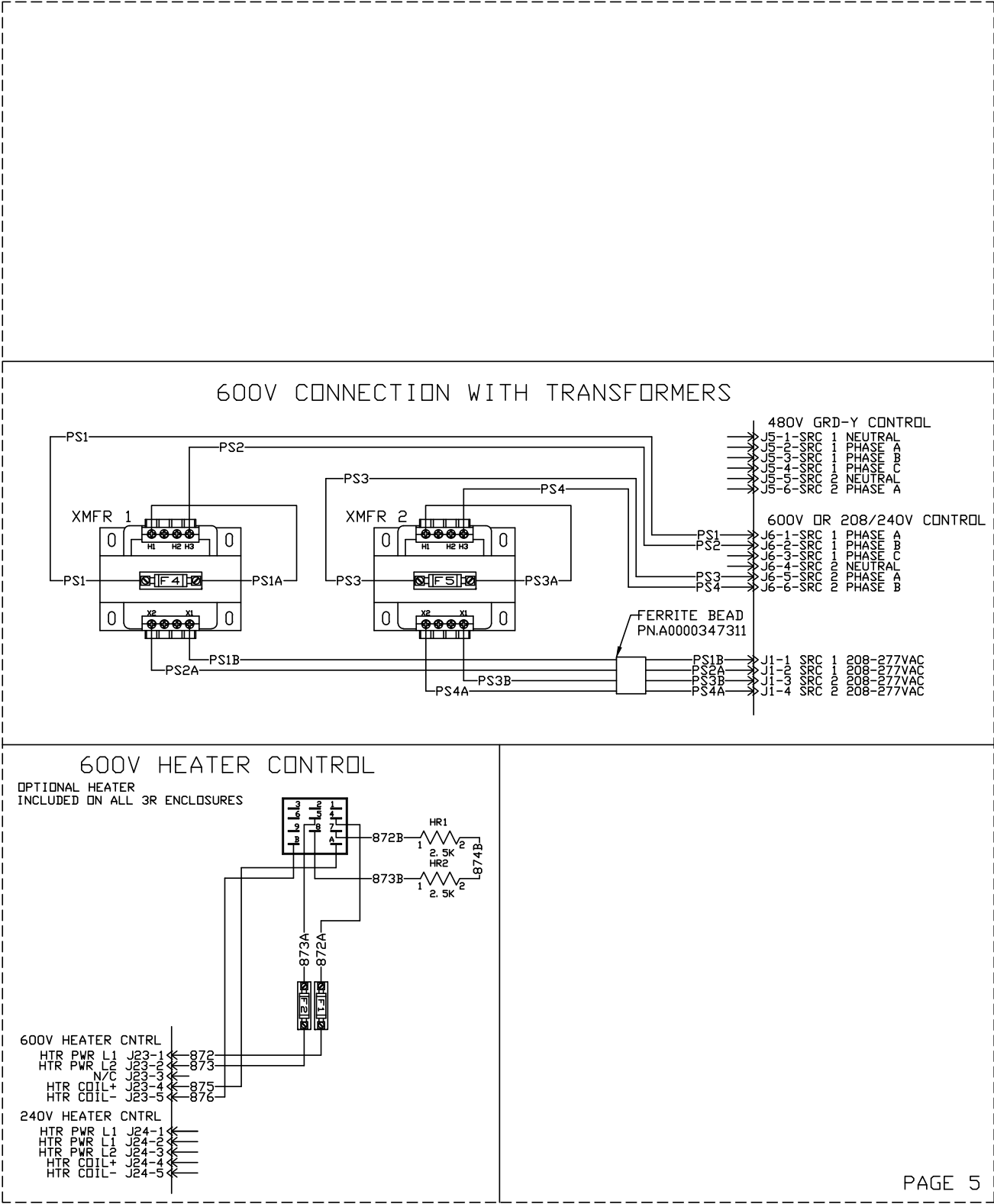
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WIRING - DIAGRAM
 TX CONTROL 600A TRANSFER SW
 DRAWING #: A0000349100

A0000349100-F SD TX Control 600A Transfer Switch (page 5 of 8)

GROUP G



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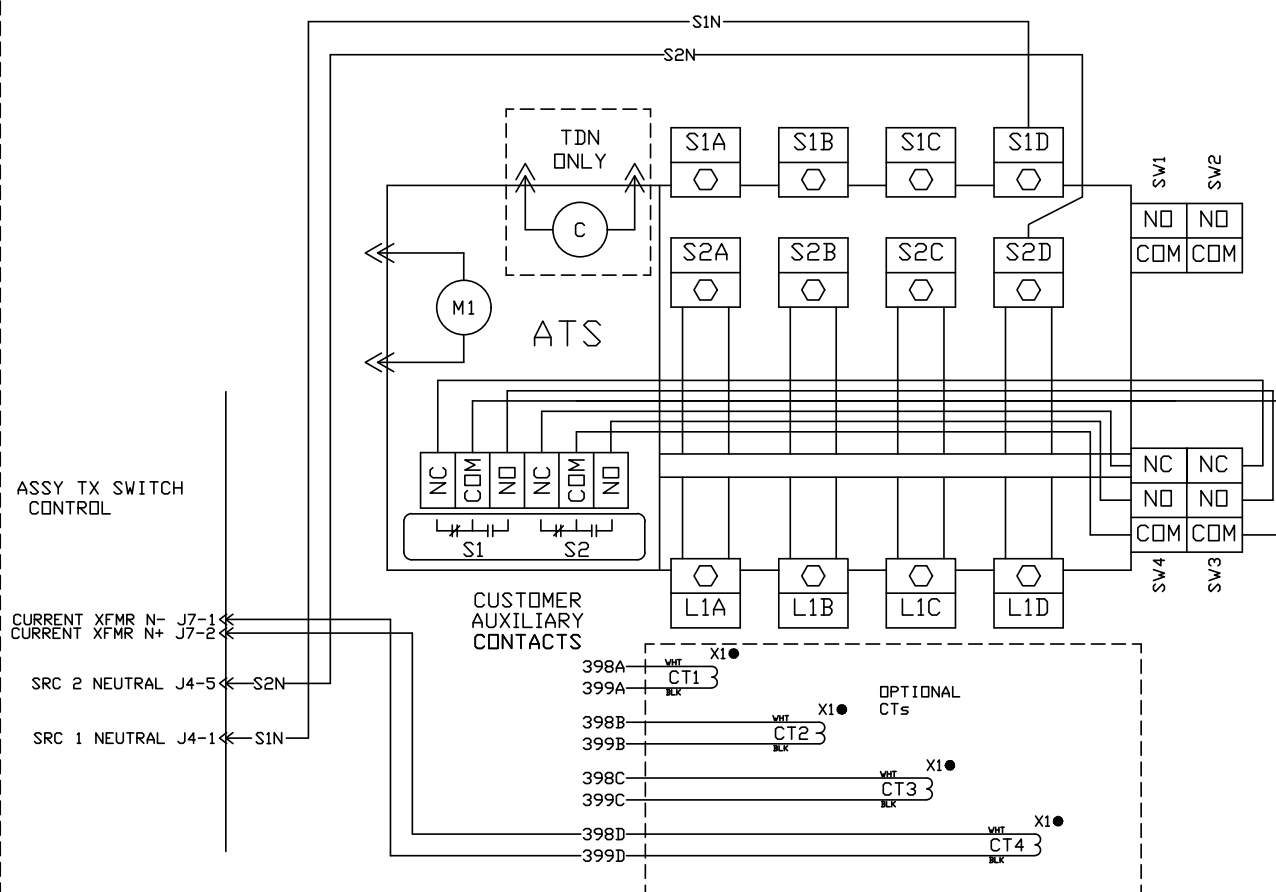
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WIRING - DIAGRAM
TX CONTROL 600A TRANSFER SW
DRAWING #: A0000349100

A0000349100-F SD TX Control 600A Transfer Switch (page 6 of 8)

GROUP G

4 POLE SWITCHED NEUTRAL CONNECTIONS



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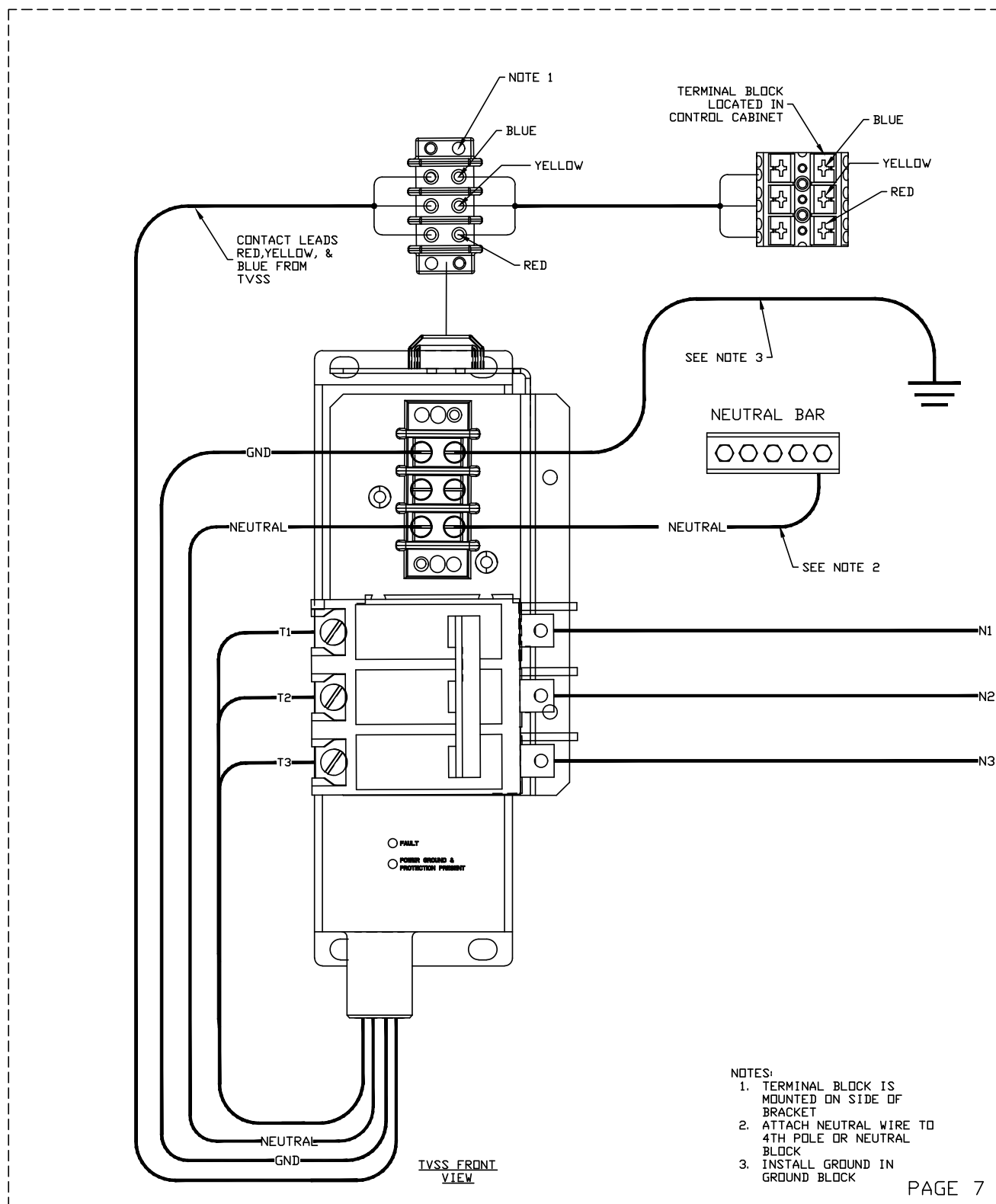
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WIRING - DIAGRAM
TX CONTROL 600A TRANSFER SW
DRAWING #: A0000349100

A0000349100-F SD TX Control 600A Transfer Switch (page 7 of 8)

GROUP G



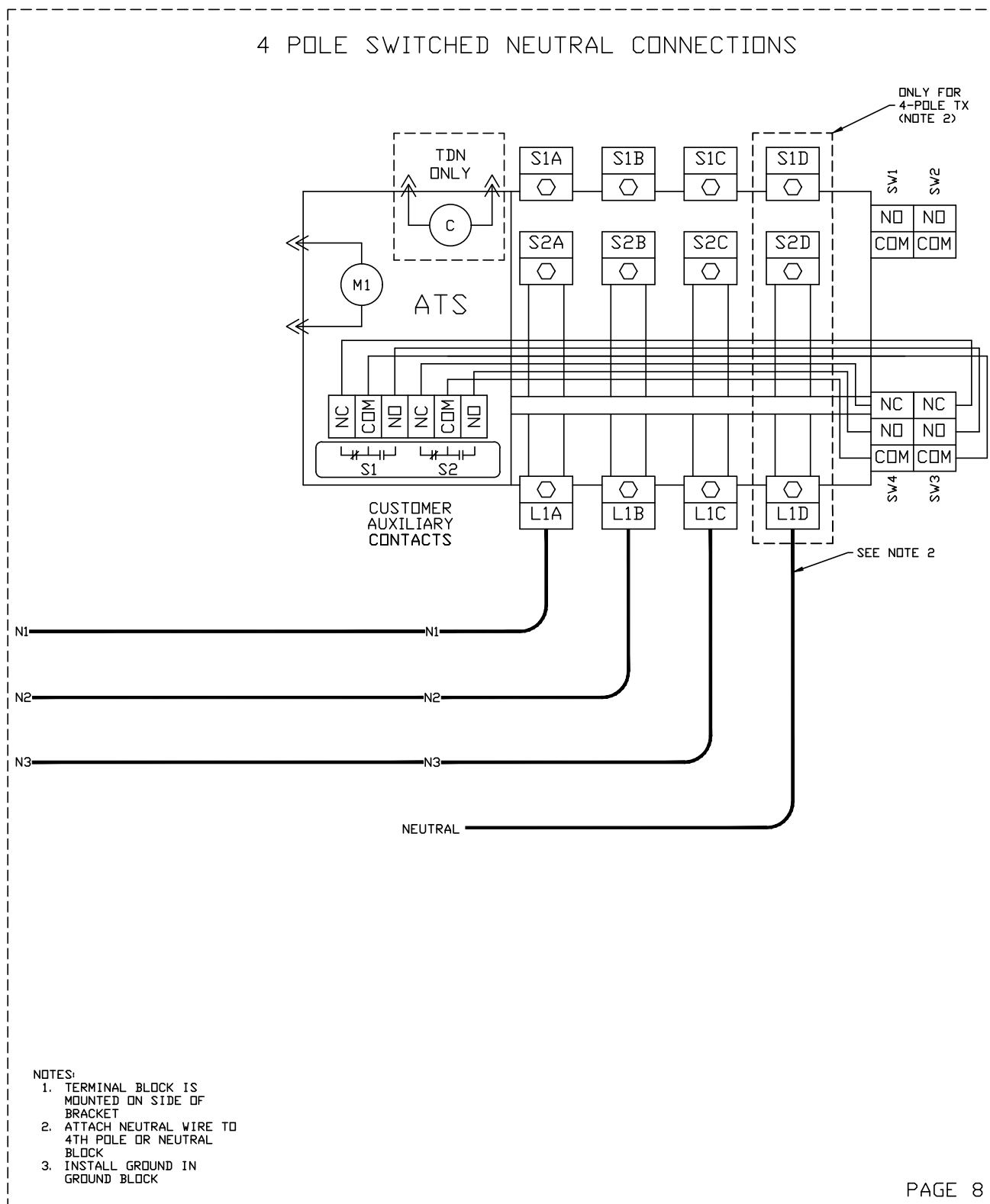
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WIRING - DIAGRAM
TX CONTROL 600A TRANSFER SW
DRAWING #: A0000349100

A0000349100-F SD TX Control 600A Transfer Switch (page 8 of 8)

GROUP G



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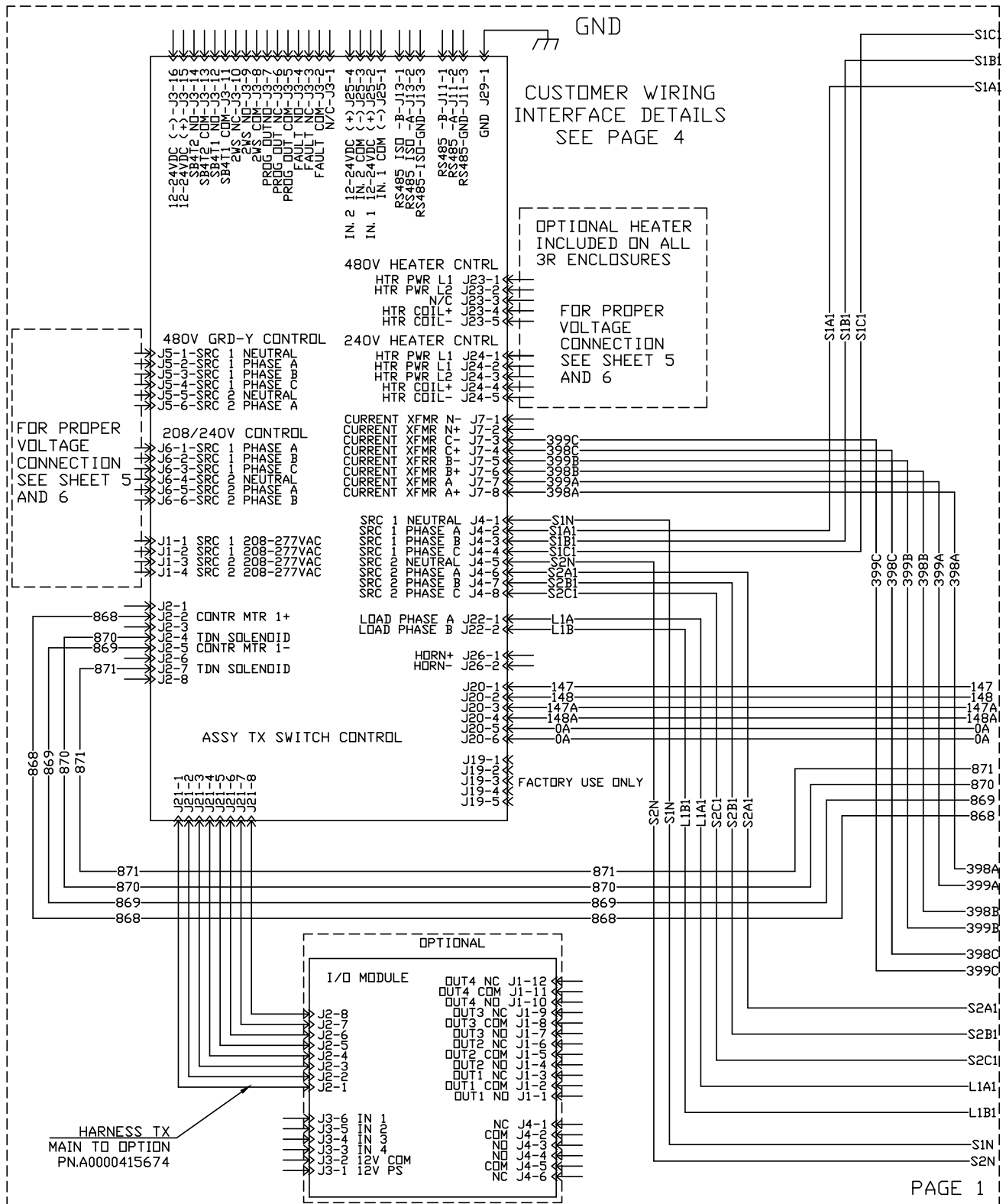
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WIRING - DIAGRAM
TX CONTROL 600A TRANSFER SW
DRAWING #: A0000349100

A0001679448-G WD TX Control 800-1000A Transfer Switch (page 1 of 12)

GROUP G

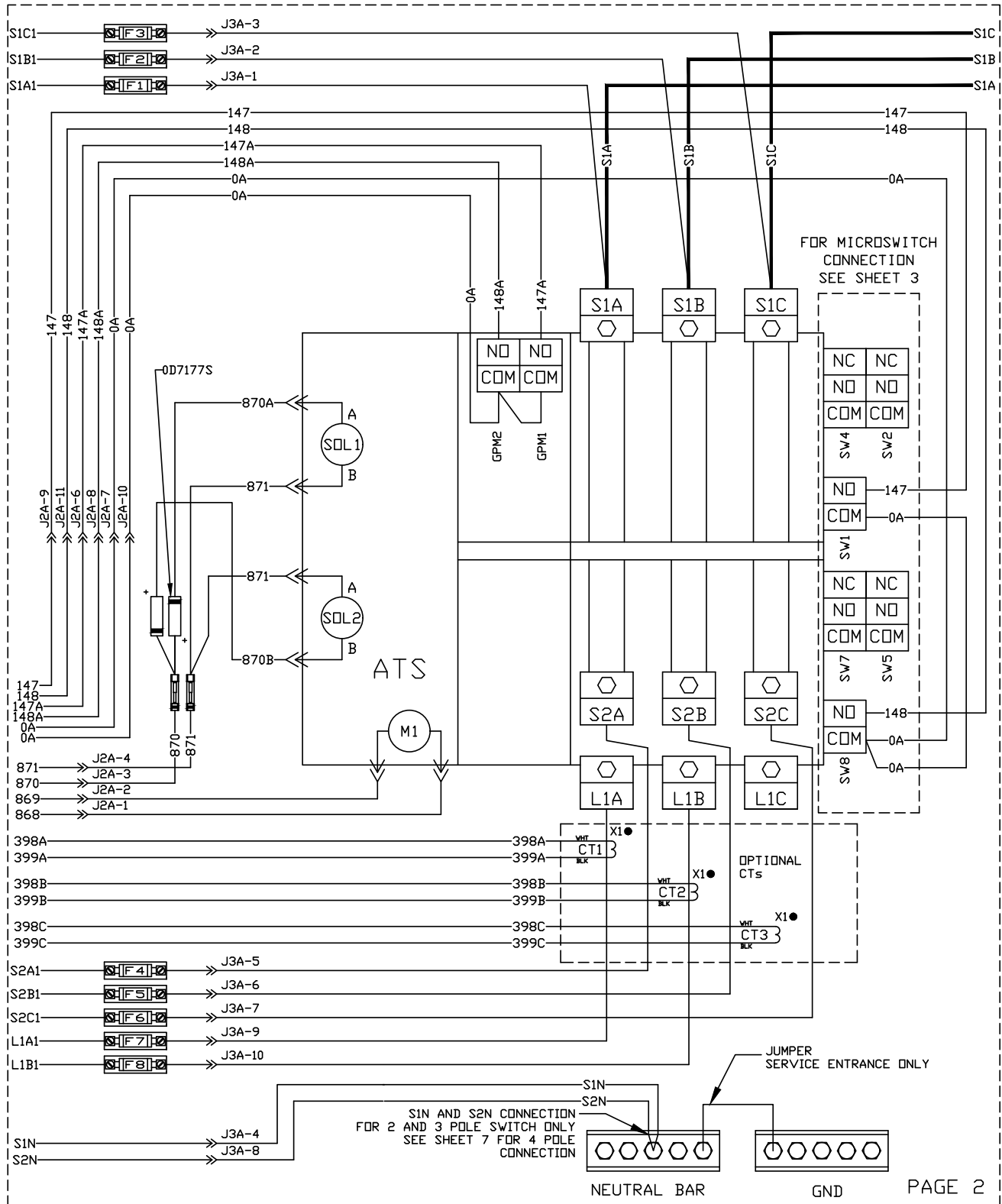


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A0001679448-G WD TX Control 800–1000A Transfer Switch (page 2 of 12)

GROUP G



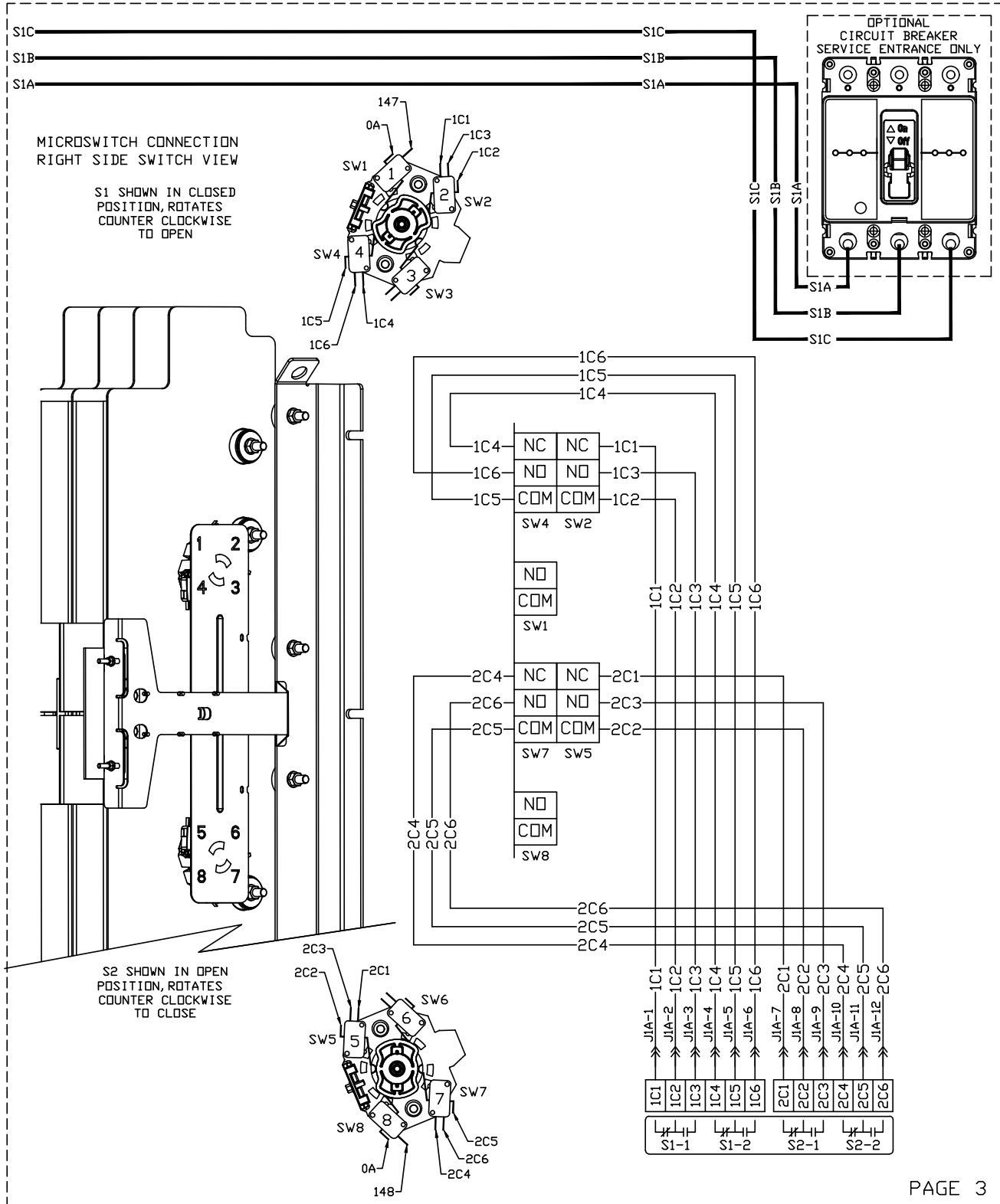
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WIRING - DIAGRAM
800-1600A TRANSFER SW
DRAWING #: A0001679448

A0001679448-G WD TX Control 800–1000A Transfer Switch (page 3 of 12)

GROUP G



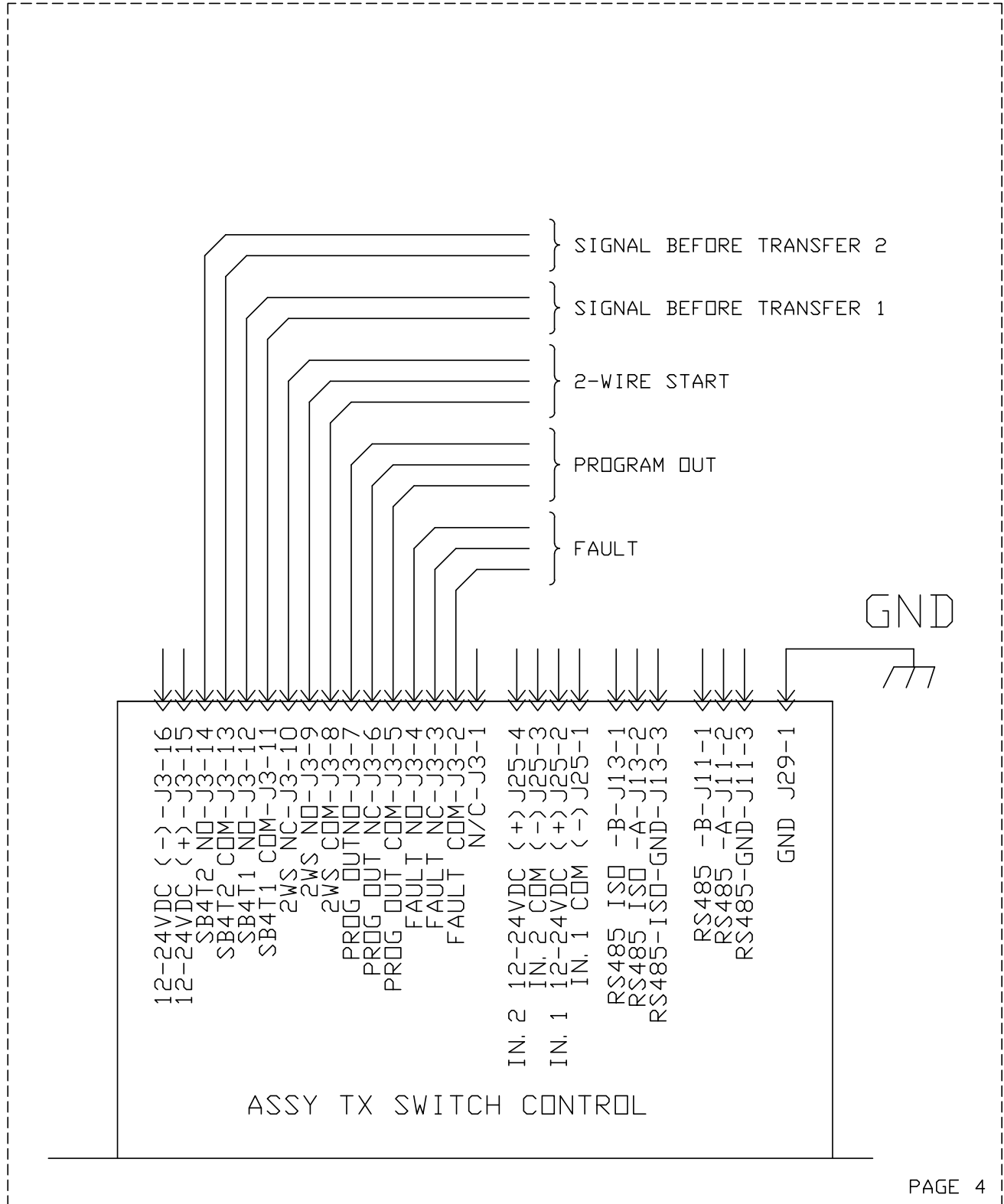
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WIRING - DIAGRAM
800-1600A TRANSFER SW
DRAWING #: A0001679448

A0001679448-G WD TX Control 800-1000A Transfer Switch (page 4 of 12)

GROUP G



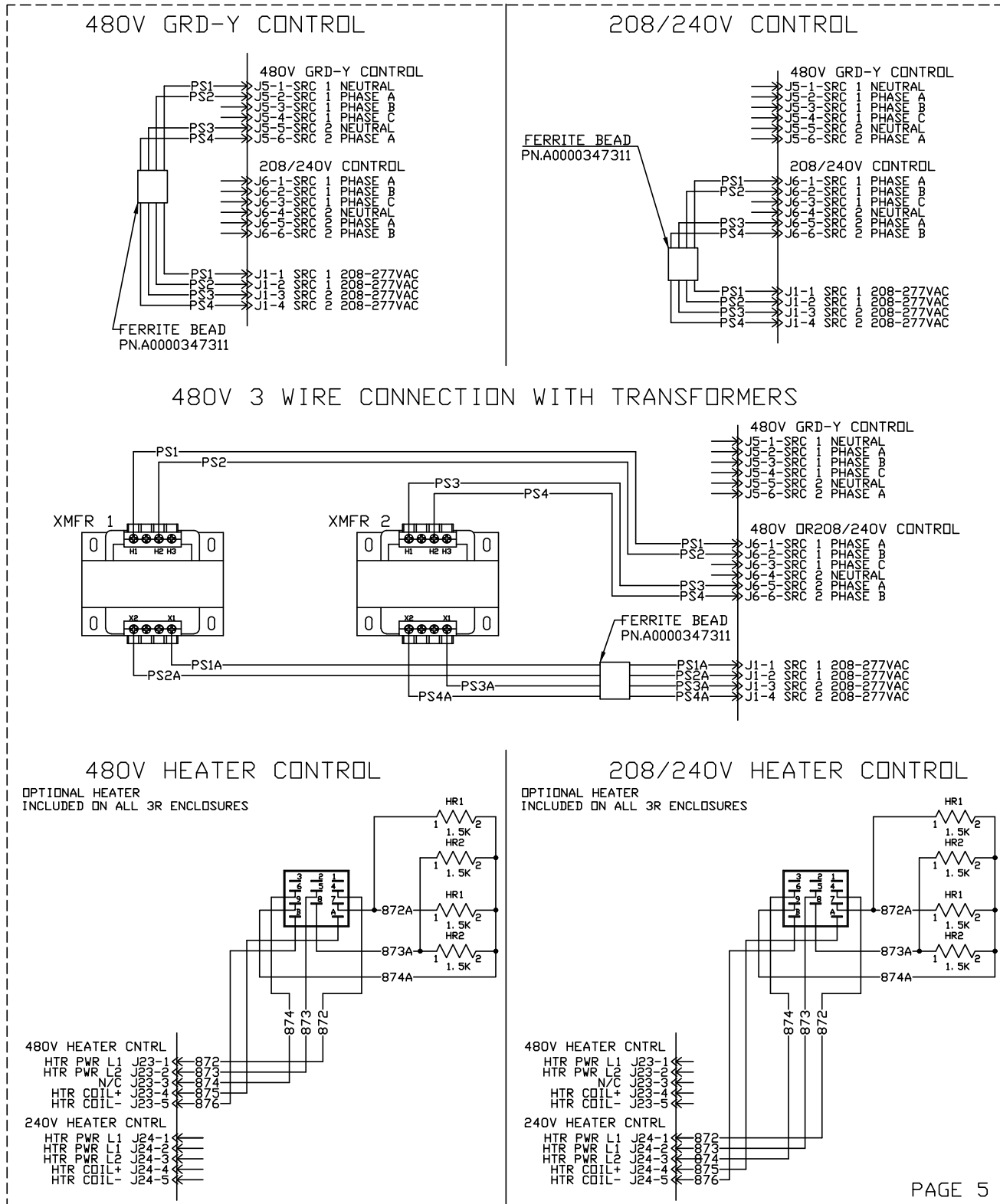
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DATE: 5/1/24

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WIRING - DIAGRAM
800-1600A TRANSFER SW
DRAWING #: A0001679448

A0001679448-G WD TX Control 800–1000A Transfer Switch (page 5 of 12)

GROUP G



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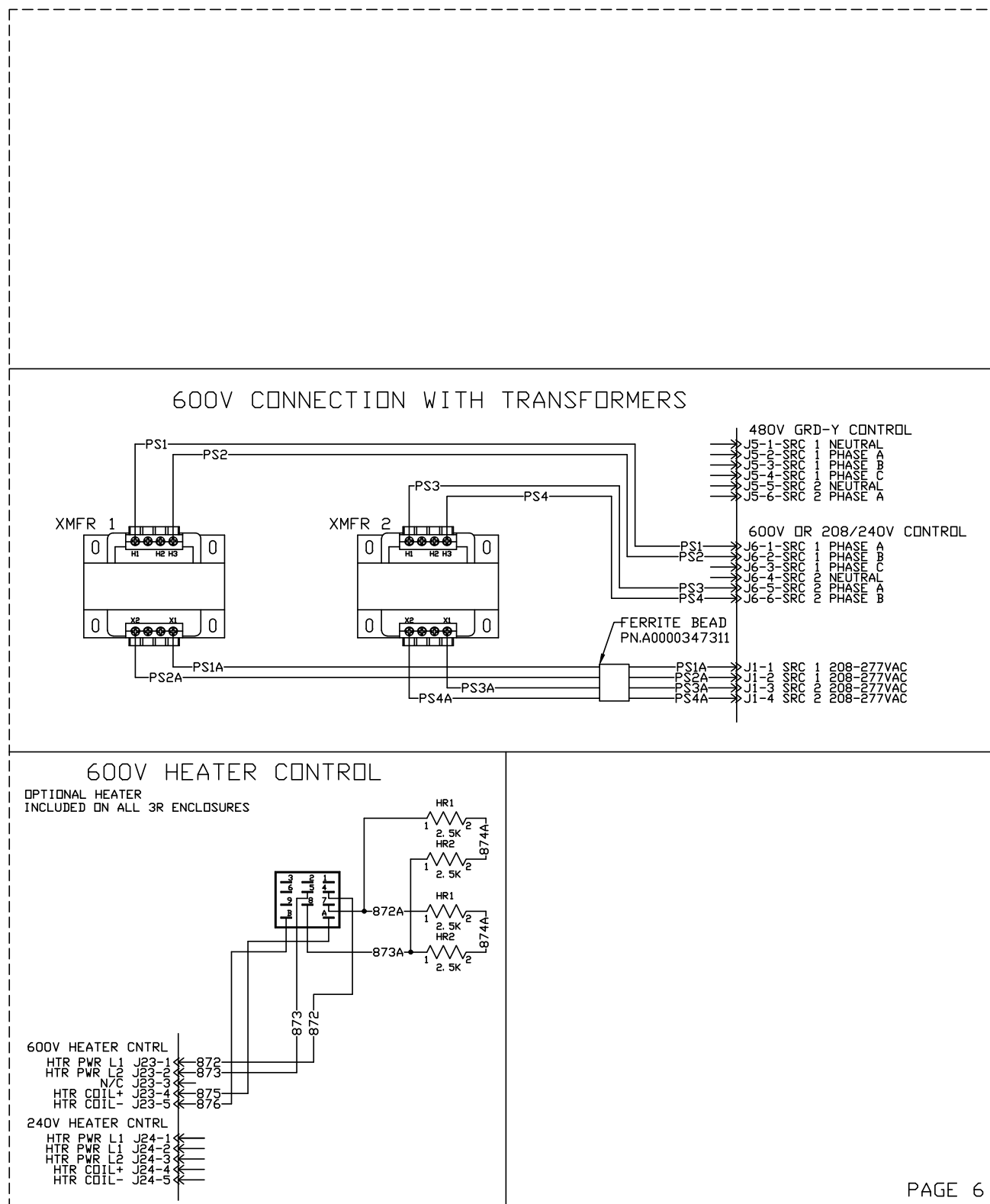
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WIRING - DIAGRAM
 800-1600A TRANSFER SW
 DRAWING #: A0001679448

A0001679448-G WD TX Control 800–1000A Transfer Switch (page 6 of 12)

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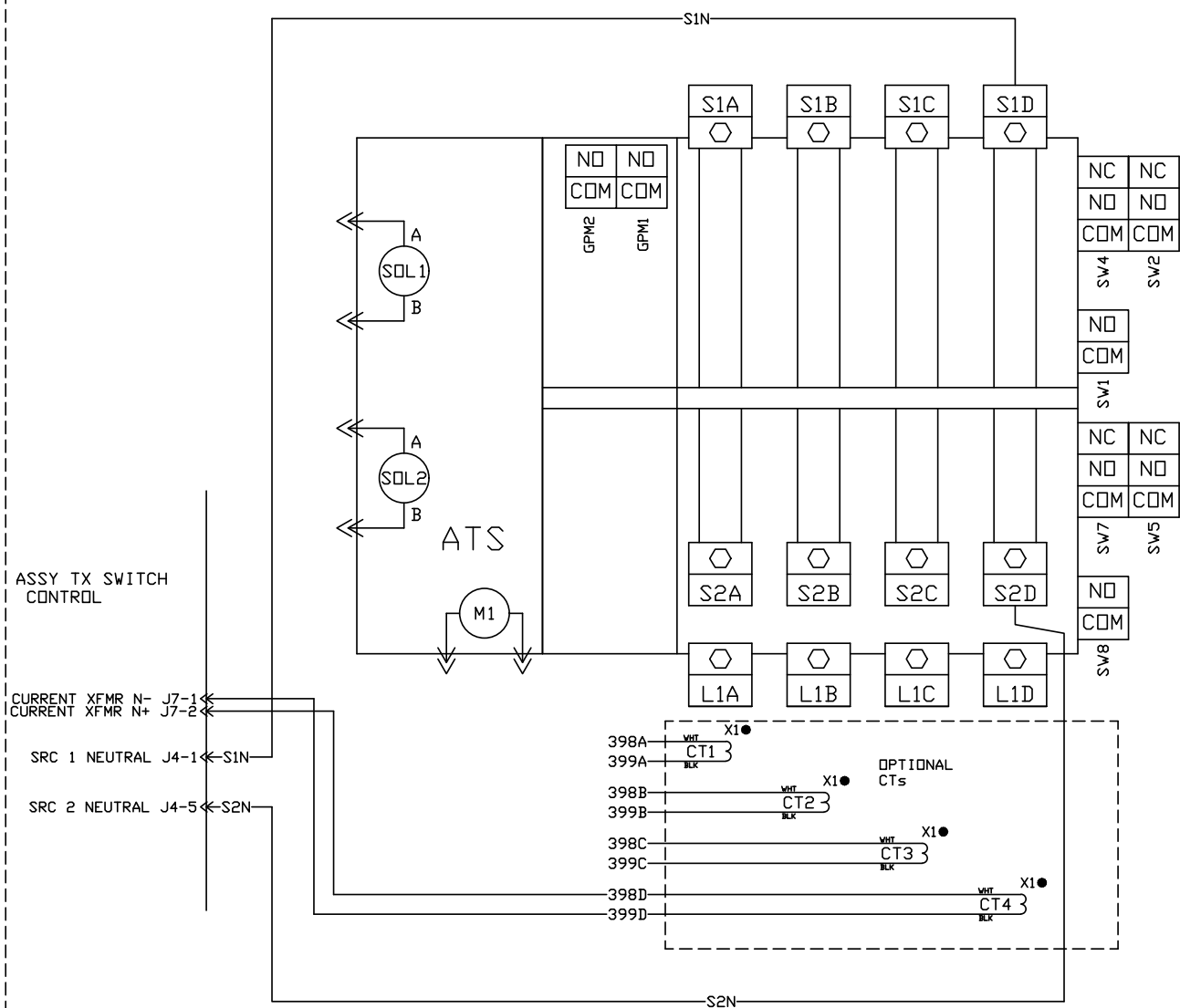
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WIRING - DIAGRAM
800-1600A TRANSFER SW
DRAWING #: A0001679448

A0001679448-G WD TX Control 800–1000A Transfer Switch (page 7 of 12)

GROUP G

4 POLE SWITCHED NEUTRAL CONNECTIONS



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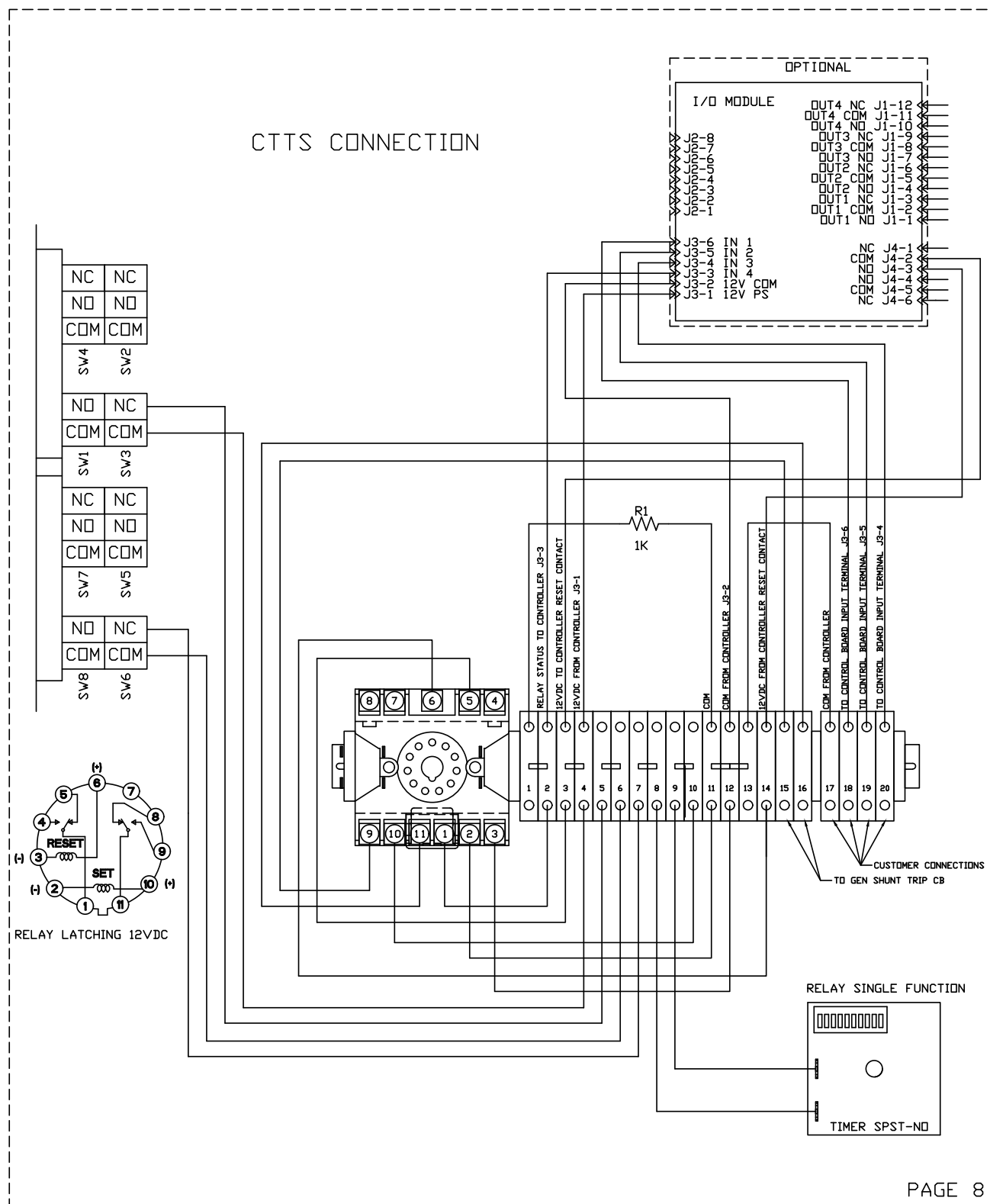
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WIRING - DIAGRAM
800-1600A TRANSFER SW
DRAWING #: A0001679448

A0001679448-G WD TX Control 800–1000A Transfer Switch (page 8 of 12)

GROUP G



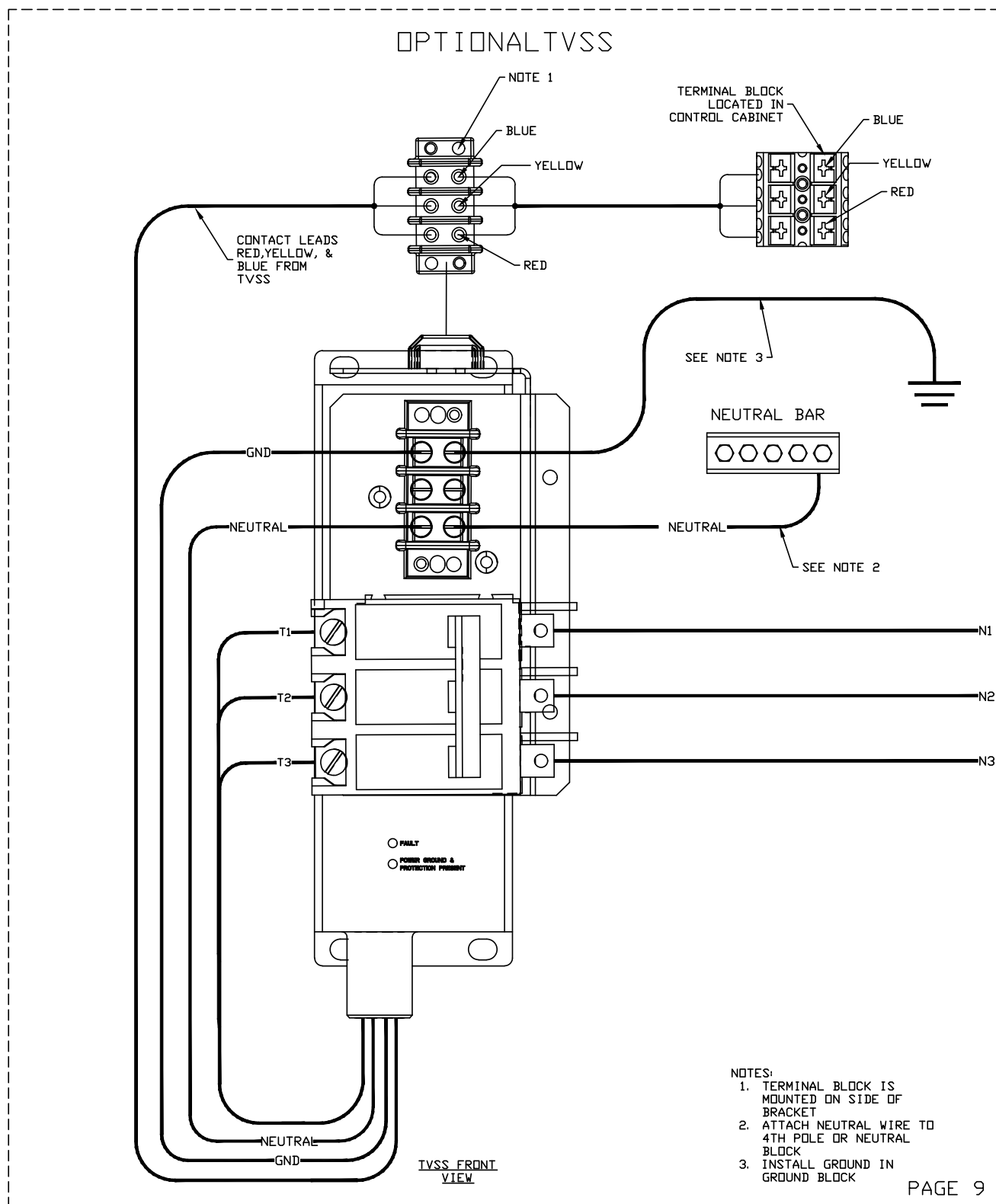
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WIRING - DIAGRAM
800-1600A TRANSFER SW
DRAWING #: A0001679448

A0001679448-G WD TX Control 800–1000A Transfer Switch (page 9 of 12)

GROUP G



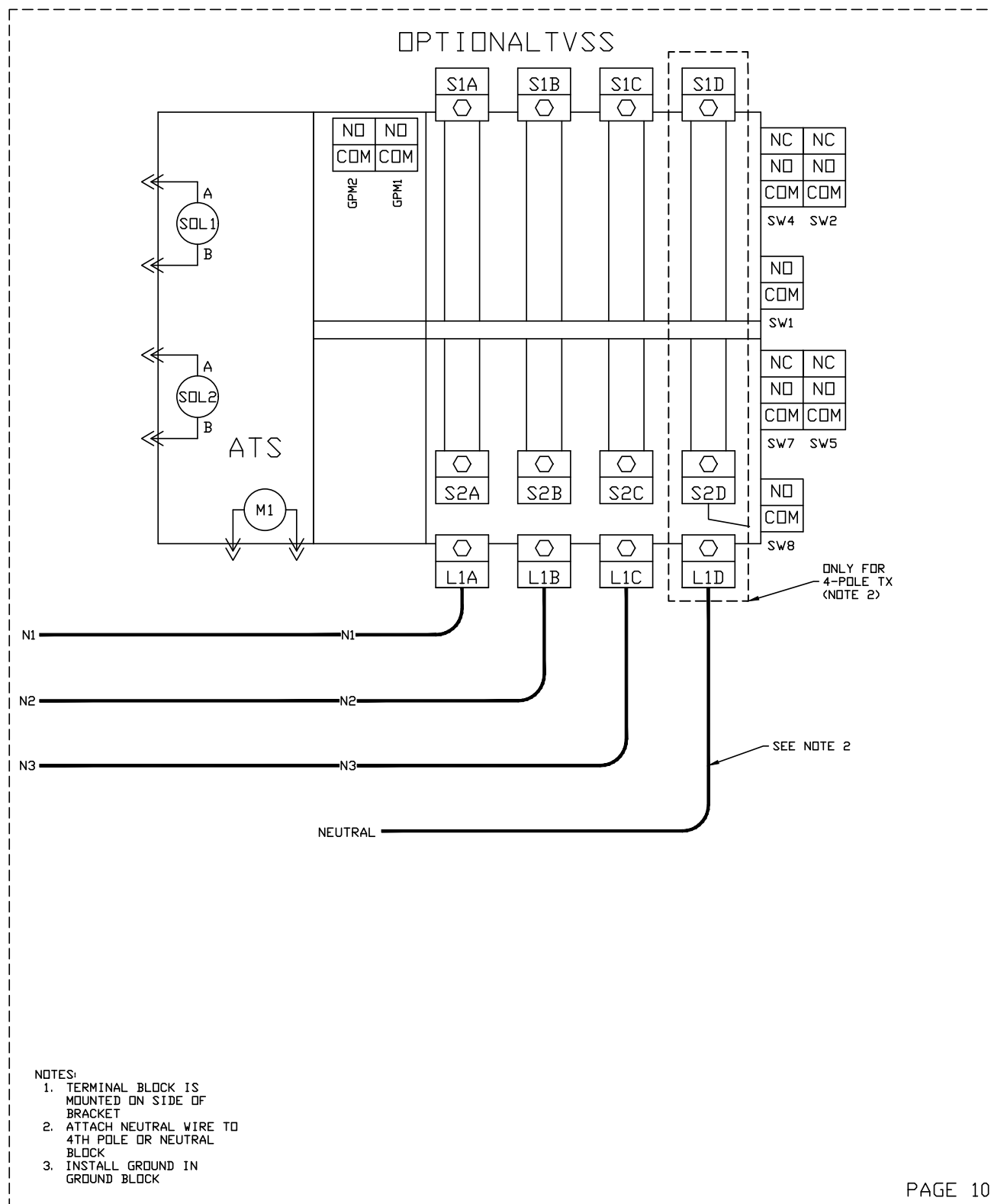
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WIRING - DIAGRAM
800-1600A TRANSFER SW
DRAWING #: A0001679448

A0001679448-G WD TX Control 800–1000A Transfer Switch (page 10 of 12)

GROUP G



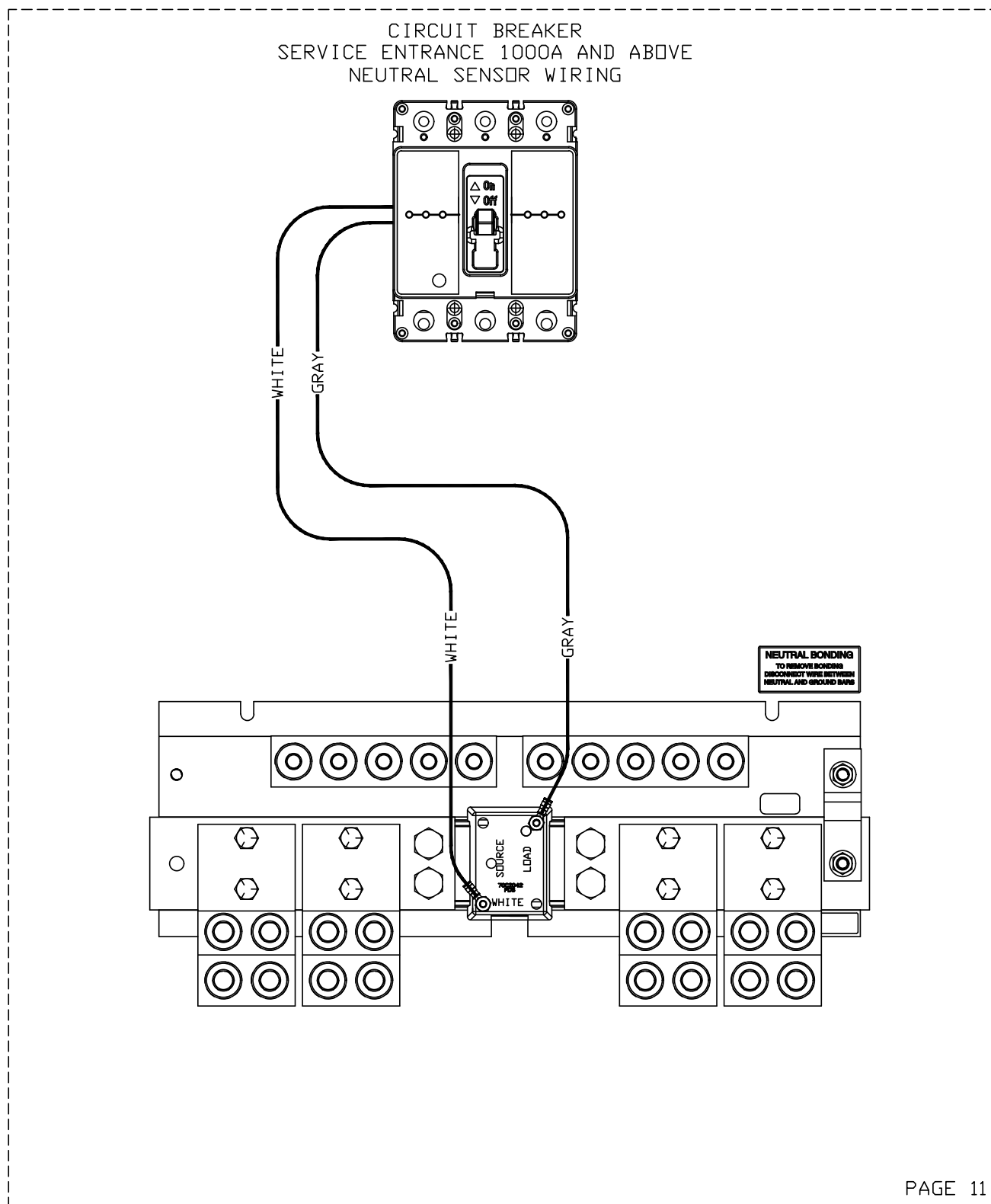
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WIRING - DIAGRAM
800-1600A TRANSFER SW
DRAWING #: A0001679448

A0001679448-G WD TX Control 800–1000A Transfer Switch (page 11 of 12)

GROUP G



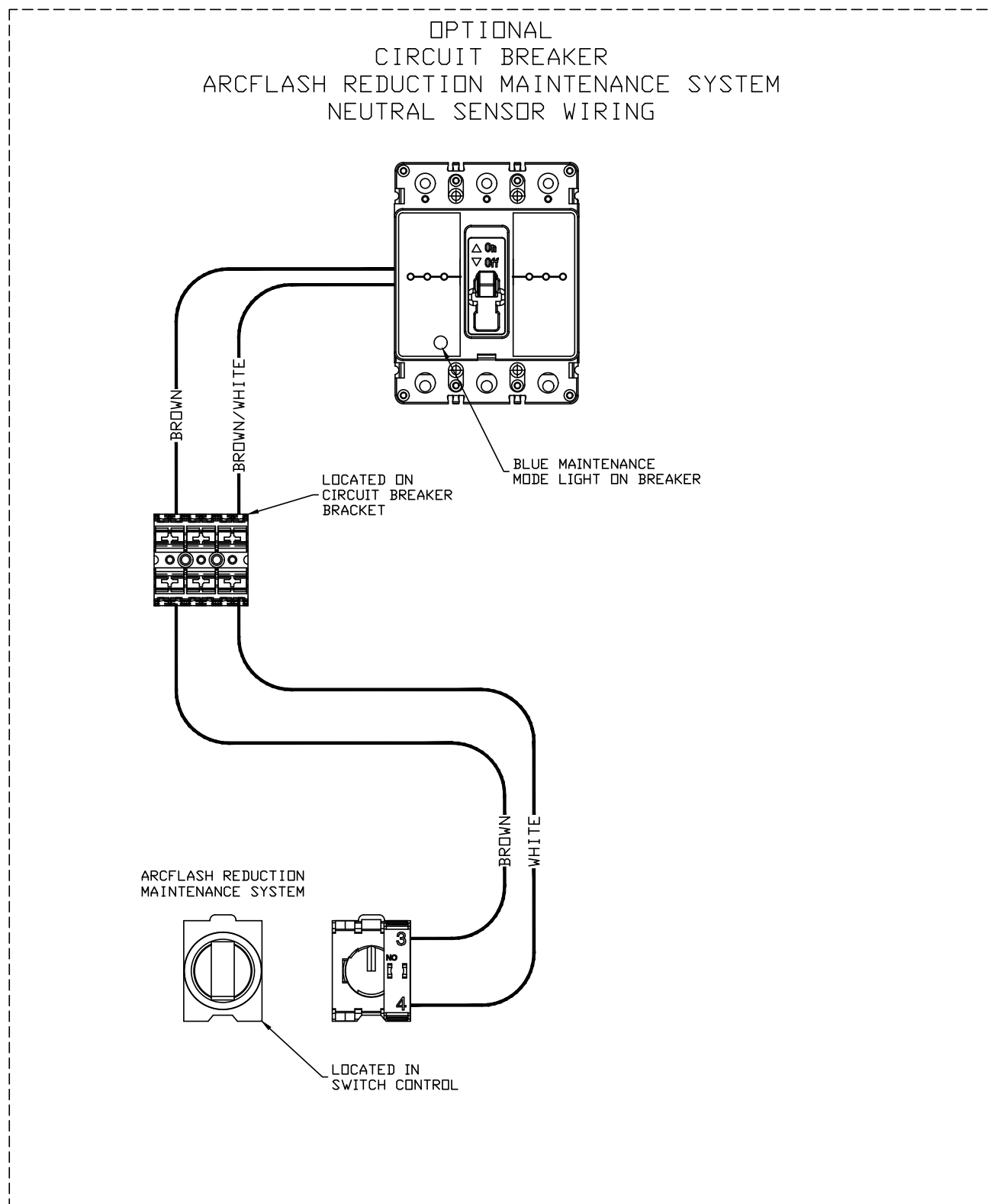
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WIRING - DIAGRAM
800-1600A TRANSFER SW
DRAWING #: A0001679448

A0001679448-G WD TX Control 800–1000A Transfer Switch (page 12 of 12)

GROUP G



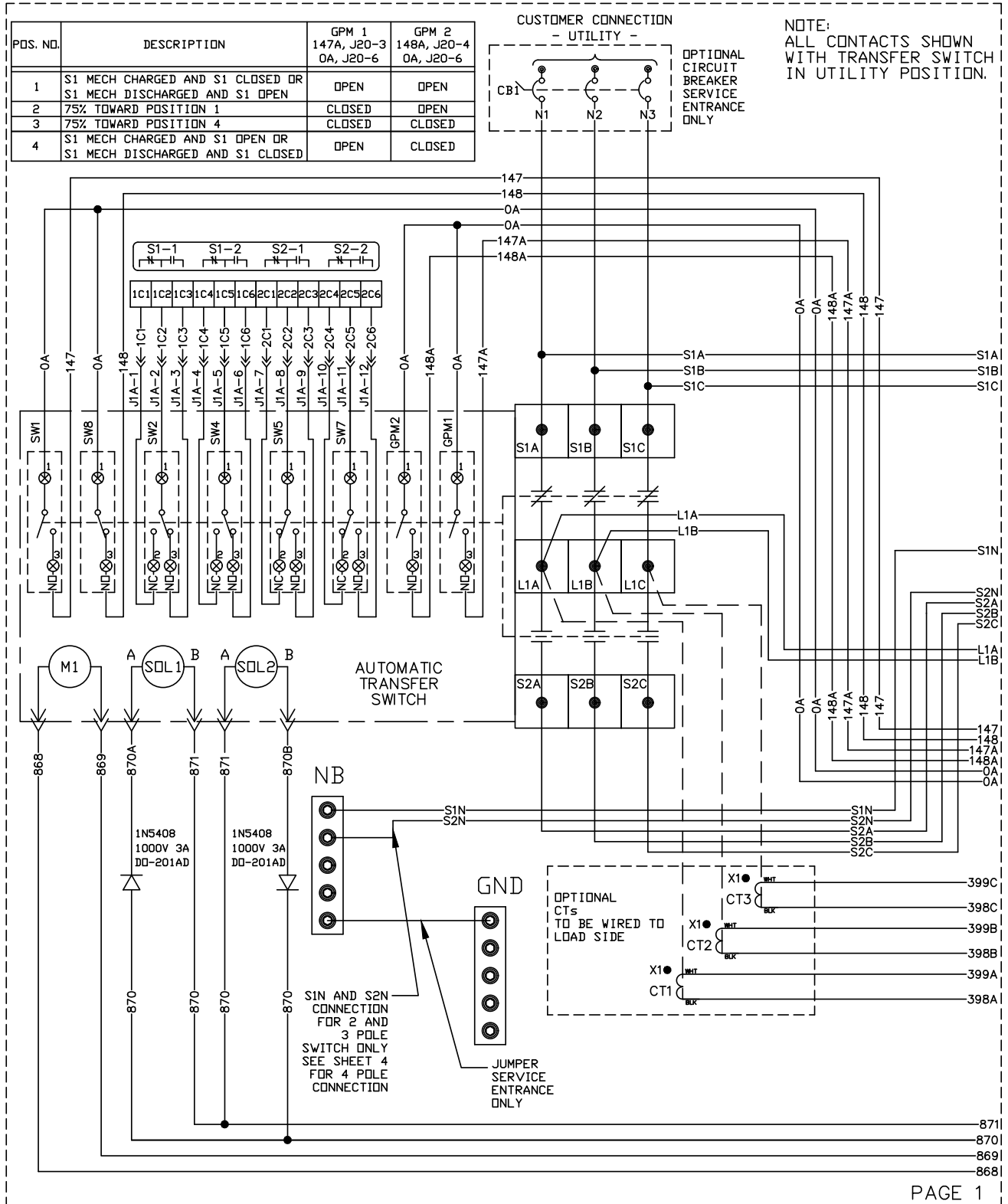
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WIRING - DIAGRAM
800-1600A TRANSFER SW
DRAWING #: A0001679448

A0001679450-G SD TX Control 800–1000A Transfer Switch (page 1 of 10)

GROUP G



PAGE 1

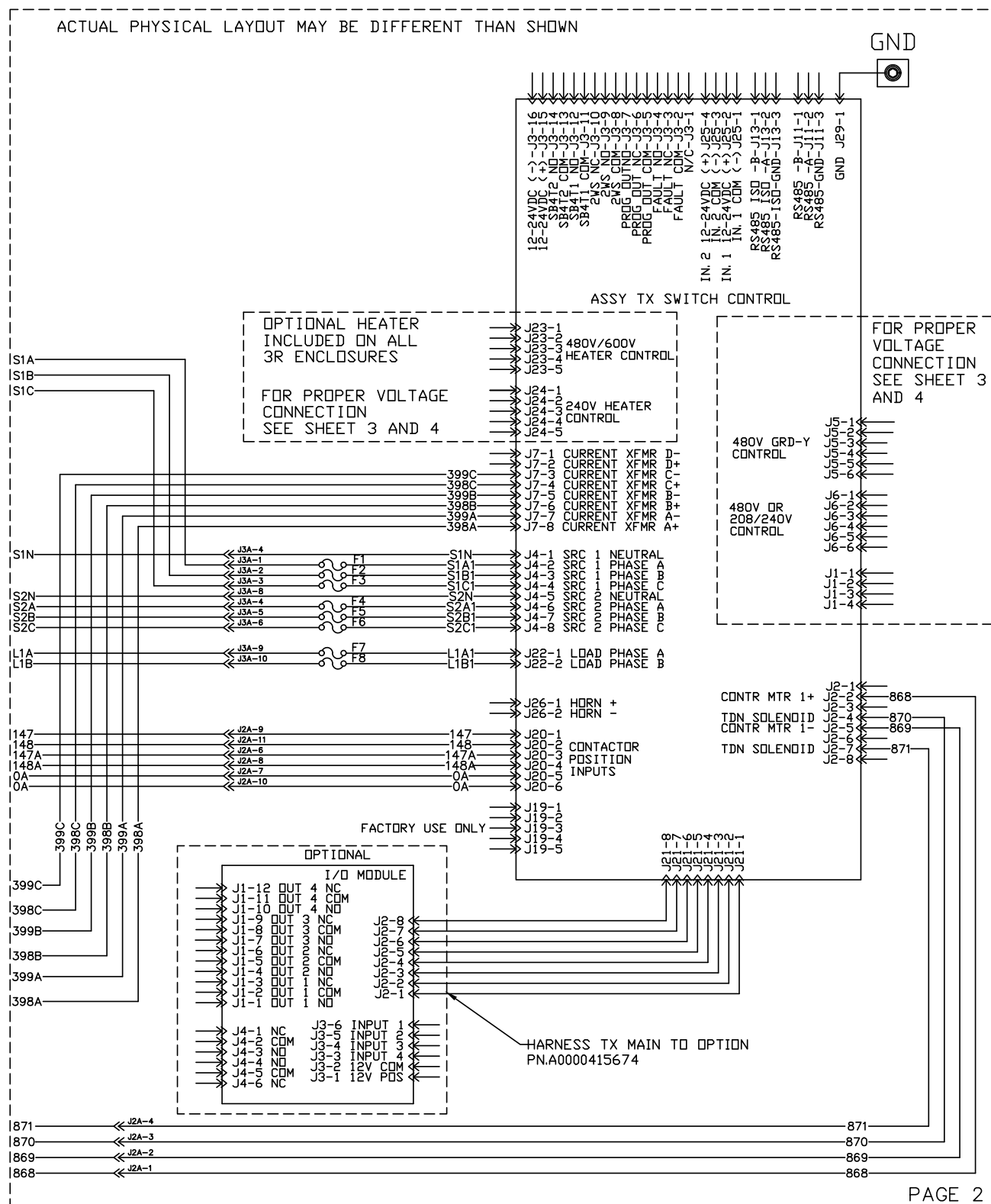
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SCHMATIC - DIAGRAM
800-1600A TRANSFER SW
DRAWING #: A0001679450

A0001679450-G SD TX Control 800-1000A Transfer Switch (page 2 of 10)

GROUP G



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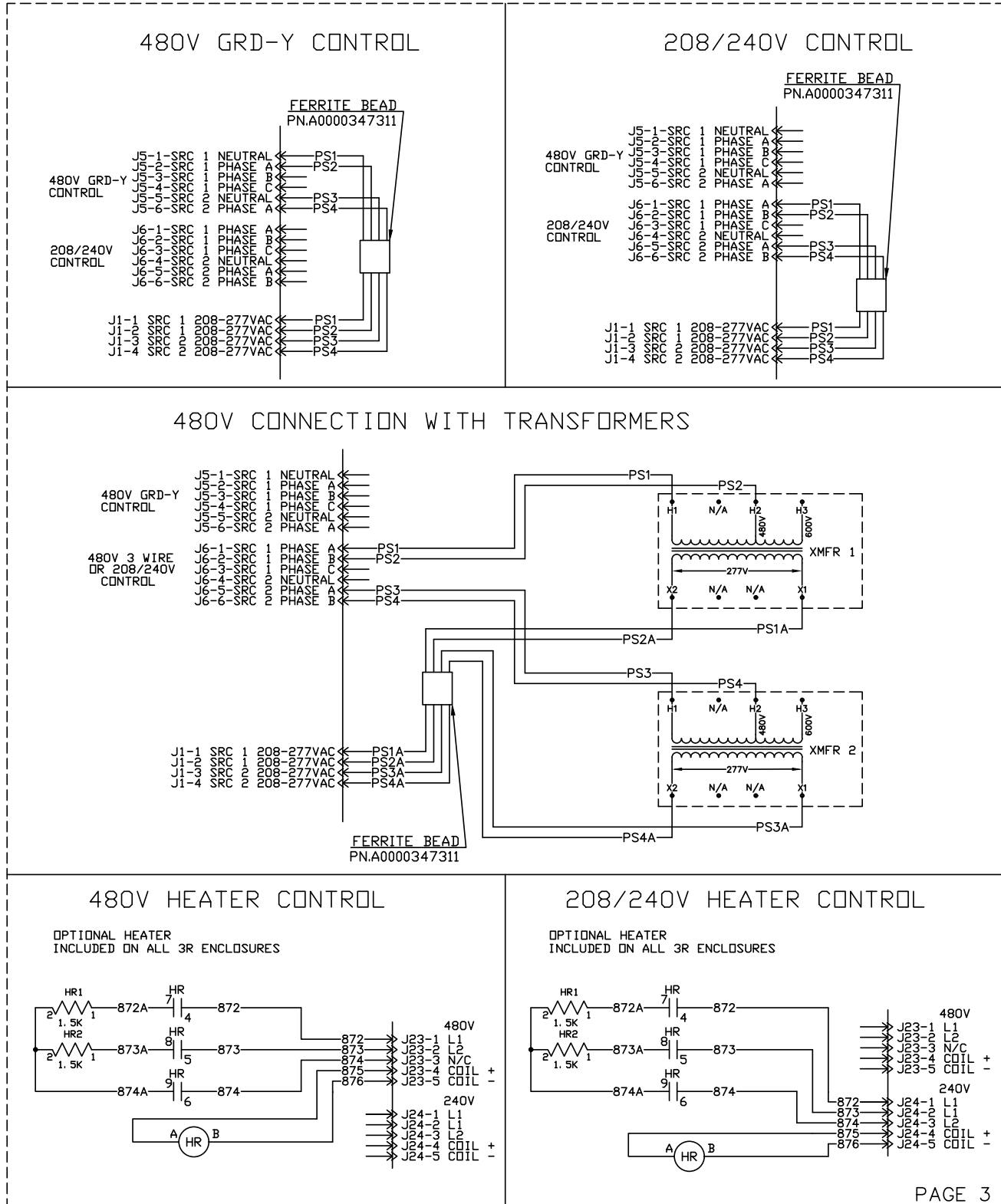
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SCHEMATIC - DIAGRAM
800-1600A TRANSFER SW
DRAWING #: A0001679450

A0001679450-G SD TX Control 800–1000A Transfer Switch (page 3 of 10)

GROUP G



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SCHEMATIC - DIAGRAM

800-1600A TRANSFER SW

DRAWING #: A0001679450

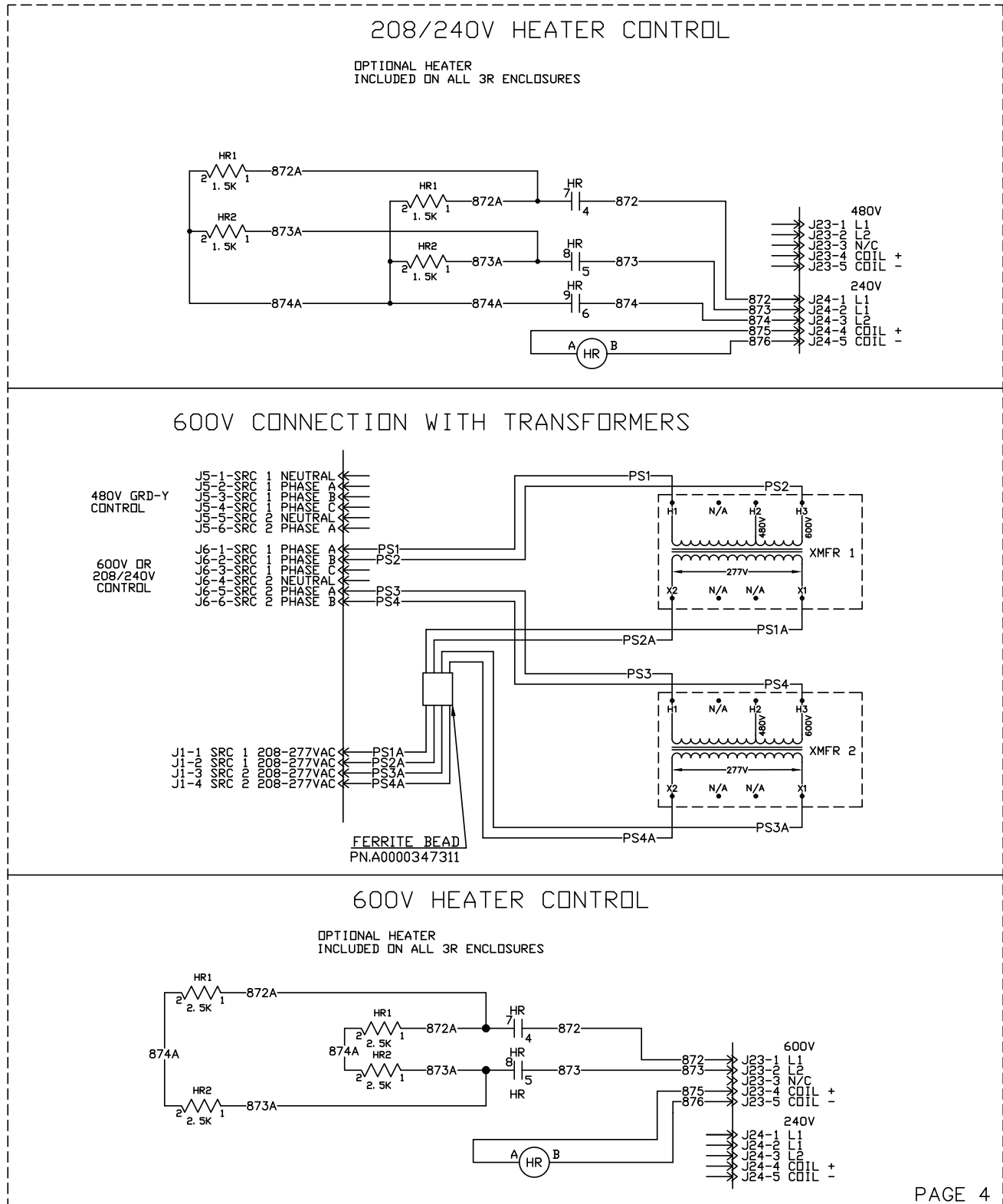
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A0001679450-G SD TX Control 800–1000A Transfer Switch (page 4 of 10)

GROUP G



PAGE 4

SCHEMATIC - DIAGRAM

800-1600A TRANSFER SW

DRAWING #: A0001679450

REVISION: CN-0084844-G

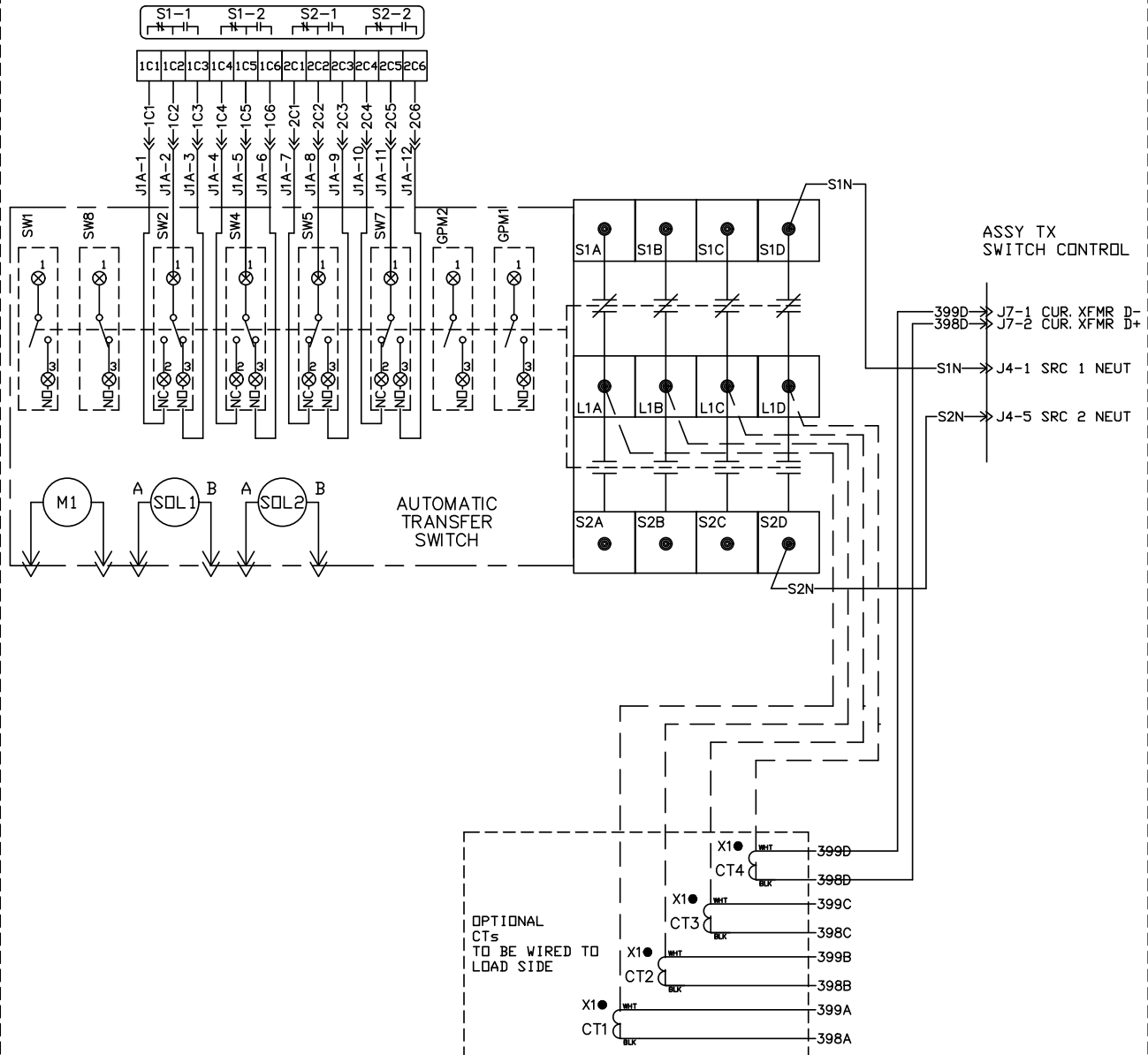
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A0001679450-G SD TX Control 800–1000A Transfer Switch (page 5 of 10)

GROUP G

4 POLE SWITCHED NEUTRAL CONNECTIONS



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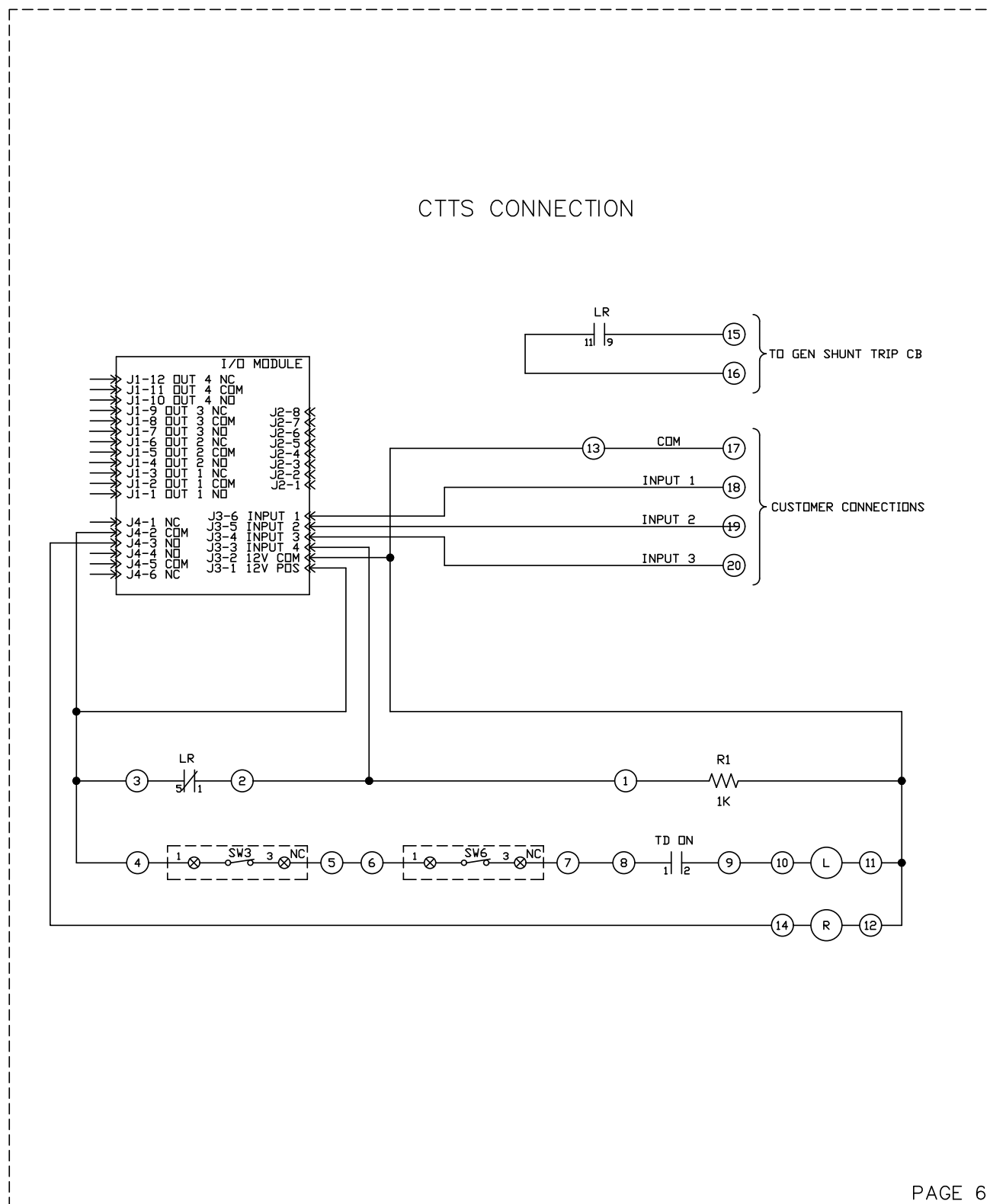
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SCHEMATIC - DIAGRAM
800-1600A TRANSFER SW
DRAWING #: A0001679450

A0001679450-G SD TX Control 800–1000A Transfer Switch (page 6 of 10)

GROUP G



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DATE: 5/1/24

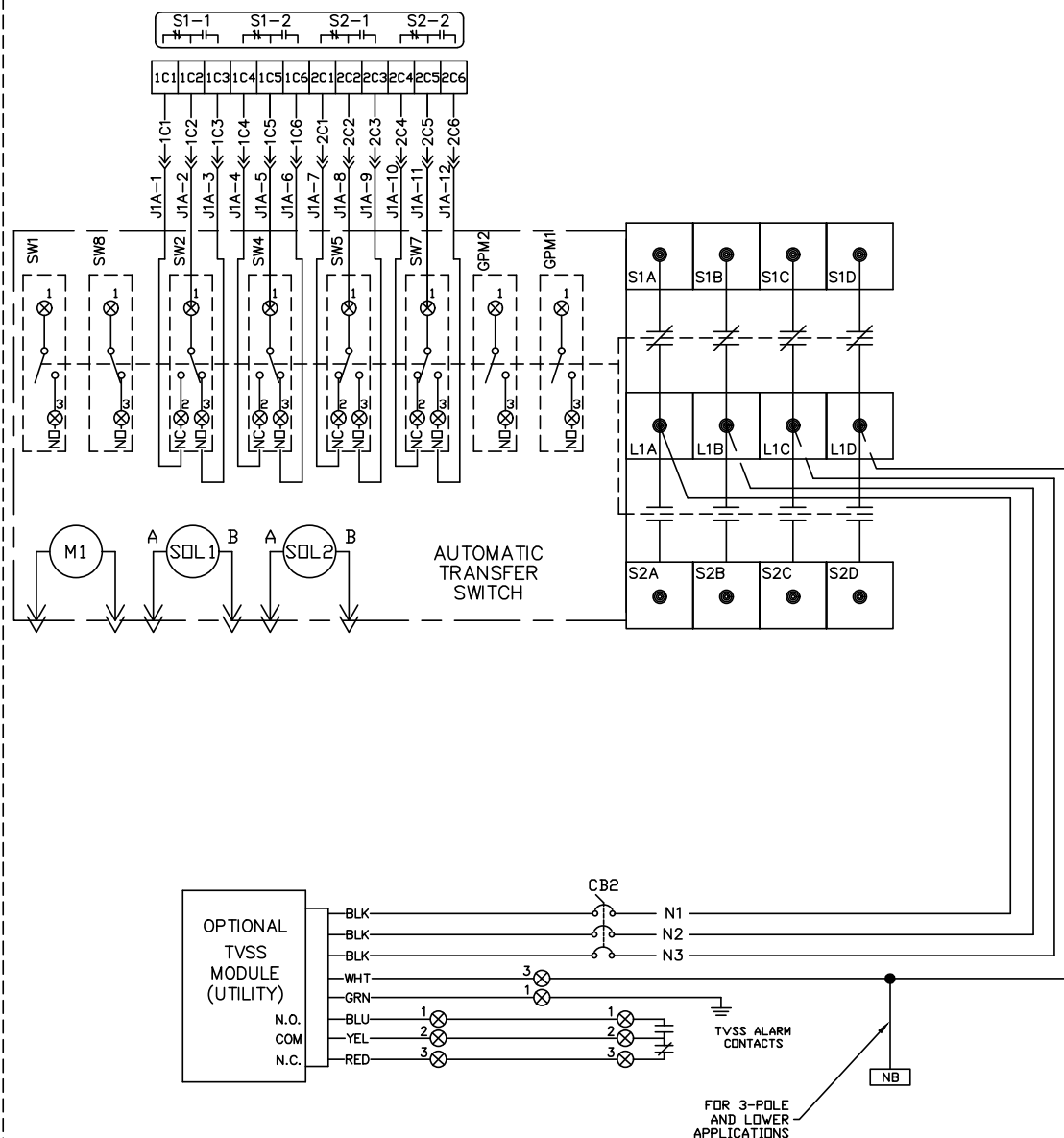
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SCHEMATIC - DIAGRAM
800-1600A TRANSFER SW
DRAWING #: A0001679450

A0001679450-G SD TX Control 800–1000A Transfer Switch (page 7 of 10)

GROUP G

3 & 4-POLE TVSS OPTION FOR TX TRANSFER SWITCH



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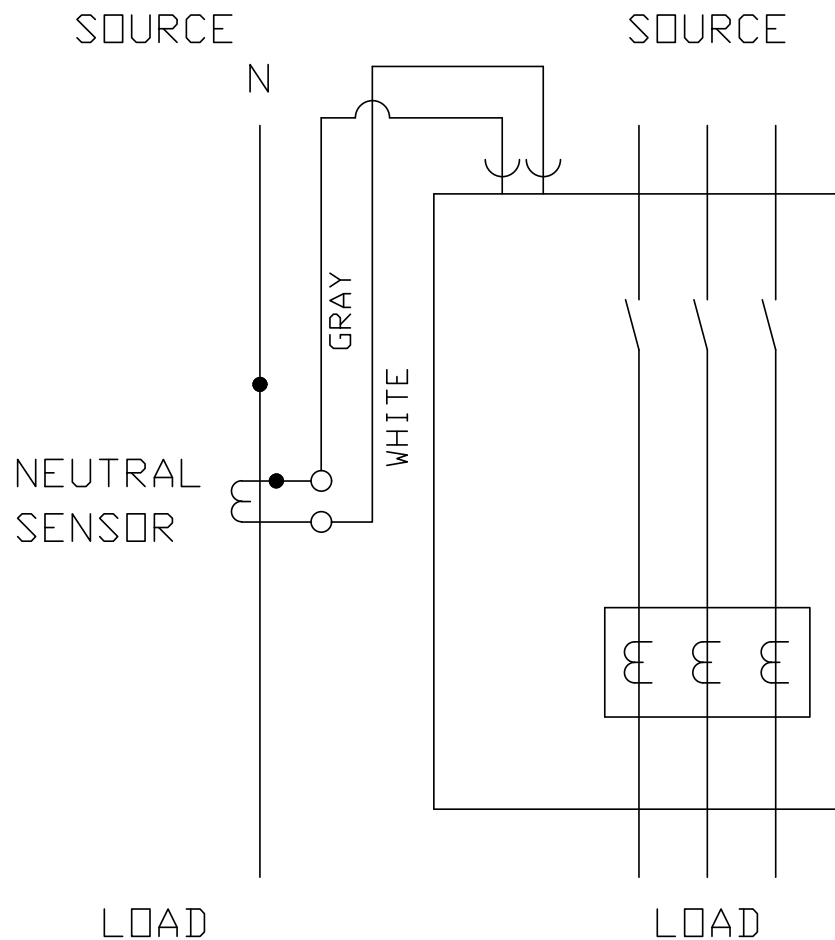
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SCHEMATIC - DIAGRAM
800-1600A TRANSFER SW
DRAWING #: A0001679450

A0001679450-G SD TX Control 800–1000A Transfer Switch (page 8 of 10)**GROUP G**

CIRCUIT BREAKER
SERVICE ENTRANCE 1000A AND ABOVE
NEUTRAL SENSOR WIRING



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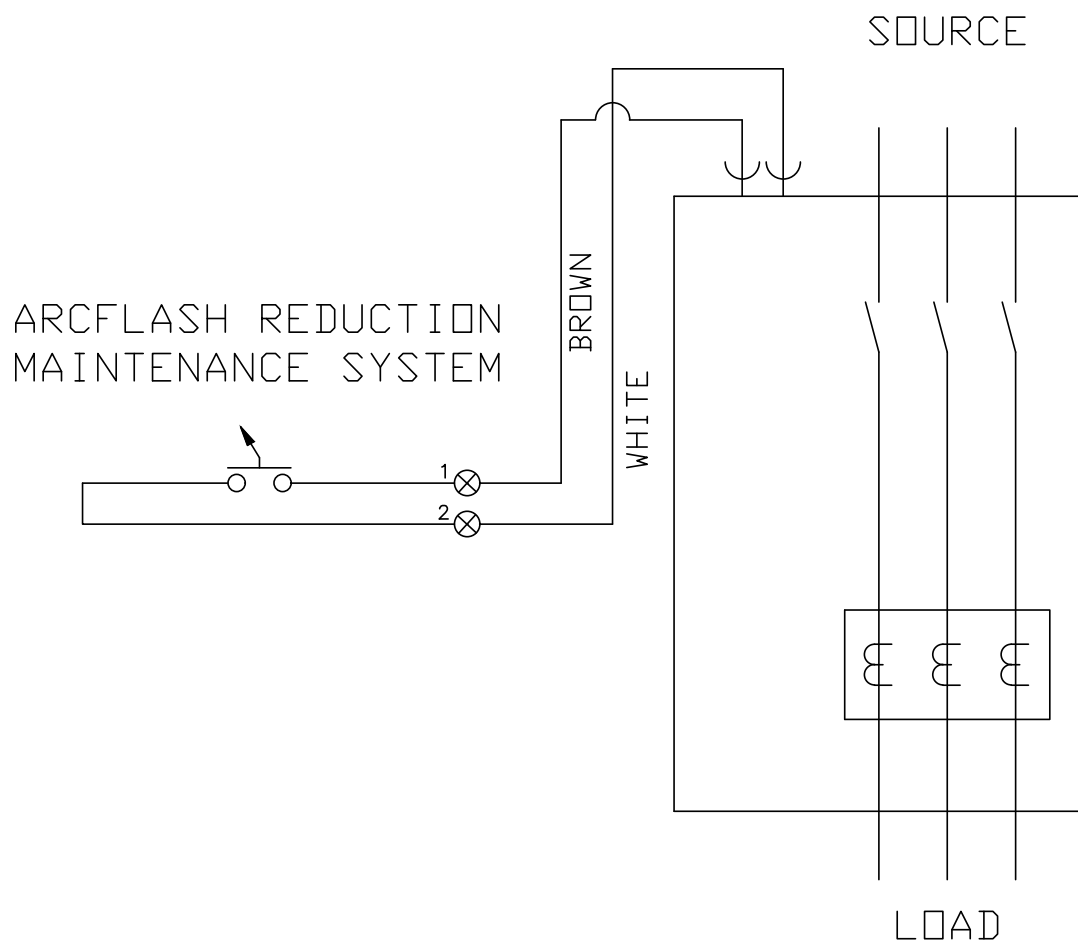
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SCHEMATIC - DIAGRAM
800-1600A TRANSFER SW
DRAWING #: A0001679450

A0001679450-G SD TX Control 800–1000A Transfer Switch (page 9 of 10)

GROUP G

OPTIONAL CIRCUIT BREAKER ARCFLASH REDUCTION MAINTENANCE SYSTEM ARMS



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SCHEMATIC - DIAGRAM
800-1600A TRANSFER SW
DRAWING #: A0001679450

A0001679450-G SD TX Control 800–1000A Transfer Switch (page 10 of 10)

GROUP G

BLANK SHEET

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SCHEMATIC - DIAGRAM
800-1600A TRANSFER SW
DRAWING #: A0001679450

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Appendix A *Permissive and Loadshed Functions*

Permissive and Loadshed functions may be programmed to any available input, but are generally to be used with the B.In 1 or B.In 2, corresponding to J25:1,2 and J25:3,4 respectively. Any of the available inputs may be programmed as Permissive.

NOTE: There is no explicit Loadshed function as it is functionally equivalent to Permissive.

To activate functionality, the input must be programmed to be associated with the *PERMISSIVE* function via the *MENU - SETTINGS - I/O* menu on the controller display. Once the function is associated with an input, transfer out of the primary source will be prevented until the input is asserted.

NOTE: The assertion of the input may be made by applying a 12/24 VDC nominal voltage across the optically-isolated inputs J25:1(-) and J25:2(+) and similarly to J25:3(-) and J25:4(+).

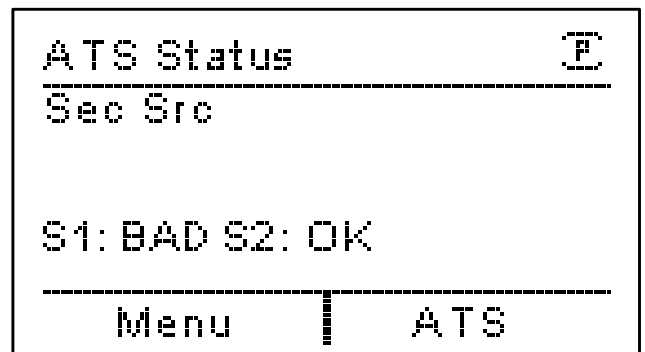
This assertion is indicated on the display by means of a P with a circle around it (in the top right corner), allowing transfer to continue. The allowable voltage range is 8-35 VDC. The impedance of this input is 16 kOhms, resulting in approximately 1-2 mA of current.

Verify appropriate signal relay or external loading is used on a general purpose relay so that minimum relay current requirements are met.

If an external voltage supply is not available, an I/O expander module may be used as a voltage source, using J3:1 (+12VDC) and J3:2 (COM). This voltage may be safely looped through the externally supplied signal level dry contacts and brought back to supply the power for any of the available inputs. Note that a maximum of 50mA may be pulled from this supply.

As noted in the manual, Loadshed is the de-assertion of the *PERMISSIVE* input while the switch is in the secondary source. On a 3-position switch, this will initiate a transfer to the OFF position.

NOTE: A 2-position switch cannot transfer to a dead source per UL standard requirement and therefore Loadshed is not programmed with any functionality.



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Figure A-1.

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Electrical Formulas

To Find	Known Values	1-phase	3-phase
Kilowatts (kW)	Volts, Current, Power Factor	$\frac{E \times I}{1000}$	$\frac{E \times I \times 1.73 \times PF}{1000}$
KVA	Volts, Current	$\frac{E \times I}{1000}$	$\frac{E \times I \times 1.73}{1000}$
Amperes	kW, Volts, Power Factor	$\frac{kW \times 1000}{E}$	$\frac{kW \times 1000}{E \times 1.73 \times PF}$
Watts	Volts, Amps, Power Factor	Volts x Amps	$E \times I \times 1.73 \times PF$
No. of Rotor Poles	Frequency, RPM	$\frac{2 \times 60 \times \text{Frequency}}{\text{RPM}}$	$\frac{2 \times 60 \times \text{Frequency}}{\text{RPM}}$
Frequency	RPM, No. of Rotor Poles	$\frac{\text{RPM} \times \text{Poles}}{2 \times 60}$	$\frac{\text{RPM} \times \text{Poles}}{2 \times 60}$
RPM	Frequency, No. of Rotor Poles	$\frac{2 \times 60 \times \text{Frequency}}{\text{Rotor Poles}}$	$\frac{2 \times 60 \times \text{Frequency}}{\text{Rotor Poles}}$
kW (required for Motor)	Motor Horsepower, Efficiency	$\frac{HP \times 0.746}{\text{Efficiency}}$	$\frac{HP \times 0.746}{\text{Efficiency}}$
Resistance	Volts, Amperes	$\frac{E}{I}$	$\frac{E}{I}$
Volts	Ohm, Amperes	$I \times R$	$I \times R$
Amperes	Ohms, Volts	$\frac{E}{R}$	$\frac{E}{R}$

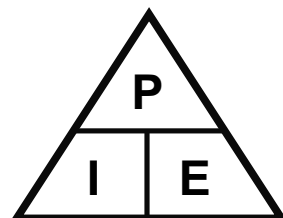
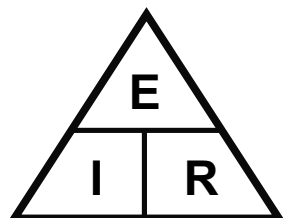
E = Volts

I = Amperes

R = Resistance (Ohms)

PF = Power Factor

Term	Symbol	Measurement
Current	I	Amps
Wattage	P	Watts
Voltage	E	Volts
Resistance	R	Ohms



003003

Constant	Shift		Result	
Voltage E	Resistance Increase	↑	Current Decrease	↓
Voltage E	Resistance Decrease	↓	Current Increase	↑
Resistance R	Voltage Decrease	↓	Current Decrease	↓
Resistance R	Voltage Increase	↑	Current Increase	↑
Current I	Resistance Decrease	↓	Voltage Decrease	↓
Current I	Resistance Increase	↑	Voltage Increase	↑
Power P	Voltage Increase	↑	Power Increase	↑
Power P	Voltage Decrease	↓	Power Decrease	↓
Power P	Current Increase	↑	Power Increase	↑
Power P	Current Decrease	↓	Power Decrease	↓

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